

8		7		6		5		4		3		2		1	
1. ALL RESISTANCE VALUES ARE IN OHMS, 0.1 WATT +/- 5%. 2. ALL CAPACITANCE VALUES ARE IN MICROFARADS. 3. ALL CRYSTALS & OSCILLATOR VALUES ARE IN HERTZ.												REV	ECN	DESCRIPTION OF REVISION	CK APPD DATE
												<REV>	<ECN>	<ECO_DESCRIPTION>	<ECODATE>
X260 MLB - CBB															
LAST_MODIFICATION=Thu Sep 10 15:22:41 2015															
PAGE		<CSA>	CONTENTS		SYNC		DATE		PAGE		<CSA>	CONTENTS		SYNC DATE	
1		1	page1		MASTER		MASTER		46		76	page78		DEVMLB 05/13/2015	
2		2	page2		J43_MLB		10/24/2012		47		78	page79		DEVMLB 05/11/2015	
3		3	page3		J43_MLB		10/24/2012		48		79	page80		DEVMLB 05/11/2015	
4		4	page4		J43_MLB		10/24/2012		49		80	page81		DEVMLB 05/11/2015	
5		5	page5		DEVMLB		05/11/2015		50		81	page82		DEVMLB 04/30/2015	
6		6	page6		DEVMLB		05/11/2015		51		82	page83		DEVMLB 05/06/2015	
7		7	page7		DEVMLB		04/14/2015		52		83	page84		DEVMLB 05/11/2015	
8		8	page8		DEVMLB		05/11/2015		53		84	page85		DEVMLB 03/16/2015	
9		9	page9		DEVMLB		05/11/2015		54		85	page86		DEVMLB 05/11/2015	
10		10	page10		DEVMLB		04/14/2015		55		86	page87		DEVMLB 05/13/2015	
11		11	page11		DEVMLB		04/14/2015		56		87	page88		DEVMLB 05/11/2015	
12		12	page12		DEVMLB		05/06/2015		57		88	page90		DEVMLB 05/11/2015	
13		13	page13		DEVMLB		05/11/2015		58		90	page91		DEVMLB 05/11/2015	
14		14	page14		DEVMLB		05/11/2015		59		91	page93		DEVMLB 05/13/2015	
15		15	page15		DEVMLB		05/11/2015		60		93	page96		DEVMLB 05/13/2015	
16		16	page16		DEVMLB		05/11/2015		61		96	page100		DEVMLB 03/16/2015	
17		17	page17		DEVMLB		05/11/2015		62		100	page102		J43_MLB 10/24/2012	
18		19	page19		DEVMLB		05/11/2015		63		102	page103		DEVMLB 04/30/2015	
19		20	page20		DEVMLB		05/04/2015		64		103	page104		J92_DEVMLB 07/08/2014	
20		22	page22		DEVMLB		04/14/2015		65		104	page105		J92_DEVMLB 07/08/2014	
21		23	page23		DEVMLB		04/14/2015		66		105	page120		J41_MLB 10/24/2012	
22		25	page25		DEVMLB		04/14/2015		67		120	page130		J92_LAYOUT_STUDY_MLB 05/18/2015	
23		30	page30		DEVMLB		04/09/2015				130	MASTER		MASTER	
24		31	page31		DEVMLB		05/11/2015				<CSA_PAGE69>	<SYNC_MASTER69>		<SYNC_DATE69>	
25		37	page37		J72_MLB		05/13/2015				<CSA_PAGE70>	<SYNC_MASTER70>		<SYNC_DATE70>	
26		38	page38		J92_DEVMLB		05/13/2015				<CSA_PAGE71>	<SYNC_MASTER71>		<SYNC_DATE71>	
27		39	page39		DEVMLB		04/14/2015				<CSA_PAGE72>	<SYNC_MASTER72>		<SYNC_DATE72>	
28		40	page40		DEVMLB		04/14/2015				<CSA_PAGE73>	<SYNC_MASTER73>		<SYNC_DATE73>	
29		48	page48		DEVMLB		05/11/2015				<CSA_PAGE74>	<SYNC_MASTER74>		<SYNC_DATE74>	
30		50	page50		DEVMLB		05/11/2015				<CSA_PAGE75>	<SYNC_MASTER75>		<SYNC_DATE75>	
31		51	page51		DEVMLB		04/14/2015				<CSA_PAGE76>	<SYNC_MASTER76>		<SYNC_DATE76>	
32		52	page52		DEVMLB		04/14/2015				<CSA_PAGE77>	<SYNC_MASTER77>		<SYNC_DATE77>	
33		53	page53		DEVMLB		05/11/2015				<CSA_PAGE78>	<SYNC_MASTER78>		<SYNC_DATE78>	
34		54	page54		DEVMLB		05/11/2015				<CSA_PAGE79>	<SYNC_MASTER79>		<SYNC_DATE79>	
35		55	page55		DEVMLB		03/16/2015				<CSA_PAGE80>	<SYNC_MASTER80>		<SYNC_DATE80>	
36		61	page61		J92_DEVMLB		07/23/2013				<CSA_PAGE81>	<SYNC_MASTER81>		<SYNC_DATE81>	
37		62	page63		DEVMLB		05/11/2015				<CSA_PAGE82>	<SYNC_MASTER82>		<SYNC_DATE82>	
38		63	page64		DEVMLB		05/11/2015				<CSA_PAGE83>	<SYNC_MASTER83>		<SYNC_DATE83>	
39		64	page67		DEVMLB		05/11/2015				<CSA_PAGE84>	<SYNC_MASTER84>		<SYNC_DATE84>	
40		67	page69		DEVMLB		03/16/2015				<CSA_PAGE85>	<SYNC_MASTER85>		<SYNC_DATE85>	
41		69	page70		<SYNC_MASTER41>		03/16/2015				<CSA_PAGE86>	<SYNC_MASTER86>		<SYNC_DATE86>	
42		70	page71		DEVMLB		05/11/2015				<CSA_PAGE87>	<SYNC_MASTER87>		<SYNC_DATE87>	
43		71	page72		DEVMLB		05/06/2015				<CSA_PAGE88>	<SYNC_MASTER88>		<SYNC_DATE88>	
44		72	page74		DEVMLB		05/11/2015				<CSA_PAGE89>	<SYNC_MASTER89>		<SYNC_DATE89>	
45		74	page76		DEVMLB		05/11/2015				<CSA_PAGE90>	<SYNC_MASTER90>		<SYNC_DATE90>	
ALIASES RESOLVED															
Table of Contents [1]															
Schematic / PCB #'s															
PART NUMBER		QTY	DESCRIPTION		REFERENCE DES		CRITICAL		BOM OPTION		DRAWING TITLE				
051-00532		1	SCHEM,MLB,X260		SCH		CRITICAL				<PART_DESCRIPTION>				
820-00244		1	PCBF,MLB,X260		PCB		CRITICAL				DRAWING NUMBER		SIZE	D	
										REVISION		<E4LABEL>			
										BRANCH		<BRANCH>			
										PAGE		1 OF 130			
										SHEET		1 OF 67			

D	8		7		6		5		4		3		2		1		CHINAFOX	
	Alternate Parts																	
	BOM Groups																	
	BOM GROUP		BOM OPTIONS				PART NUMBER		ALTERNATE FOR PART NUMBER		BOM OPTION		REF DES		COMMENTS:			
	MLB_COMMON		ALTERNATE,COMMON,MLB_MISC,MLB_DEBUG:ENG,MLB_PROGPARTS,BUILD:CBB				376S00074		376S0855				ALL		Toshiba alt for Diodes dual			
	MLB_MISC		CAM_FREQ:24M,CAM_XTAL:NO,SSDRAM:4GBIT,S3X:B0,UPC:A1,PCH24M:SUS,EQ:4CH,PMIC:A0D,BSSB_GP10				376S1129		376S0855				ALL		NXP alt for Diodes dual			
	MLB_DEBUG:ENG		DEBUGLED				376S1089		376S1128				ALL		NXP alt for Diodes single			
							138S0703		138S0648		128S00012		ALL		Murata alt to Taiyo Yuden			
							372S0186		372S0185				ALL		NXP alt to Diodes			
							376S1053		376S0604				ALL		Diodes alt to Fairchild			
							107S0249		107S0251		138S00044		ALL		TFT alt to Cytac			
							128S00044		128S00041		138S00072		ALL		Rubycon alt to Sanyo			
							740S00004		740S0134				ALL		Kemet alt to Sanyo			
							740S00005		740S0190		132S00012		ALL		Polytronics alt to Wayon			
							128S0296		128S0487		152S00028		ALL		Sanyo alt to NEC			
							128S00012		128S0487		353S00712		ALL		ROHM alt to NEC			
							128S0397		128S0325		353S00714		ALL		Kemet alt to Sanyo			
							107S00070		107S0085		337S00174		ALL		TDK alt to Murata			
							376S00007		376S1179		138S00035		ALL		AOZ alt to Vishay			
							376S1080		376S0820		152S00343		ALL		Diodes alt to Onsemi			
							152S00311		152S2037		155S00155		ALL		Cytac new alt to Cytac			
							152S00098		152S00224		155S0741		ALL		Vishay alt to Cytac			
							128S00021		128S00020		155S00165		ALL		ROHM alt to NEC			
							138S0786		138S0705		155S0166		ALL		Samsung alt to Murata			
	CPU DRAM SPD Straps																	
	BOM GROUP		BOM OPTIONS				PART NUMBER		ALTERNATE FOR PART NUMBER		BOM OPTION		REF DES		COMMENTS:			
	DRAM:SAM_8GB		RAMCFG0_L,DRAM_TYPE:SAMSUNG_8GB				128S00007		128S00015				ALL		NEC alt to Sanyo			
	DRAM:HYN_8GB		RAMCFG0_L,RAMCFG1_L,DRAM_TYPE:HYNIX_8GB				128S00009		128S00015				ALL		Kemet alt to Sanyo			
	DRAM:ELP_8GB		DRAM_TYPE:ELPIDA_8GB				128S00011		128S00010				ALL		Kemet alt to Sanyo			
							128S00026		128S00010				ALL		NEC alt to Sanyo			
							128S00031		128S00010				ALL		ROHM alt to Sanyo			
							138S00073		138S00047				ALL		Murata alt to Taiyo			
							138S0772		138S00013				ALL		Taiyo alt to Murata			
							132S00012		132S0401				ALL		Multi alt to Taiyo			
							152S00028		152S1751				ALL		Murata alt to Taiyo Yuden			
							353S00712		353S2216				ALL		Onsemi alt to TI			
							353S00714		353S2208				ALL		ONsemi alt to TI			
							337S00174		337S00175		S3X:B0		ALL		SIX B0 alt to AO			
							138S00077		138S00035				ALL		Taiyo alt to Murata			
							152S00343		152S1682				ALL		Murata alt to Cytac			
							155S00155		155S0441				ALL		TDK alt to Murata			
							155S0741		155S0361				ALL		Murata alt to TDK			
							155S00165		155S0382				ALL		Taiyo alt to TDK			
							155S00166		155S0391				ALL		TDK alt to Murata			
							155S0706		155S0302				ALL		Taiyo alt to Murata			
							197S00053		197S00050				ALL		Ryoceca alt to TSC			
							197S00055		197S00050				ALL		Murata alt to TSC			
							311S00004		311S0370				ALL		Onsemi alt to NXP			
							333S00030		333S00016				ALL		Hynix alt to Micron			
							338S00175		338S1264				ALL		Broadcom alt packaging			
							107S00020		107S0276				ALL		TFT alt to Cytac			
							107S00021		107S0284				ALL		TFT alt to Cytac			
							353S00773		353S4296				ALL		Semtech alt to TI			
							311S00057		311S0612				ALL		Diodes alt to NXP			
							155S00011		155S00008				ALL		Murata alt to TDK			
							138S0739		138S0706				ALL		Samsung alt to Murata			
							138S0945		138S0706				ALL		Ryoceca alt to Murata			
							138S00090		138S0943				ALL		Taiyo alt to Murata			
							138S00087		138S1086				ALL		Taiyo alt to Murata			
							107S00086		107S00056				ALL		TFT alt to Cytac			
							152S00347		152S00220				ALL		Chilisin alt to Cytac			
							138S00032		138S0831				ALL		Taiyo alt to Murata			
							138S00049		138S0831				ALL		Ryoceca alt to Murata			
							152S00362		152S1129				ALL		Cytac alt to Murata			
							128S00039		128S00038				ALL		NEC alt to Kemet			
	CPU DRAM CFG Chart																	
	VENDOR		CFG 1		CFG 0		SIZE		CFG 3		CFG 2							
	HYNIX		0		0		2GB		0		0							
	SAMSUNG		1		0		4GB QDP		0		1							
	MICRON		0		1		4GB DDP		1		0							
	ELPIDA		1		1		8GB		1		1							
	SSD Configs																	
	BOM GROUP		BOM OPTIONS				PART NUMBER		ALTERNATE FOR PART NUMBER		BOM OPTION		REF DES		COMMENTS:			
	SSD:128GB		NAND:64GB,CAPACITY1				SSD:128GB_SAND		CAPACITY0,CAPACITY1		NAND:128GB_SAND,CAPACITY0,CAPACITY1,SANDISK_1Y							
	SSD:256GB		NAND:128GB,CAPACITY0,CAPACITY1				SSD:512GB		NAND:256GB,CAPACITY2									
	SSD:256GB_SAND		NAND:128GB_SAND,CAPACITY0,CAPACITY1,SANDISK_1Y				SSD:512GB_SAND		NAND:256GB_SAND,CAPACITY2,SANDISK_1Y									
	SSD Parts																	
	PART NUMBER		QTY		DESCRIPTION		REFERENCE DES		CRITICAL		BOM OPTION							
	337S00175		1		IC,S3-X,B0		U8600		CRITICAL		S3X:B0							
	333S00017		1		IC,LPDDR3-1600,128MX8,1.8V,25NM,276 POP		U8600_POP_DRAM		CRITICAL		SSDRAM:4GBIT							
	335S1030		2		NAND,T0G DDR2,2CH,64GB,1YNM,3.3V,LGA60		U9100,U9120		CRITICAL		NAND:64GB							
	335S00050		2		NAND,T0G DDR2,2CH,128GB,1YNM,3.3V,LGA60		U9100,U9120		CRITICAL		NAND:128GB							
	335S00051		2		NAND,T0G DDR2,2CH,256GB,1YNM,3.3V,LGA60		U9100,U9120		CRITICAL		NAND:256GB							
	335S00121		2		NAND,T0G DDR2,2CH,128GB,1YNM,3.3V,LGA60		U9100,U9120		CRITICAL		NAND:128GB_SAND							
	335S00122		2		NAND,T0G DDR2,2CH,256GB,1YNM,3.3V,LGA60		U9100,U9120		CRITICAL		NAND:256GB_SAND							
	Module Parts																	
	PART NUMBER		QTY		DESCRIPTION		REFERENCE DES		CRITICAL		BOM OPTION							
	337S00207		1		IC,CPU,SKT,SRZEN,FQ0,D1,2/2,1.1,5W,.85,B1515		U0500		CRITICAL		CPU:1.1GHZ							
	337S00208		1		IC,CPU,SKT,SRZEN,FQ0,D1,2/2,1.2,5W,.85,B1515		U0500		CRITICAL		CPU:1.2GHZ							
	337S00209		1		IC,CPU,SKT,SRZEN,FQ0,D1,2/2,1.3,5W,.85,B1515		U0500		CRITICAL		CPU:1.3GHZ							
	946-3892		1		J11/J13 MLB DYMAX ADHESIVE 29993-SC 0.4G		GLUE		CRITICAL									
	825-00162		1		LABEL,BARCODE,2D,1D,CONFIG,MLB,X260		LABEL											
	DRAM Parts																	
	PART NUMBER		QTY		DESCRIPTION		REFERENCE DES		CRITICAL		BOM OPTION							
	333S00052		2		IC,SDRAM,LPDDR-1866,32GBIT,253B FBGA		U2300,U2500		CRITICAL		DRAM_TYPE:SAMSUNG_8GB							
	333S00053		2		IC,SDRAM,LPDDR-1866,32GBIT,253B FBGA		U2300,U2500		CRITICAL		DRAM_TYPE:HYNIX_8GB							
	333S00054		2		IC,SDRAM,LPDDR-1866,32GBIT,253B FBGA		U2300,U2500		CRITICAL		DRAM_TYPE:ELPIDA_8GB							
	BOM Configuration [2]																	
	DRAWING NUMBER		REVISION		BRANCH		PAGE		SHEET									
	<SCH_NUM>		<E4LABEL>		<BRANCH>		2 OF 130		2 OF 67									
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	BOM Configuration [2]																	
	DRAWING NUMBER						REVISION						BRANCH					
	<SCH_NUM>						<E4LABEL>						<BRANCH>					
	PAGE						SHEET											
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													</					

BOM GROUP	BOM OPTIONS
MLB_COMMON	ALTERNATE,COMMON,MLB_MISC,MLB_DEBUG:ENG,MLB_PROGPARTS,BUILD:CBB
MLB_MISC	CAM_FREQ:24M,CAM_XTAL:NO,SSDRAM:4GBIT,S3X:B0,UPC:A1,PCH24M:SUS,EQ:4CH,PMIC:A0D,BSSB_GP10
MLB_DEBUG:ENG	DEBUGLED

BOM GROUP	BOM OPTIONS
DRAM:SAM_8GB	RAMCFG0_L,DRAM_TYPE:SAMSUNG_8GB
DRAM:HYN_8GB	RAMCFG0_L,RAMCFG1_L,DRAM_TYPE:HYNIX_8GB
DRAM:ELP_8GB	DRAM_TYPE:ELPIDA_8GB

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
335S00084	1	IC,SPI SERIAL FLASH,8MBIT,3.0V,CSP,8P	U3090	CRITICAL	UPCROM:BLANK
341S00466	1	IC,NVM (VXXX) CBB, X260	U3090	CRITICAL	UPCROM:PROG
338S1231	1	IC,SMC12-B1,40MHZ/50MKIPS MCU,7K7,168BGA	U5000	CRITICAL	SMC:BLANK
341S00431	1	IC,SMC-B1,EXTERNAL (V2.33A2) PROTO 1,X260	U5000	CRITICAL	SMC:PROG
335S1009	1	64 MBIT SPI QUAD I/O FLASH,CSP,3.3V,QUAD IO	U6100	CRITICAL	BOOTROM_WIN:BLANK
335S1010	1	64 MBIT SPI QUAD I/O FLASH,CSP,3.3V,QUAD IO	U6100	CRITICAL	BOOTROM_MAC:BLANK
335S1029	1	64 MBIT SPI QUAD I/O FLASH,CSP,3.3V,QUAD IO	U6100	CRITICAL	BOOTROM_MIC:BLANK
341S00465	1	IC,EFI ROM (VXXX) CBB,X260	U6100	CRITICAL	BOOTROM:PROG

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
337S00207	1	IC,CPU,SKT,SRZEN,FQ0,D1,2/2,1.1,5W,.85,B1515	U0500	CRITICAL	CPU:1.1GHZ
337S00208	1	IC,CPU,SKT,SRZEN,FQ0,D1,2/2,1.2,5W,.85,B1515	U0500	CRITICAL	CPU:1.2GHZ
337S00209	1	IC,CPU,SKT,SRZEN,FQ0,D1,2/2,1.3,5W,.85,B1515	U0500	CRITICAL	CPU:1.3GHZ
946-3892	1	J11/J13 MLB DYMAX ADHESIVE 29993-SC 0.4G	GLUE	CRITICAL	
825-00162	1	LABEL,BARCODE,2D,1D,CONFIG,MLB,X260	LABEL		

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
333S00052	2	IC,SDRAM,LPDDR-1866,32GBIT,253B FBGA	U2300,U2500	CRITICAL	DRAM_TYPE:SAMSUNG_8GB
333S00053	2	IC,SDRAM,LPDDR-1866,32GBIT,253B FBGA	U2300,U2500	CRITICAL	DRAM_TYPE:HYNIX_8GB
333S00054	2	IC,SDRAM,LPDDR-1866,32GBIT,253B FBGA	U2300,U2500	CRITICAL	DRAM_TYPE:ELPIDA_8GB

CPU DRAM CFG Chart

VENDOR	CFG 1	CFG 0
HYNIX	0	0
SAMSUNG	1	0
MICRON	0	1
ELPIDA	1	1

SIZE	CFG 3	CFG 2
2GB	0	0
4GB QDP	0	1
4GB DDP	1	0
8GB	1	1

BOM GROUP	BOM OPTIONS
SSD:128GB	NAND:64GB,CAPACITY1
SSD:256GB	NAND:128GB,CAPACITY0,CAPACITY1
SSD:512GB	NAND:256GB,CAPACITY2
SSD:256GB_SAND	NAND:128GB_SAND,CAPACITY0,CAPACITY1,SANDISK_1Y
SSD:512GB_SAND	NAND:256GB_SAND,CAPACITY2,SANDISK_1Y

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
337S00175	1	IC,S3-X,B0	U8600	CRITICAL	S3X:B0
333S00017	1	IC,LPDDR3-1600,128MX8,1.8V,25NM,276 POP	U8600_POP_DRAM	CRITICAL	SSDRAM:4GBIT
335S1030	2	NAND,T0G DDR2,2CH,64GB,1YNM,3.3V,LGA60	U9100,U9120	CRITICAL	NAND:64GB
335S00050	2	NAND,T0G DDR2,2CH,128GB,1YNM,3.3V,LGA60	U9100,U9120	CRITICAL	NAND:128GB
335S00051	2	NAND,T0G DDR2,2CH,256GB,1YNM,3.3V,LGA60	U9100,U9120	CRITICAL	NAND:256GB
335S00121	2	NAND,T0G DDR2,2CH,128GB,1YNM,3.3V,LGA60	U9100,U9120	CRITICAL	NAND:128GB_SAND
335S00122	2	NAND,T0G DDR2,2CH,256GB,1YNM,3.3V,LGA60	U9100,U9120	CRITICAL	NAND:256GB_SAND

Module Parts

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
353S00660	1	IC,CD3215,USB PWR SWITCH,A1,6X6MM,BGA96	U3100	CRITICAL	UPC:A1
338S00168	1	IC,PMU,P650839A0D,7X7MM,BGA168	U7800	CRITICAL	PMIC:A0D

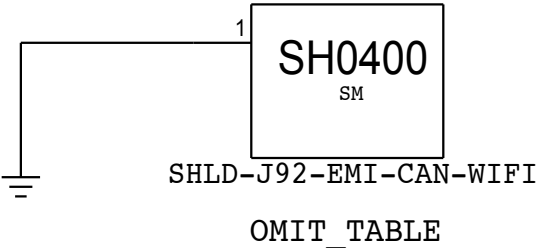
Board IDs

BOM GROUP	BOM OPTIONS
BUILD:PROTO1	BOARDID0_L
BUILD:CBB	BOARDID1_L

Board ID should start all-high (first build), and each new build increment stuffing in a binary fashion (ID0 is the LSB). IDX is high if not set low in the BOM table.

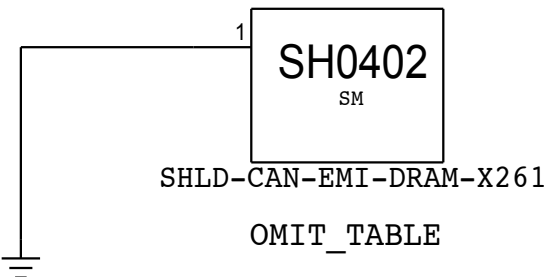
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WIFI EMI CAN



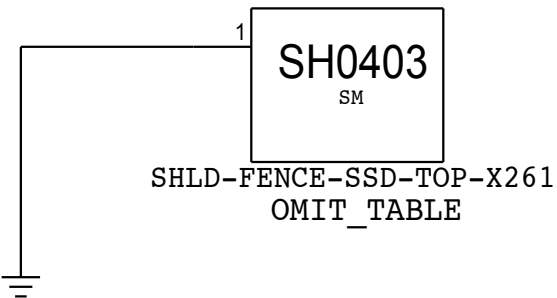
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806-7064	1	CAN,EMI,WIFI,X261	SH0400	CRITICAL	

DRAM EMI CAN

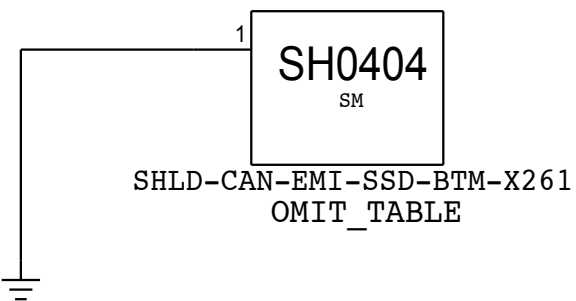


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806-04848	1	CAN,EMI,DRAM,X260	SH0402	CRITICAL	

SSD EMI CANS

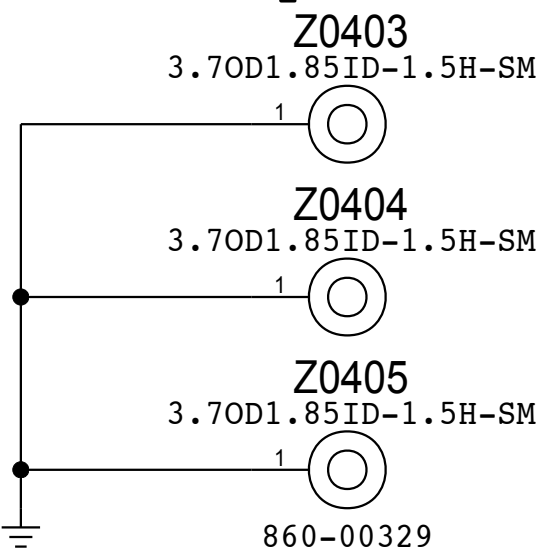


PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
806-00889	1	FENCE,EMI,SSD,TOP,X261	SH0403	CRITICAL	
870-00878	1	TAPE,CONDUCTIVE,SSD,REEL,X261	SSD_TAPE_TOP	CRITICAL	

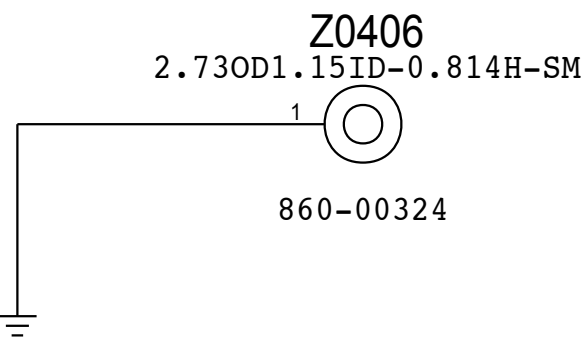


PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
806-05622	1	FENCE,EMI,SSD,BTM,X260	SH0404	CRITICAL	
870-01557	1	TAPE,CONDUCTIVE,SSD,REEL,X260	SSD_TAPE_BTM	CRITICAL	

CPU Heat Spreader Bosses



USB-C BTB Connector Boss



PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
870-00998	1	MYLAR,SHIM,MLB,X261	MYLAR_SHIM	CRITICAL	



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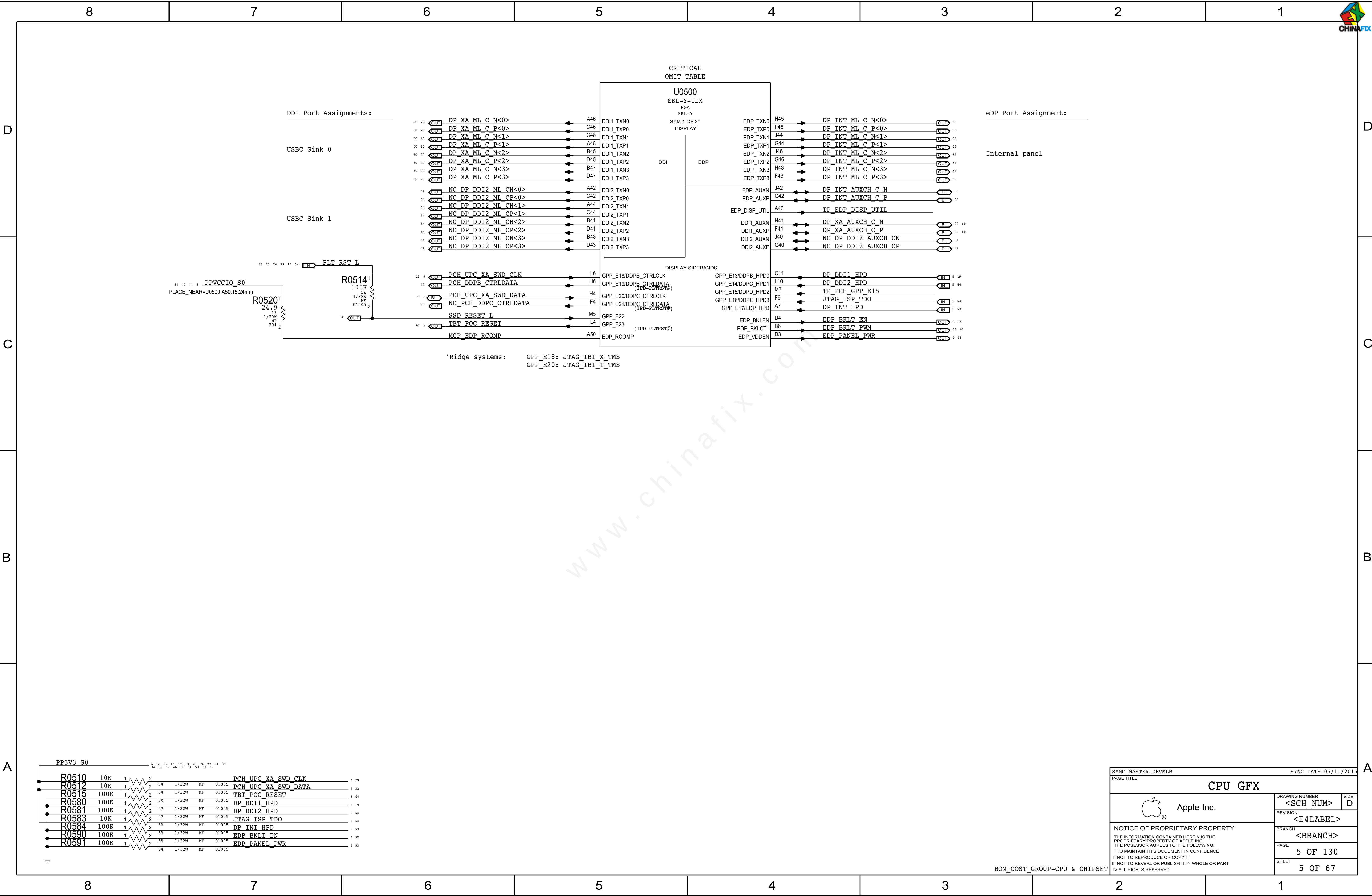
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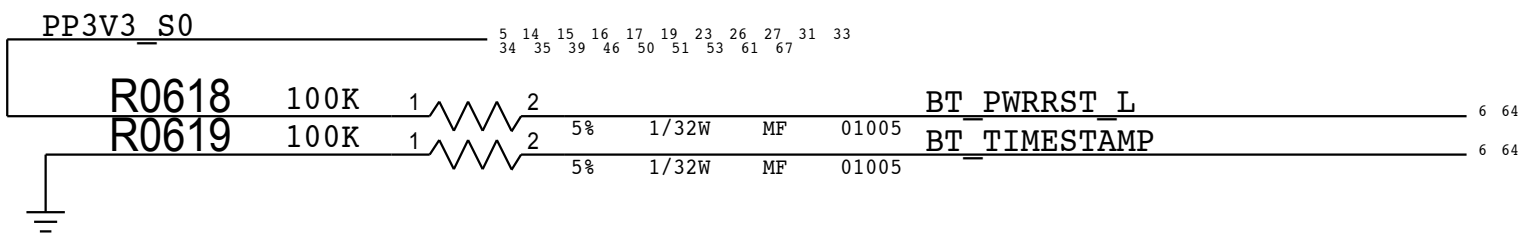
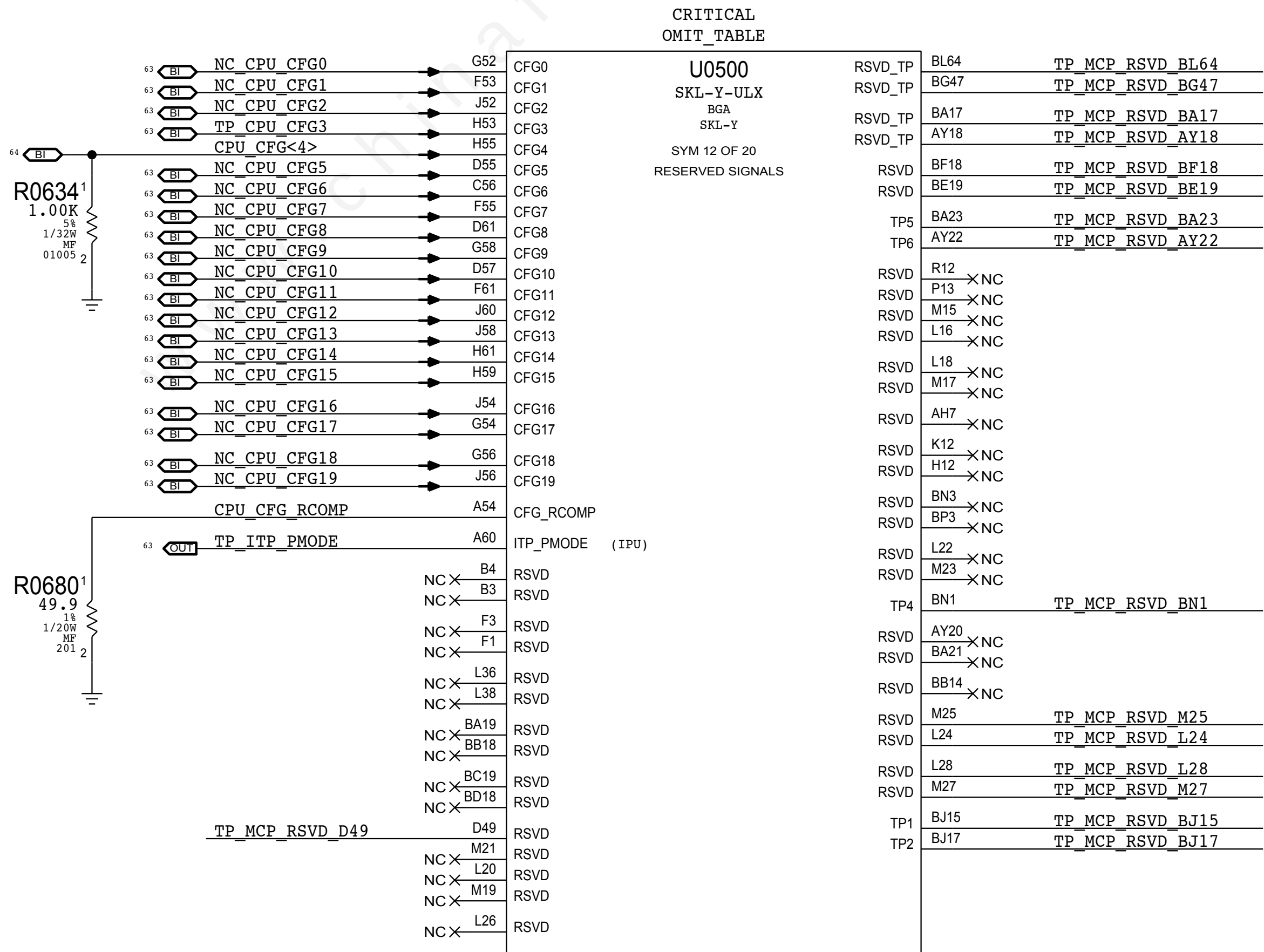
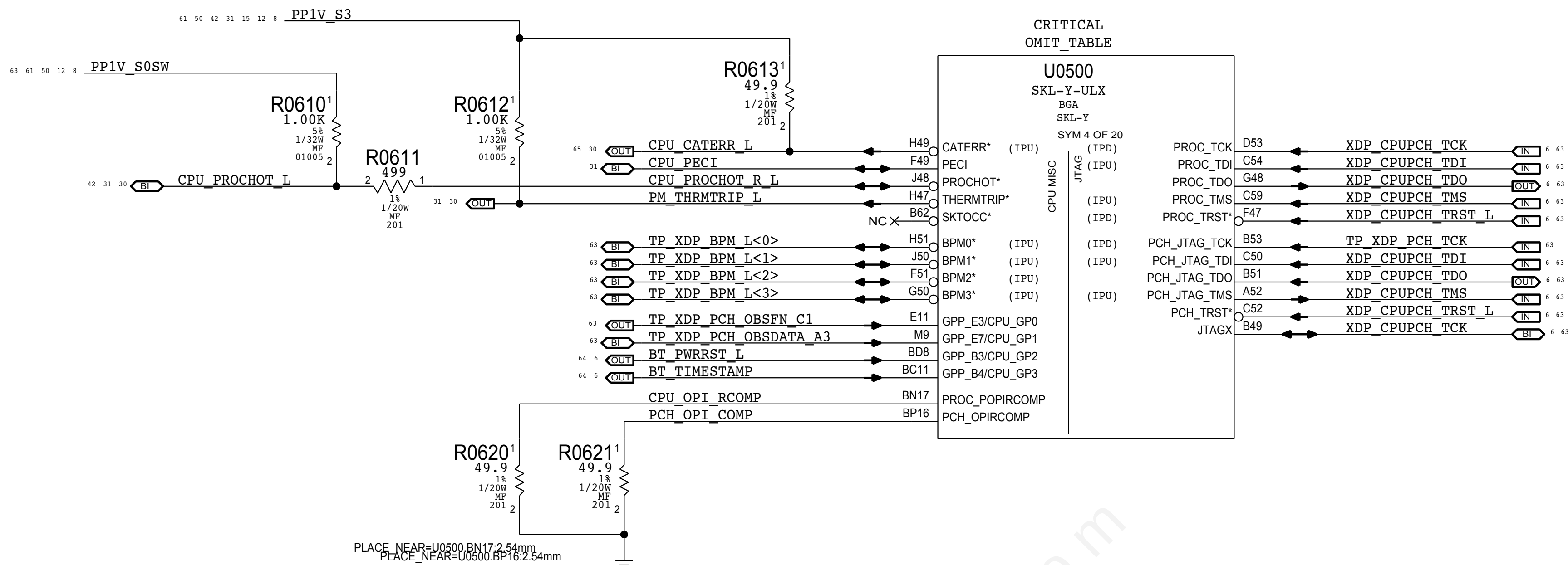
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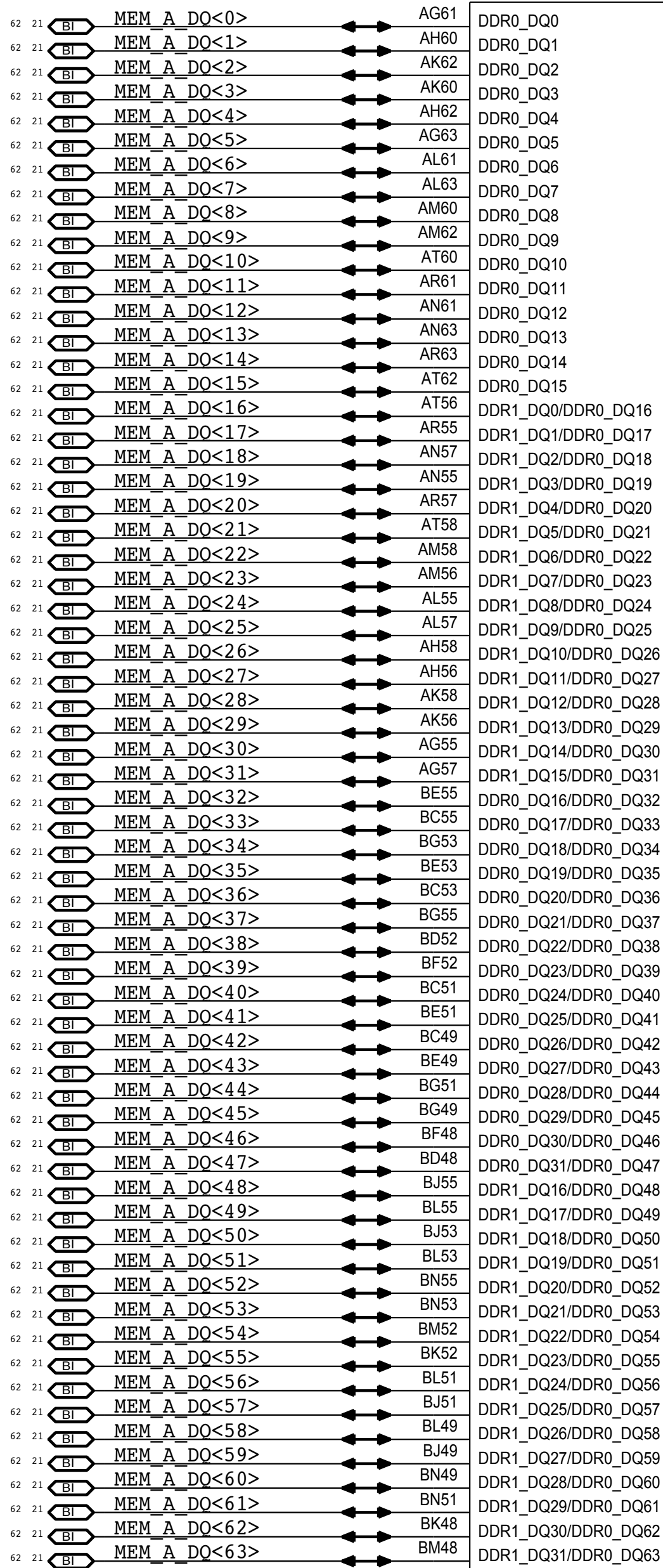
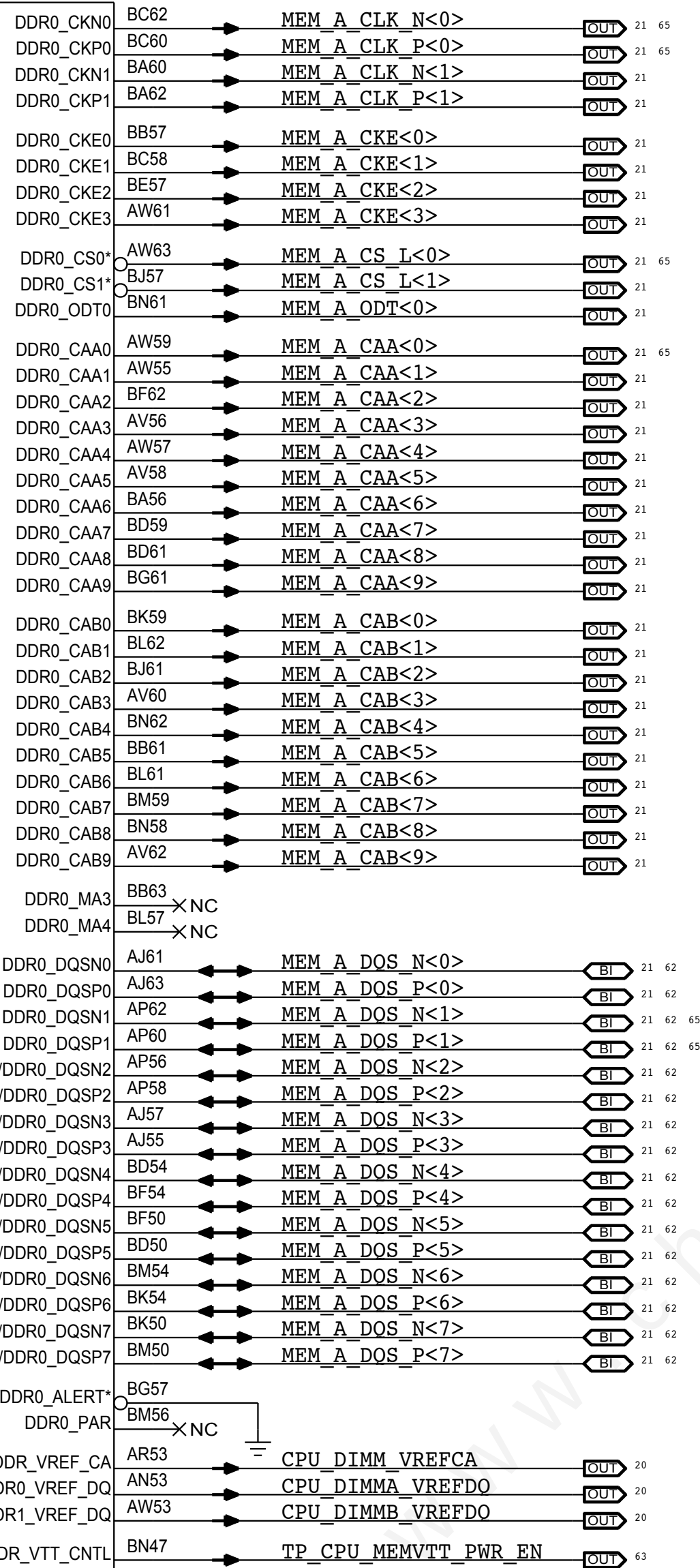
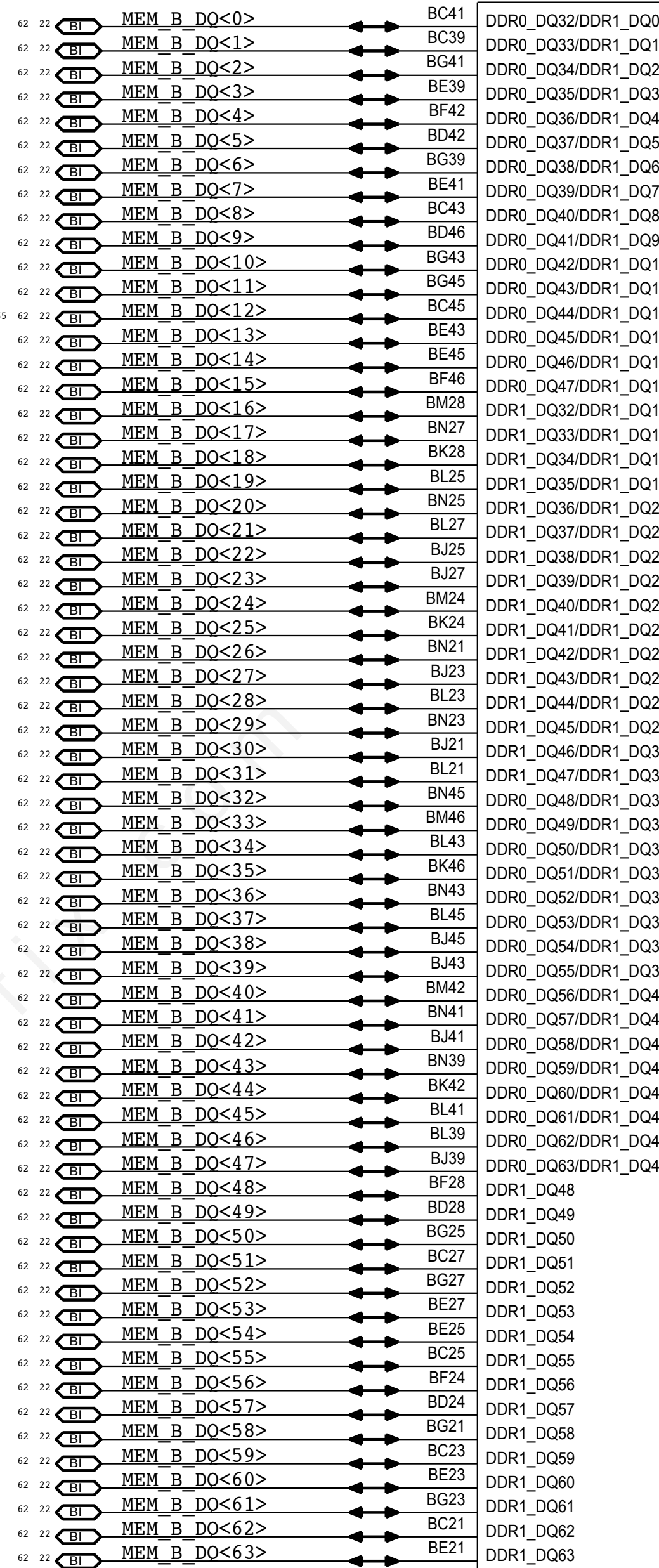
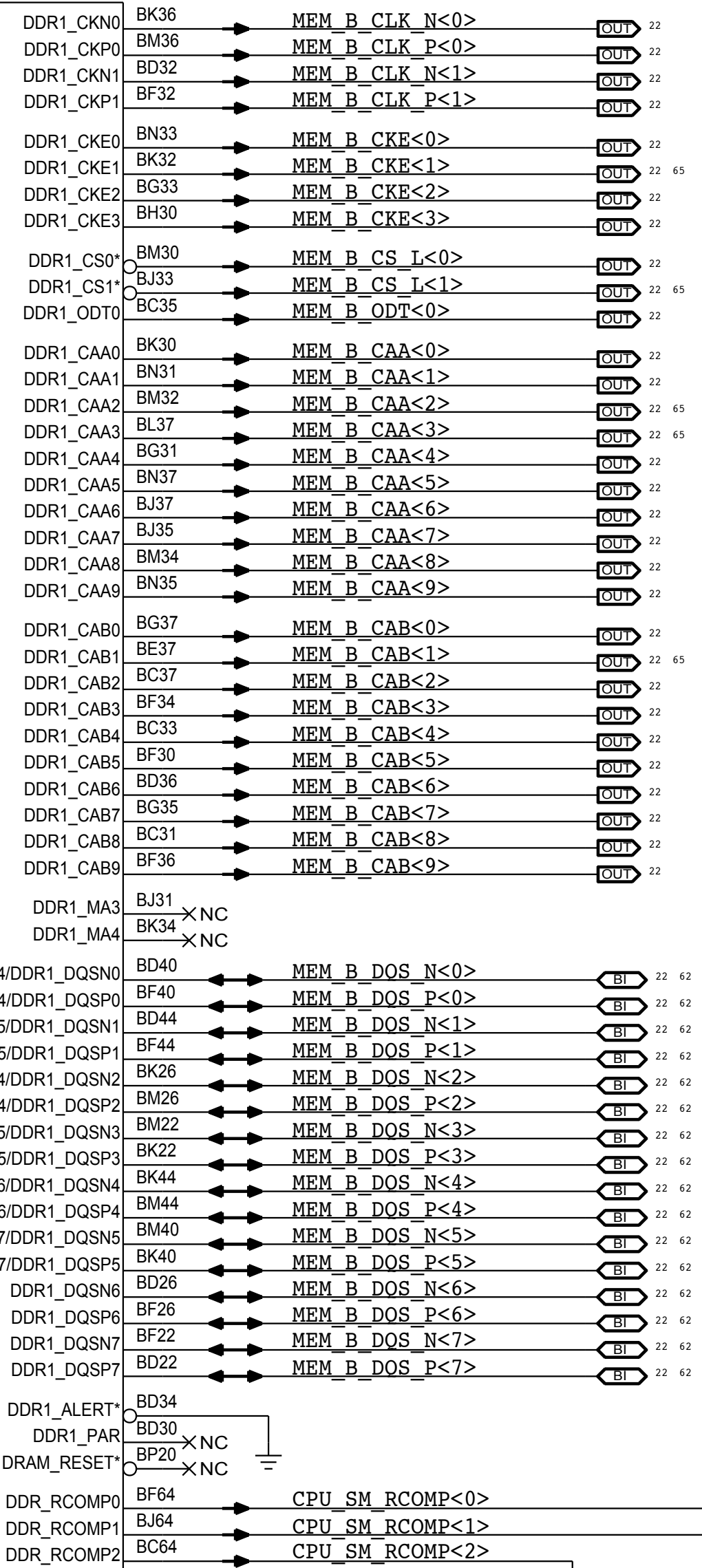
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CRITICAL
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SKL-Y-ULX
BGA
SYM 2 OF 20
DDR CH - ACRITICAL
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SKL-Y-ULX
BGA
SYM 3 OF 20
DDR CH - BPLACE_NEAR=U0500,BC64:12.7mm
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PLACE_NEAR=U0500,BF64:12.7mm

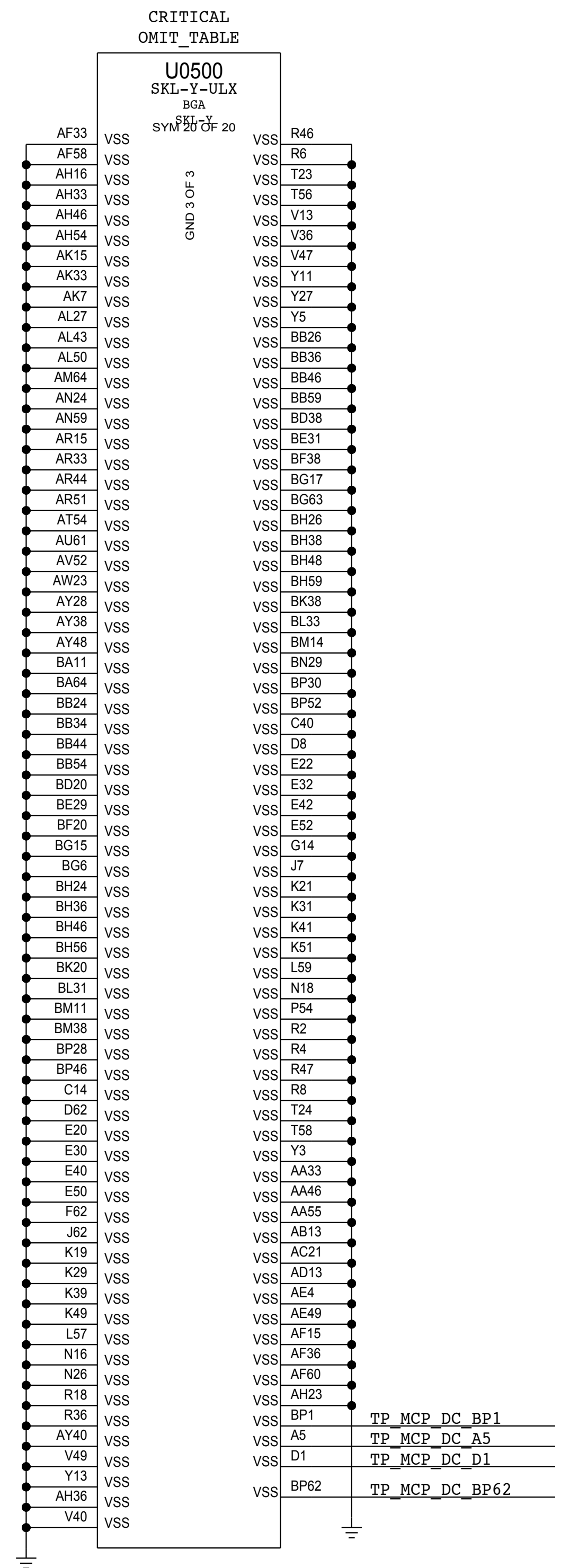
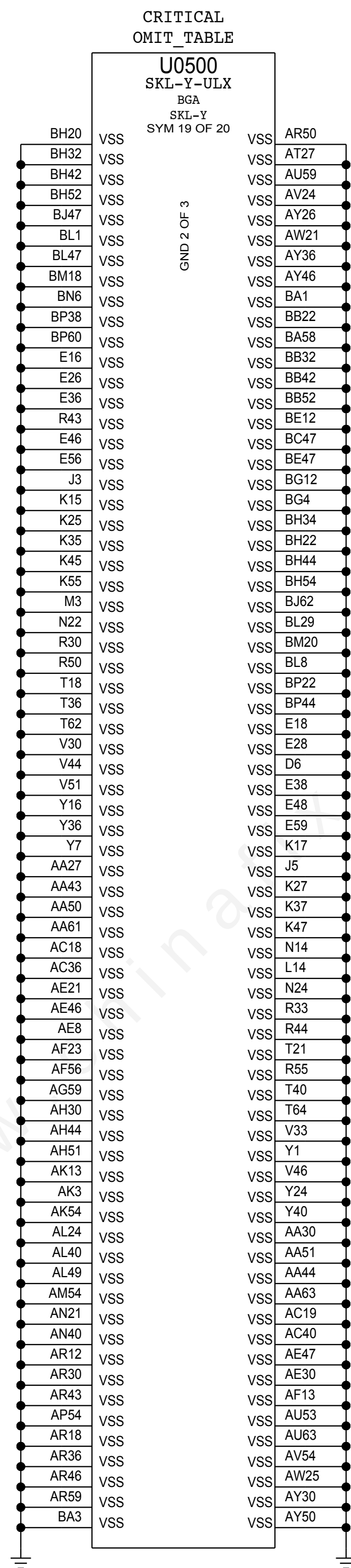
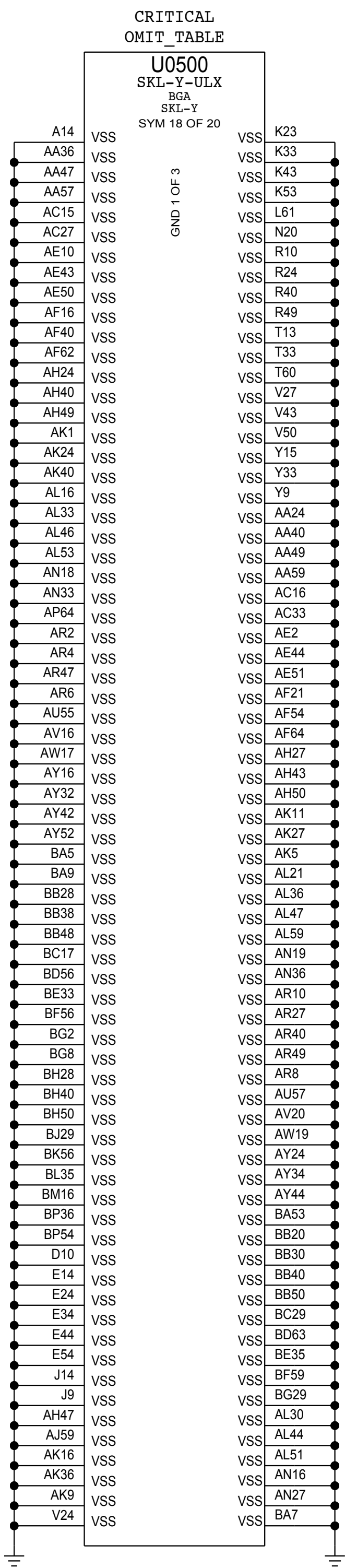
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BOM_COST_GROUP=CPU & CHIPSET

NOTE: Aliases not used on CPU supply outputs to avoid any extraneous connections.





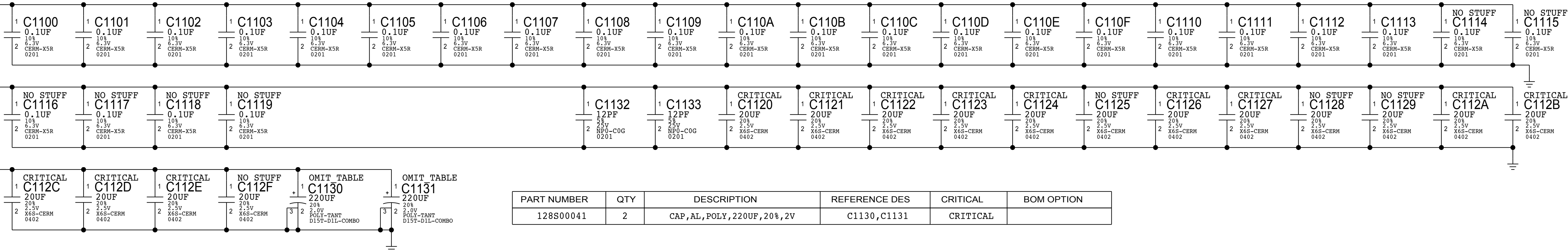
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All Intel recommendations from Intel doc #543016 Skylake U and Y Platform Design Guide Rev 1.3 unless stated otherwise
Intel's placeholders are no longer required per 4/20/15 email from Srin

CPU VCORE Decoupling

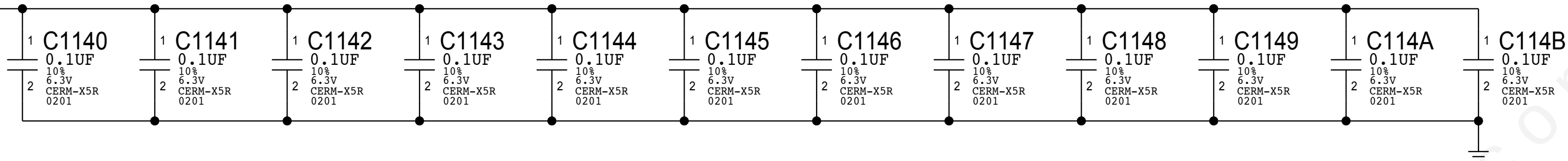
Intel recommendation (Table 52-2): 20x 0.1uF 0201 STUFF, 8x 10uF 0402, 6x 47uF 6.3V 0805 STUFF (2x NO STUFF)
Apple implementation : 20x 0.1uF 0201 STUFF (6x NO STUFF), 16x 20uF 0402, 3x 270uF 2V B2



PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
128S00041	2	CAP,AL,POLY,220UF,20%,2V	C1130,C1131	CRITICAL	

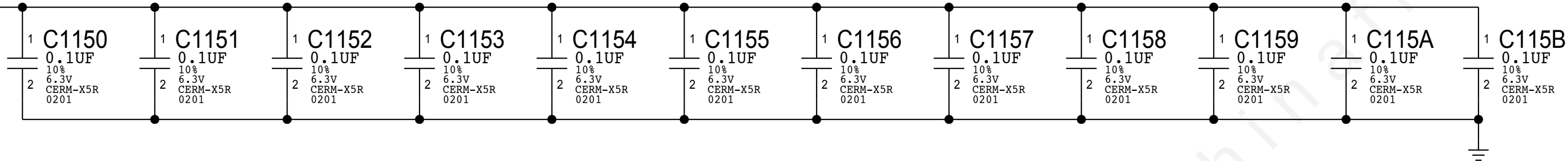
CPU VCORE G0 Decoupling

Intel recommendation (Table 52-2): 12x 0.1uF 0201
Apple implementation : 12x 0.1uF 0201



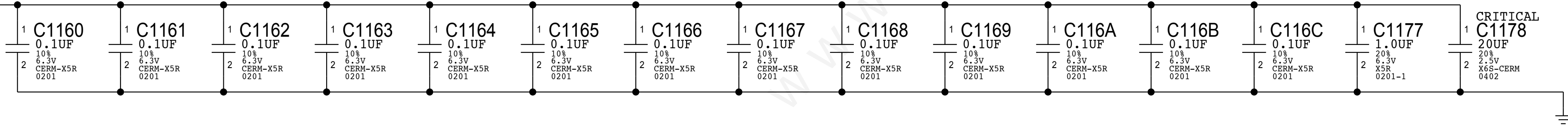
CPU VCORE G1 Decoupling

Intel recommendation (Table 52-2): 12x 0.1uF 0201
Apple implementation : 12x 0.1uF 0201



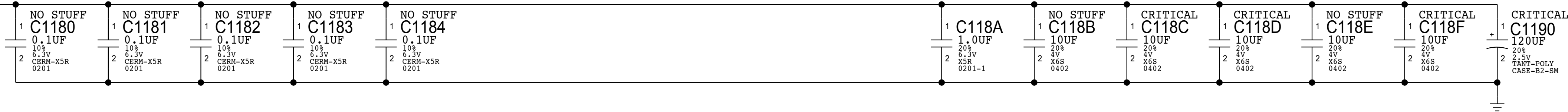
CPU VCCIO Decoupling

Intel recommendation (Table 52-2): 13x 0.1uF 0201 STUFF (10x NO STUFF), 1x 1uF 0402, 1x 22uF 0805
Apple implementation : 13x 0.1uF 0201 STUFF (10x NO STUFF), 1x 1uF 0201, 1x 20uF 0402



CPU VCCSA Decoupling

Intel recommendation (Table 52-2): 1x 1uF 0201, 4x 22uF 0603
Apple implementation : 1x 1uF 0201, 5x 0.1uF NO STUFF, 5x 20uF 0402, 1x 270uF 2V B2

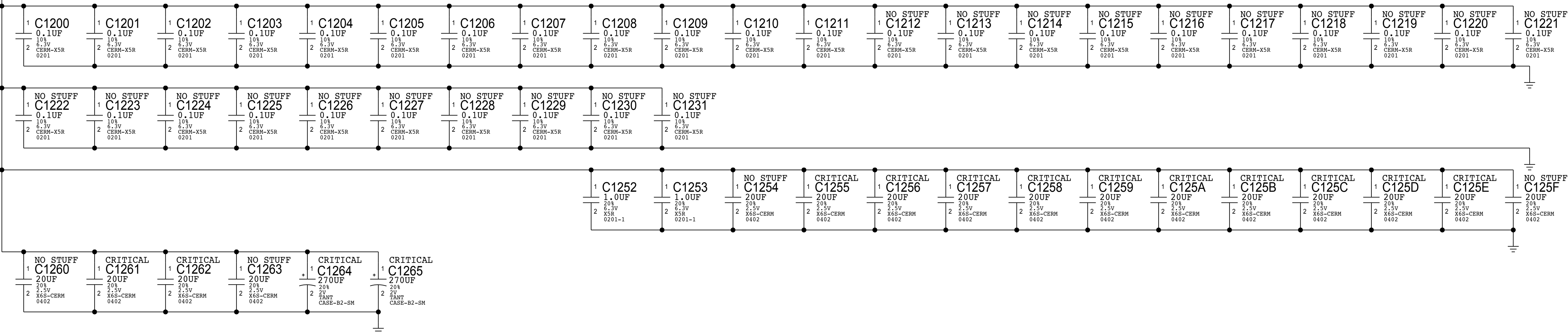




All Intel recommendations from Intel doc #543016 Skylake U and Y Platform Design Guide Rev 1.3 unless stated otherwise
Intel's placeholders are no longer required per 4/20/15 email from Srini

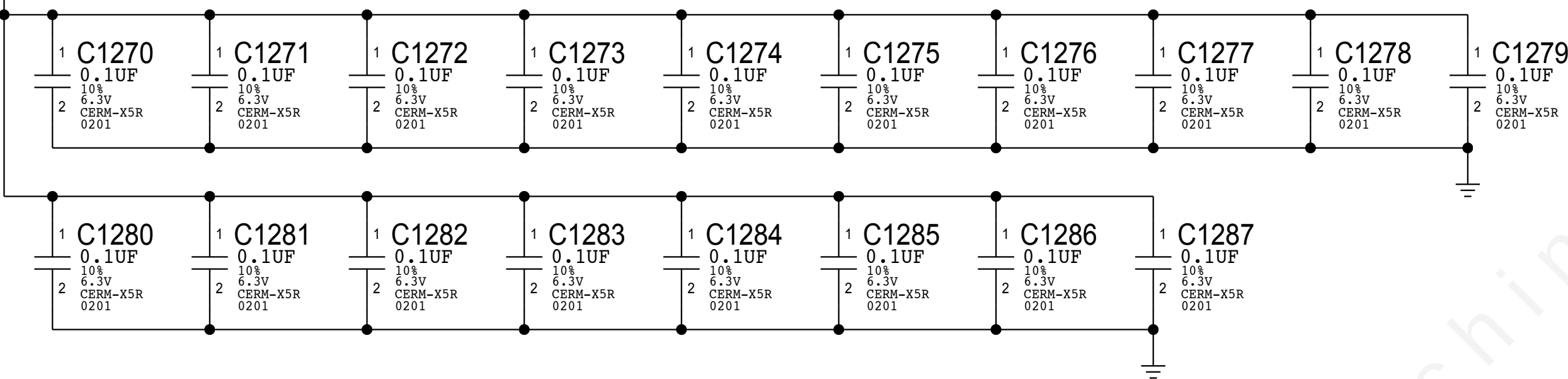
CPU GT Decoupling

Intel recommendation (Table 52-2): 12x 0.1uF 0201 STUFF, 2x 1uF 0402, 2x 10uF 0402 NO STUFF, 9x 47uF 6.3V 0805
Apple implementation : 12x 0.1uF 0201 STUFF (20x NO STUFF), 2x 1uF 0201, 16x 20uF 0402, 2x 270uF 2V B2

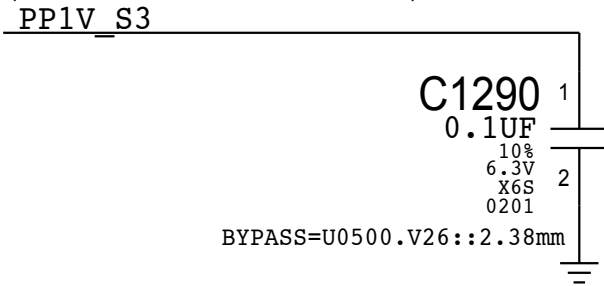


CPU VDDQ DECOUPLING

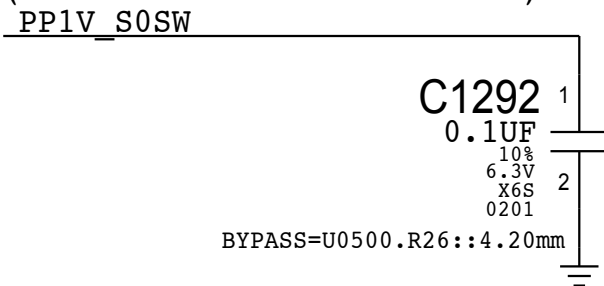
Intel recommendation (Table 52-2): 18x 0.1uF 0201
Apple implementation : 18x 0.1uF 0201



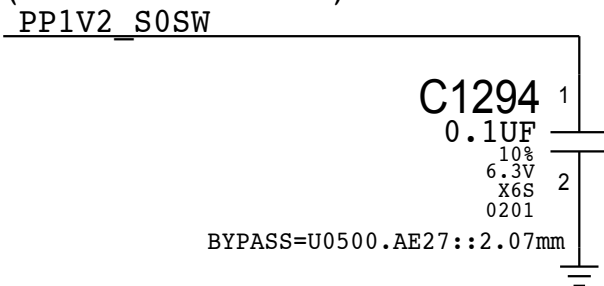
CPU VCCST BYPASS
(CPU 1.0V SUSTAIN PWR)



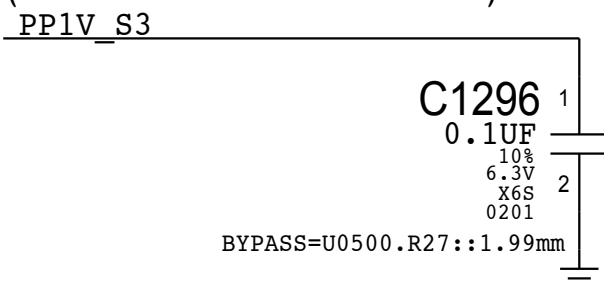
CPU VCCSTG BYPASS
(CPU 1.0V SUSTAIN GATED PWR)



CPU VCCPLL OC BYPASS
(CPU 1.2V PLL PWR)



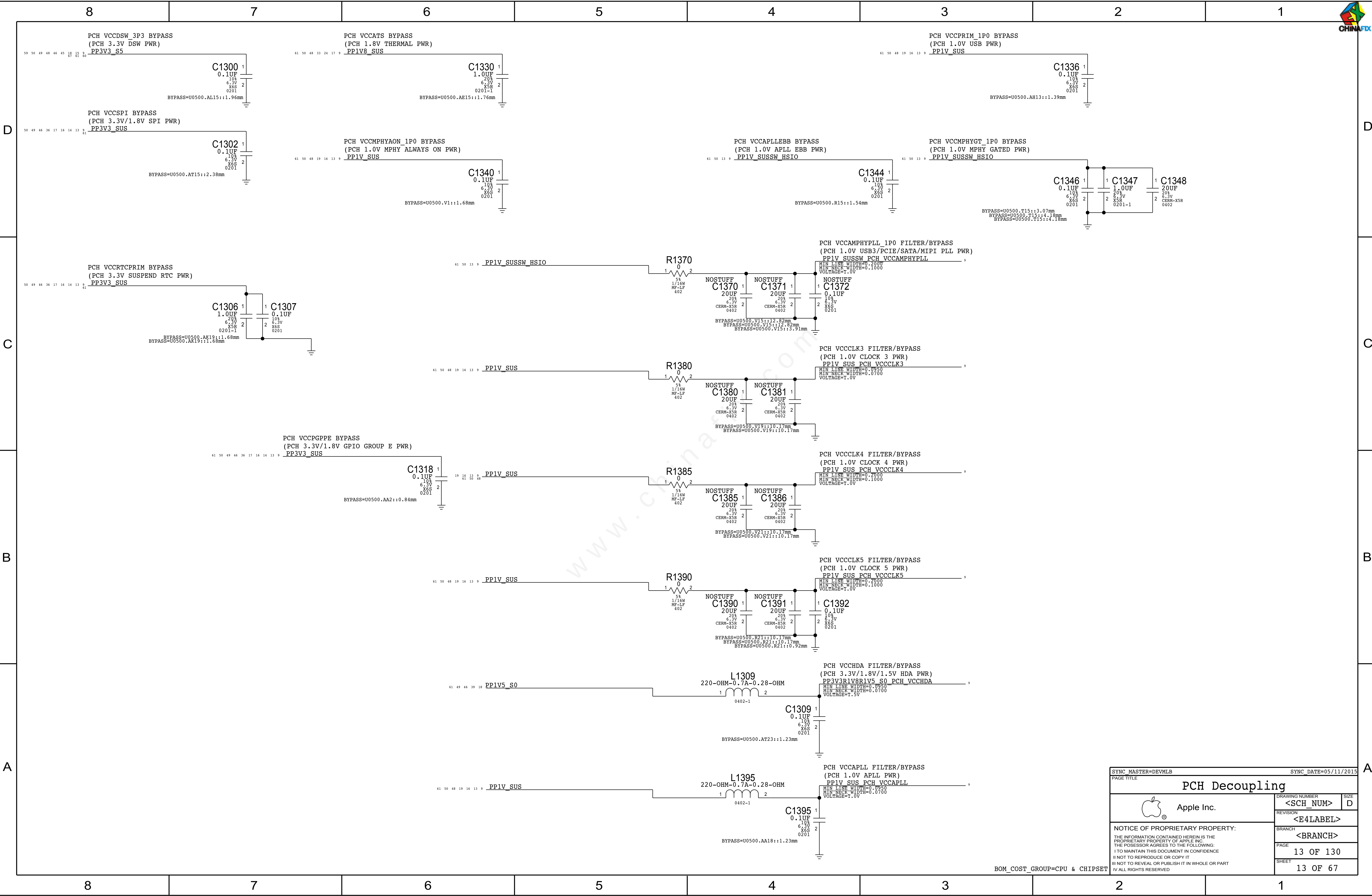
CPU VCCPLL BYPASS
(CPU 1.0V DIGITAL PLL PWR)



NOTE: Bypass calcs based on 0.1mm trace width with worst case stackup and 500pH added for power/ground via pair.

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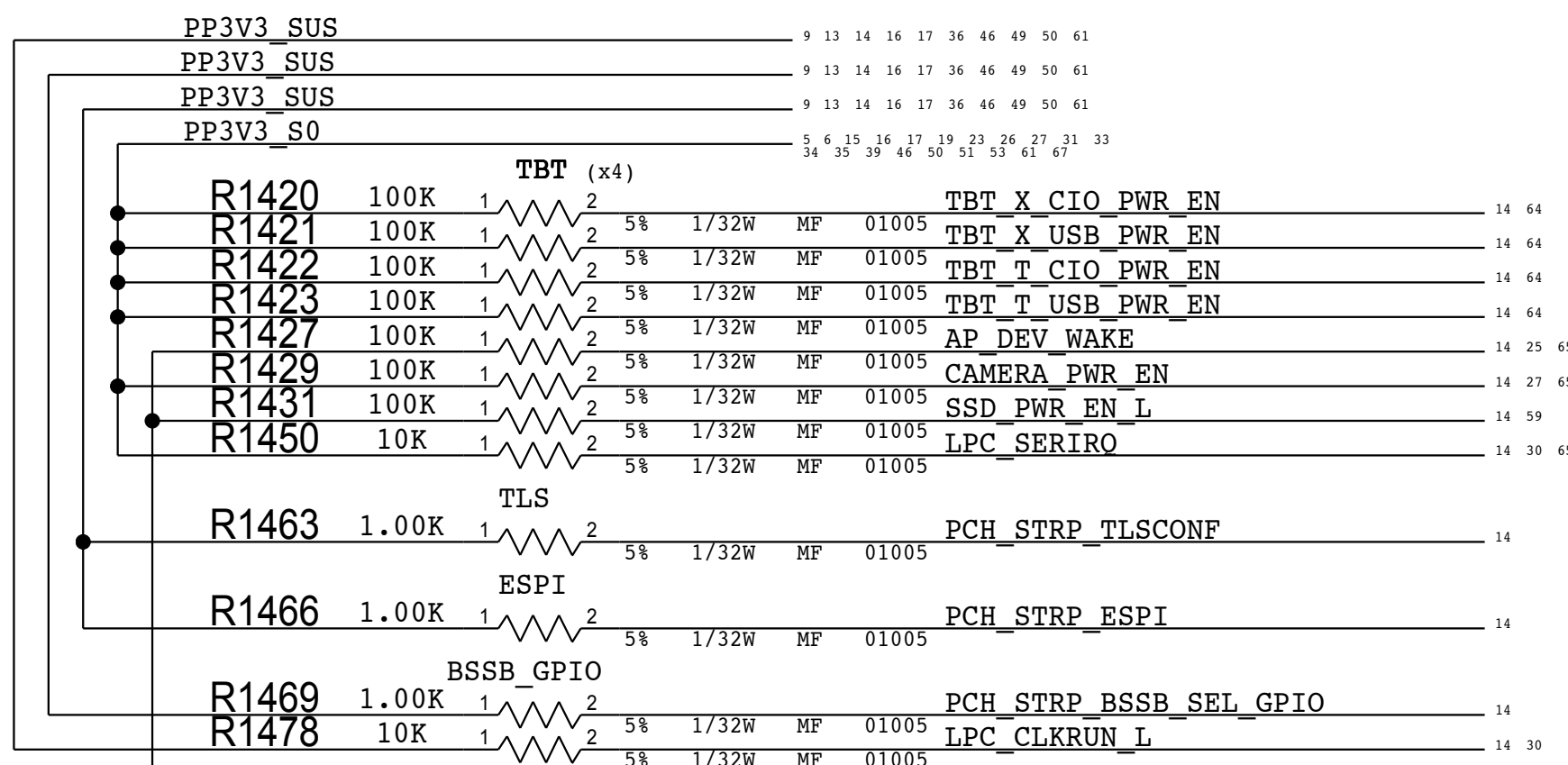
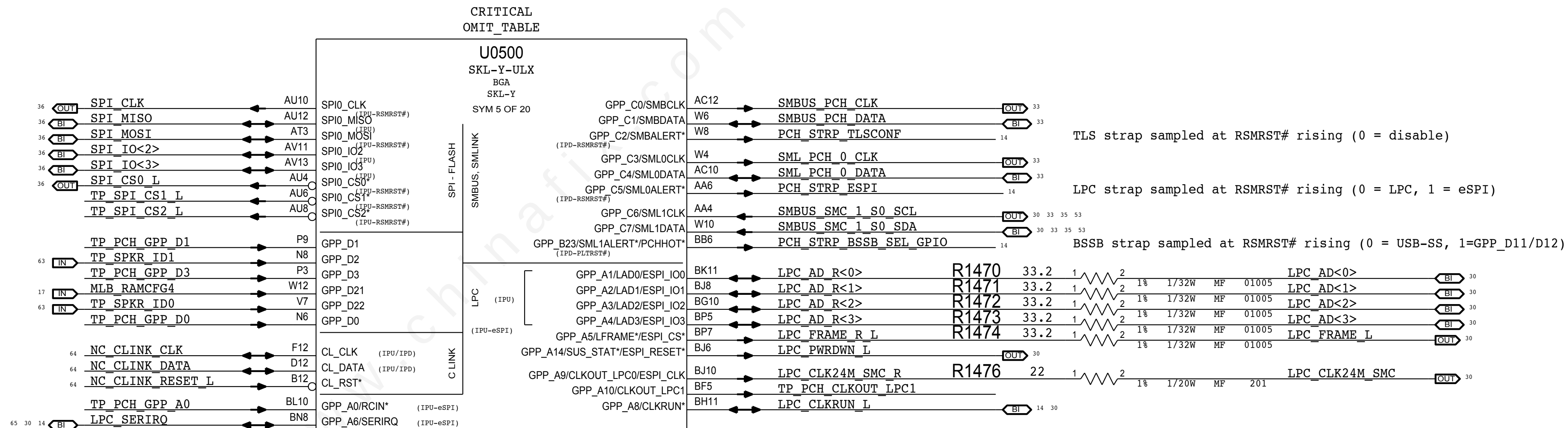
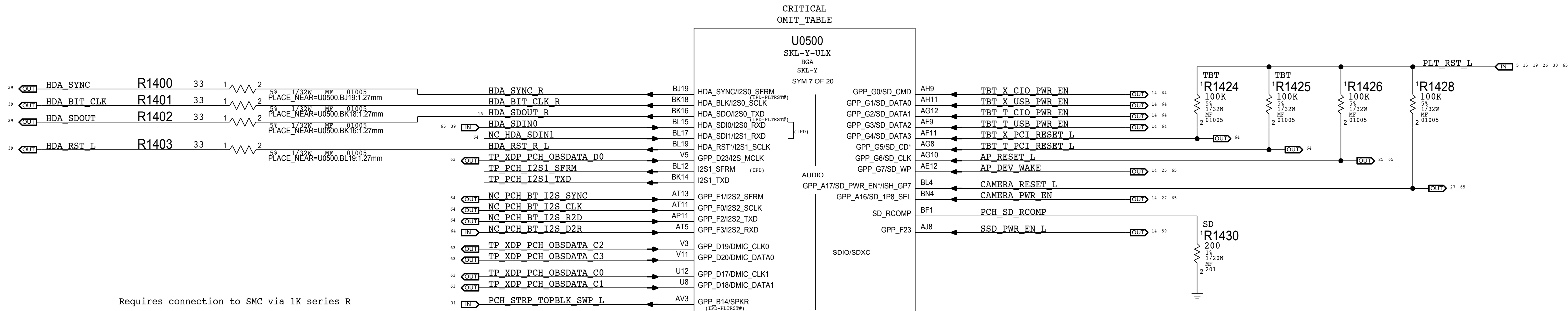
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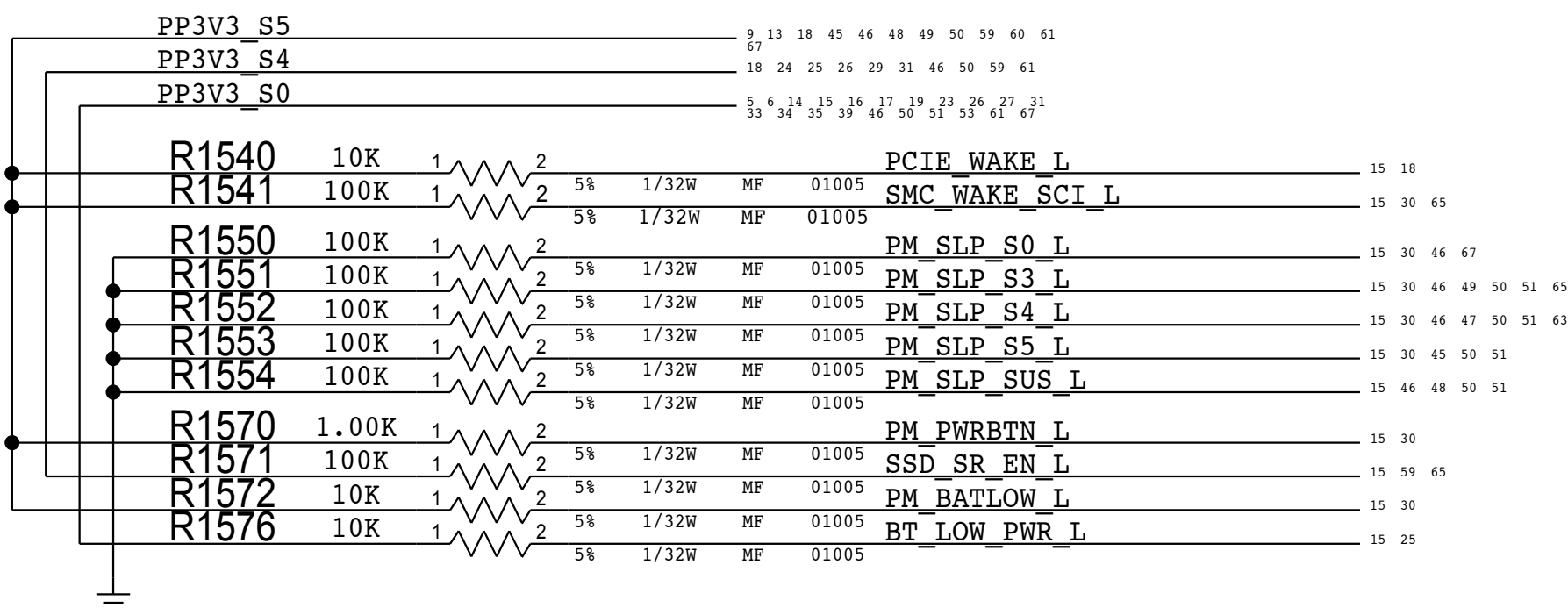
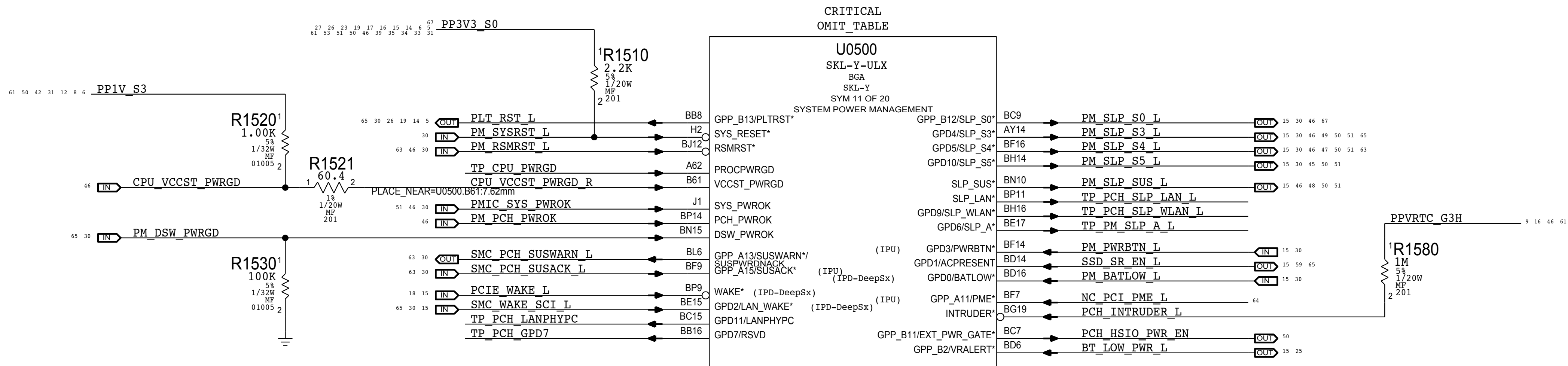
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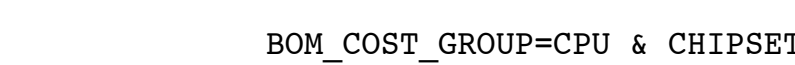
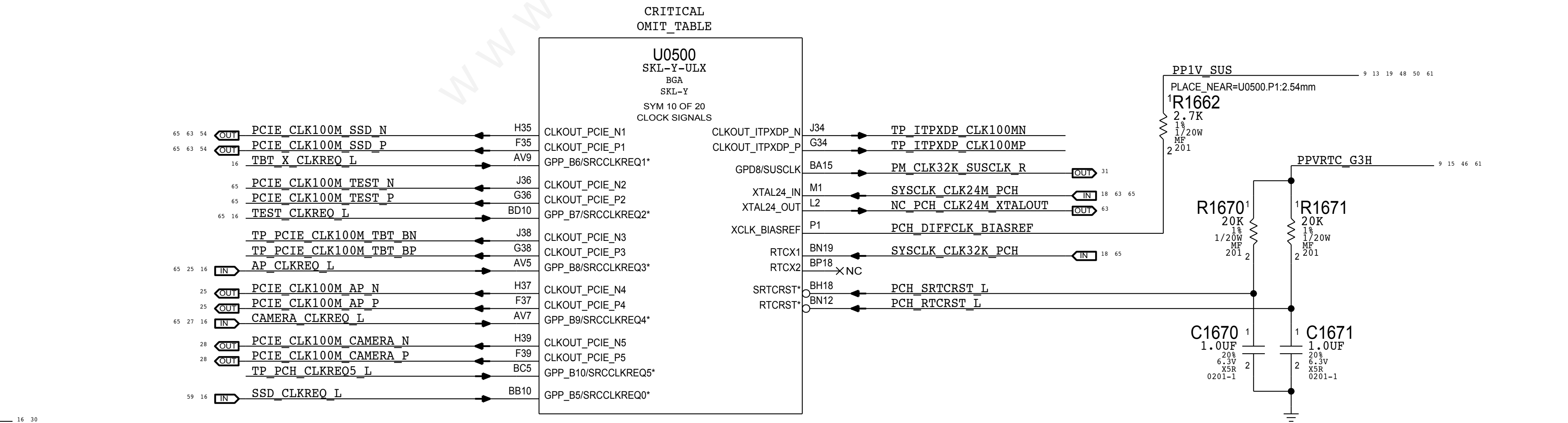
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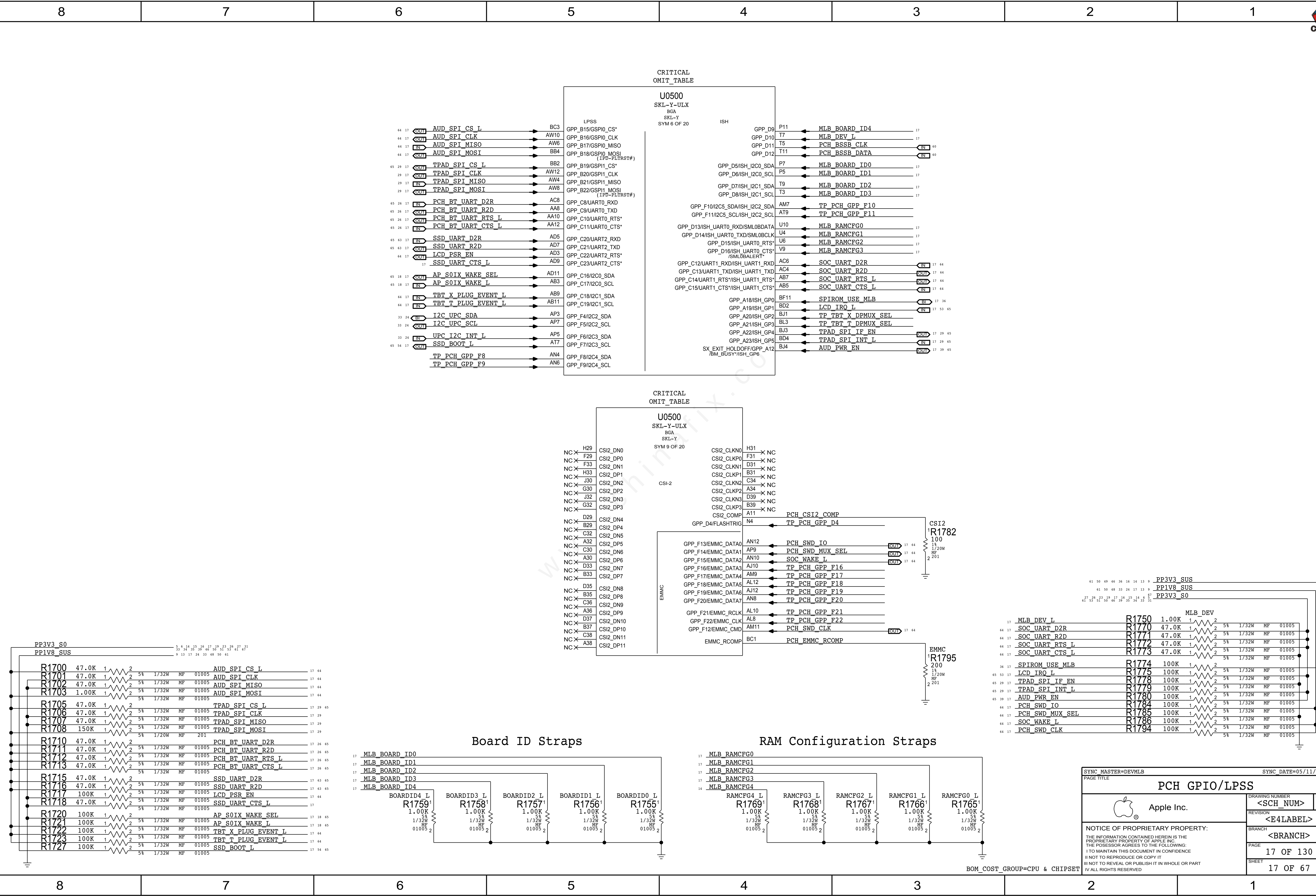
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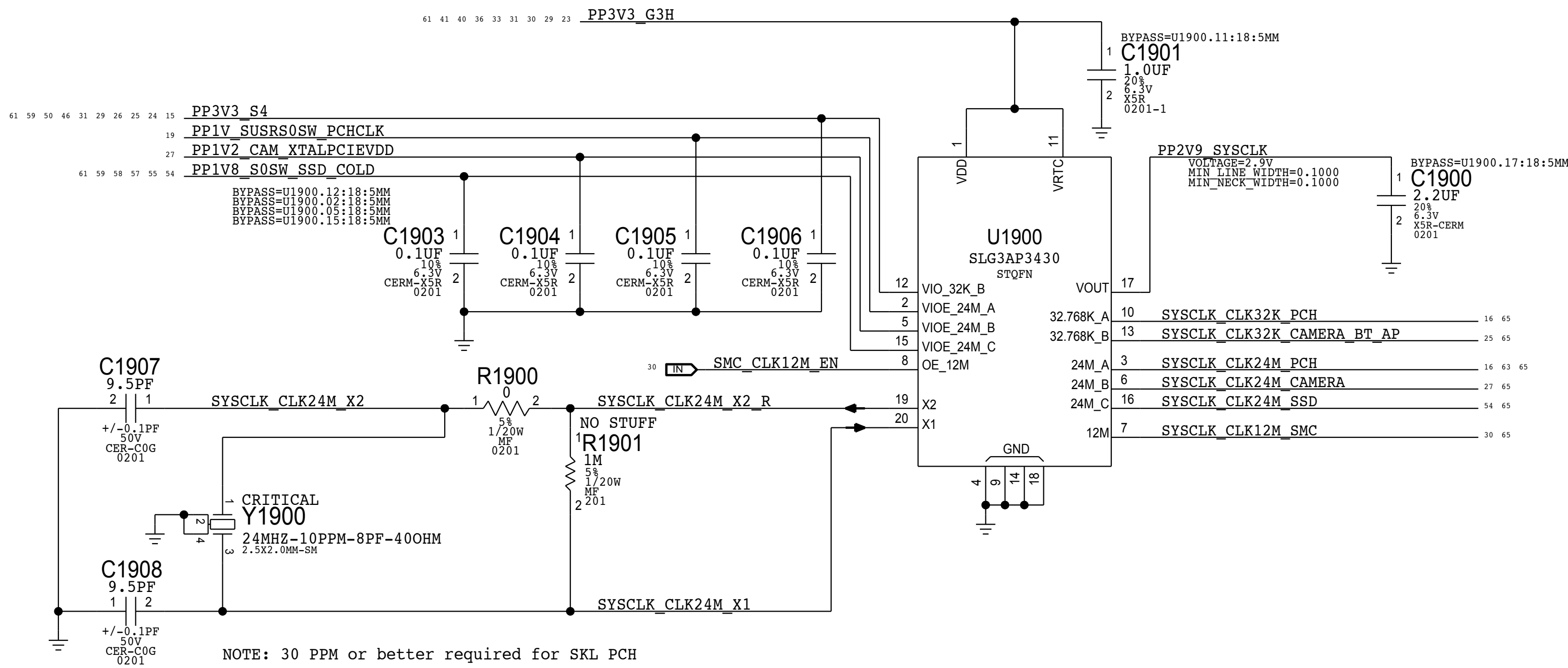
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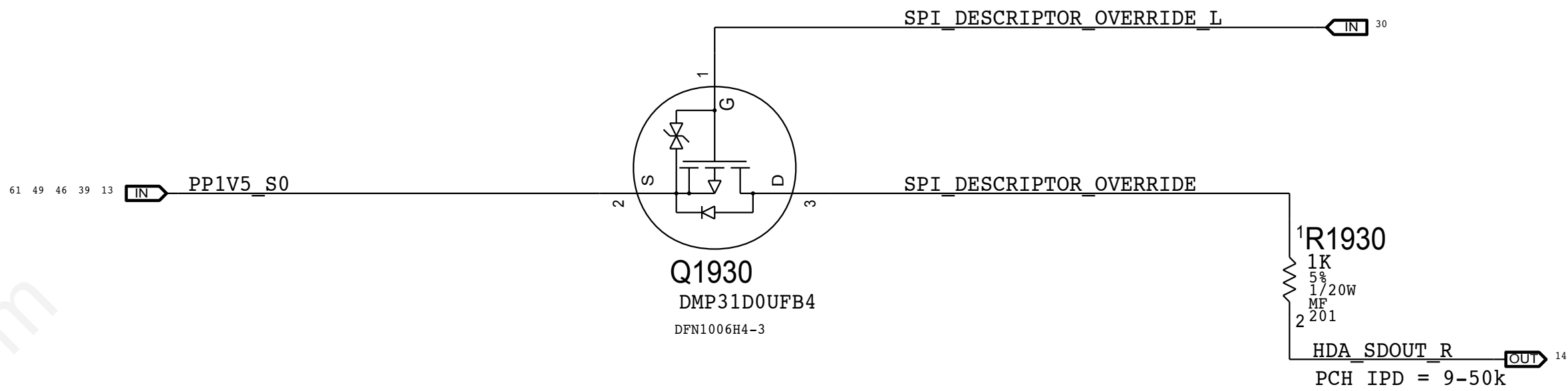
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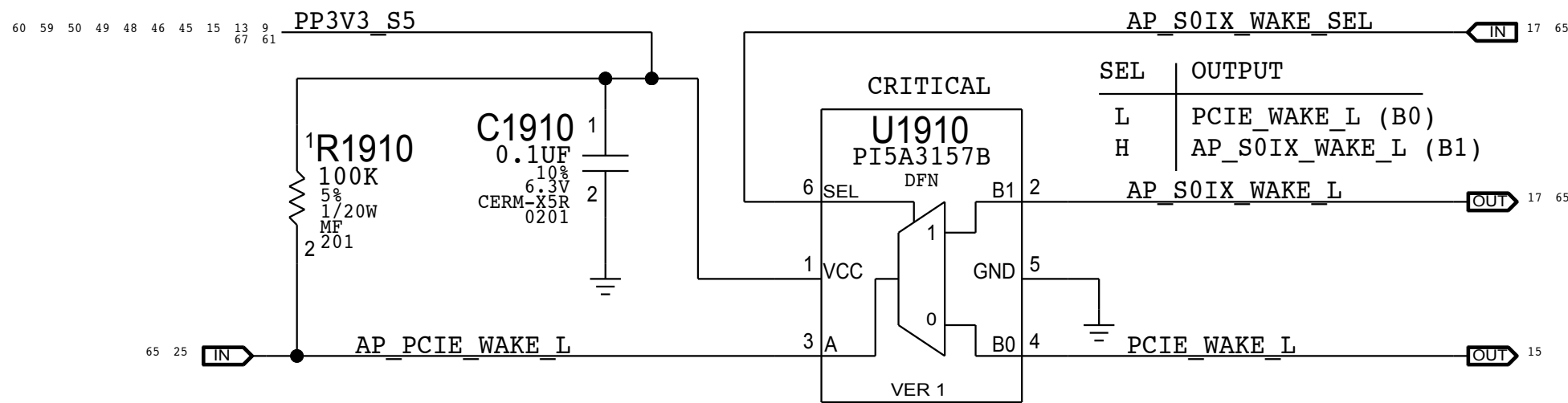
System 32kHz / 12MHz / 24MHz Clock Generator



PCH ME Disable Strap



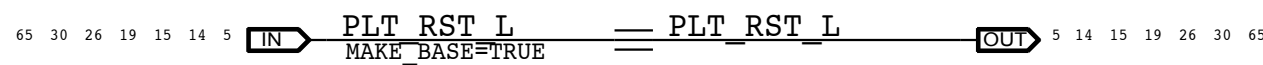
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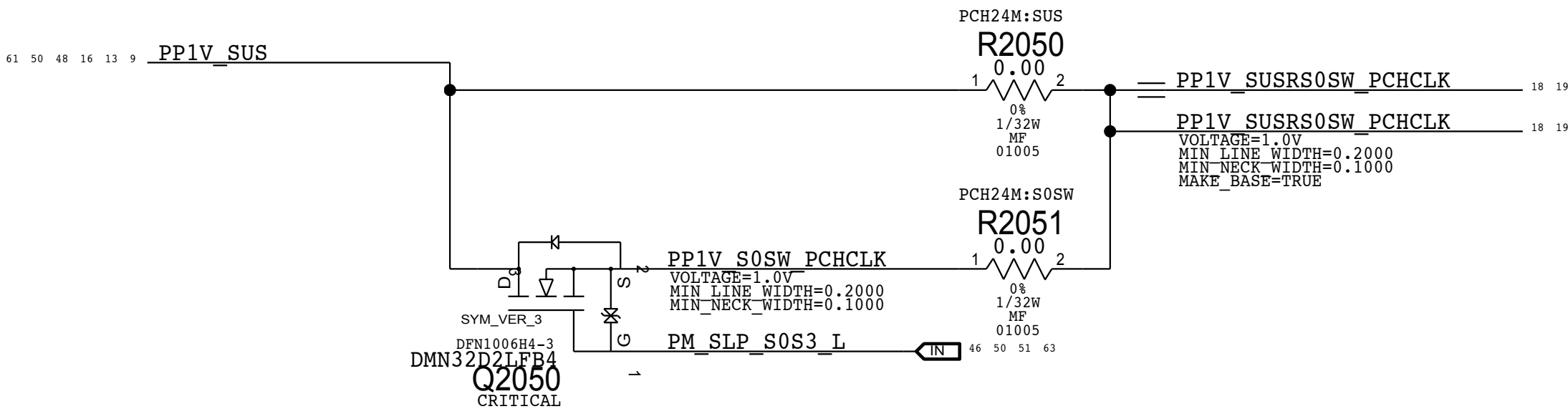


Platform Reset Connections

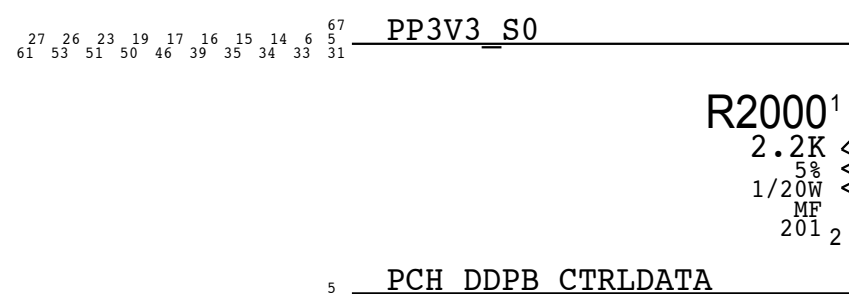
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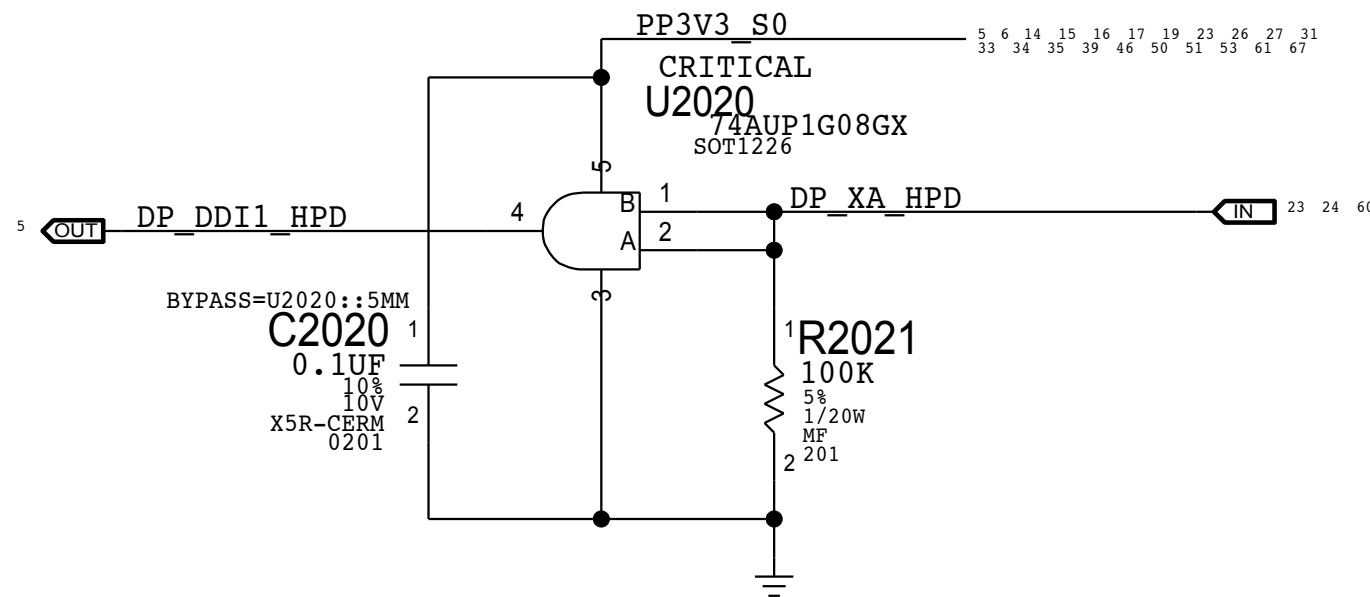
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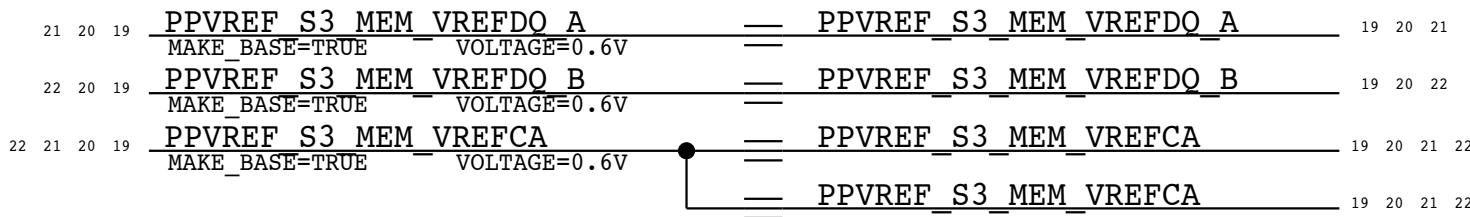
DP DDP Straps



HPD S0 Isolation

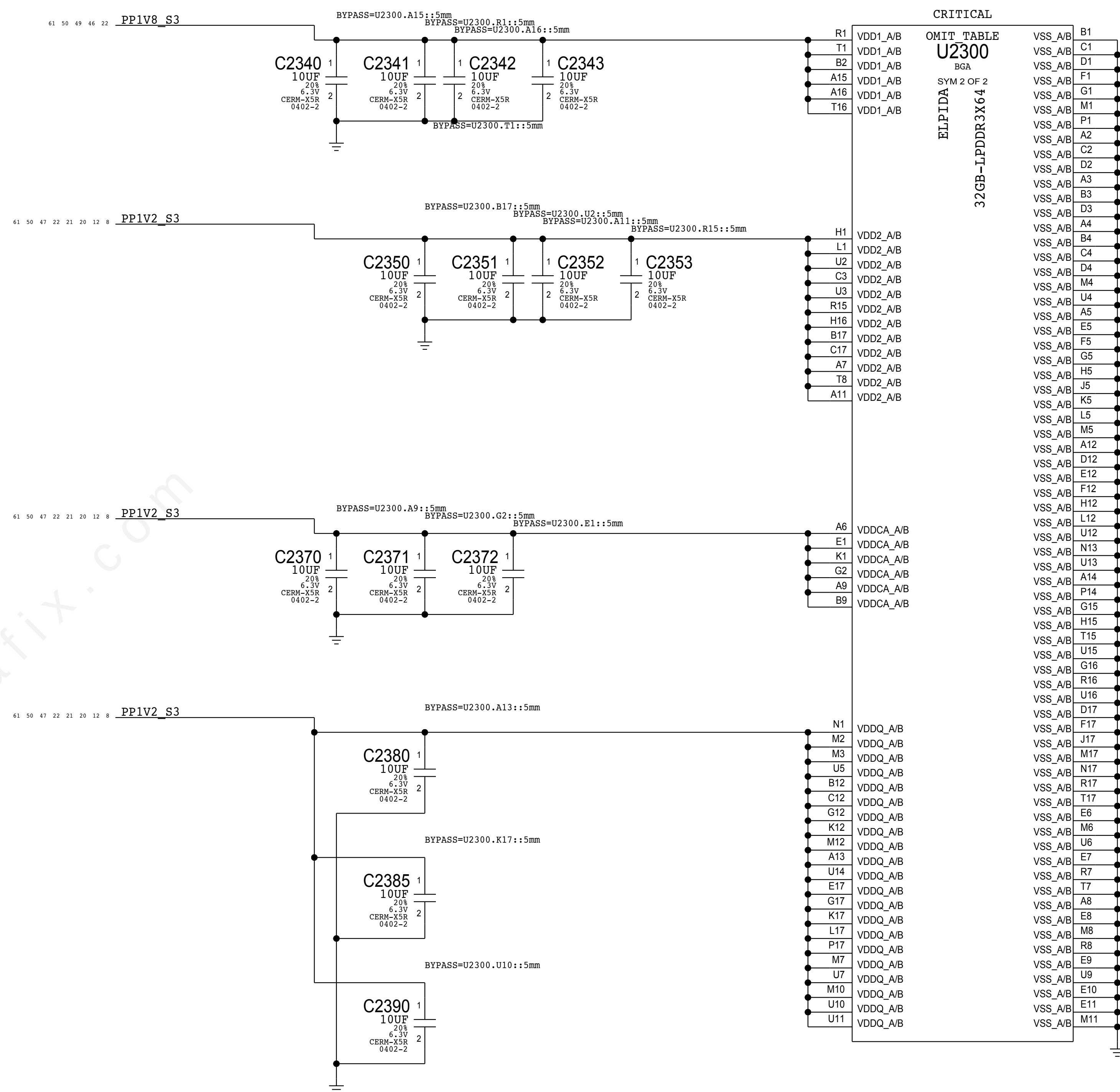
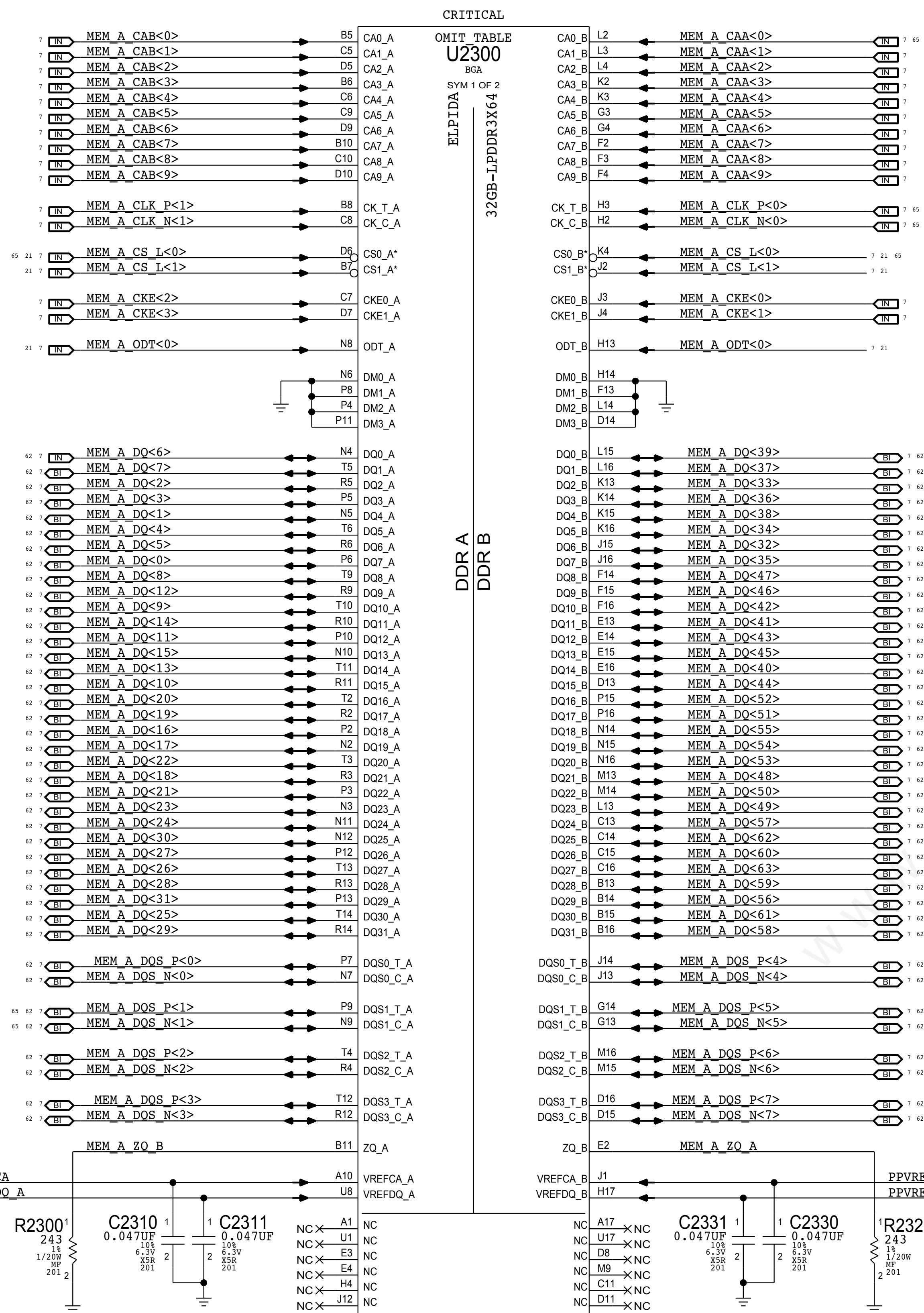


LPDDR3 Alias Support



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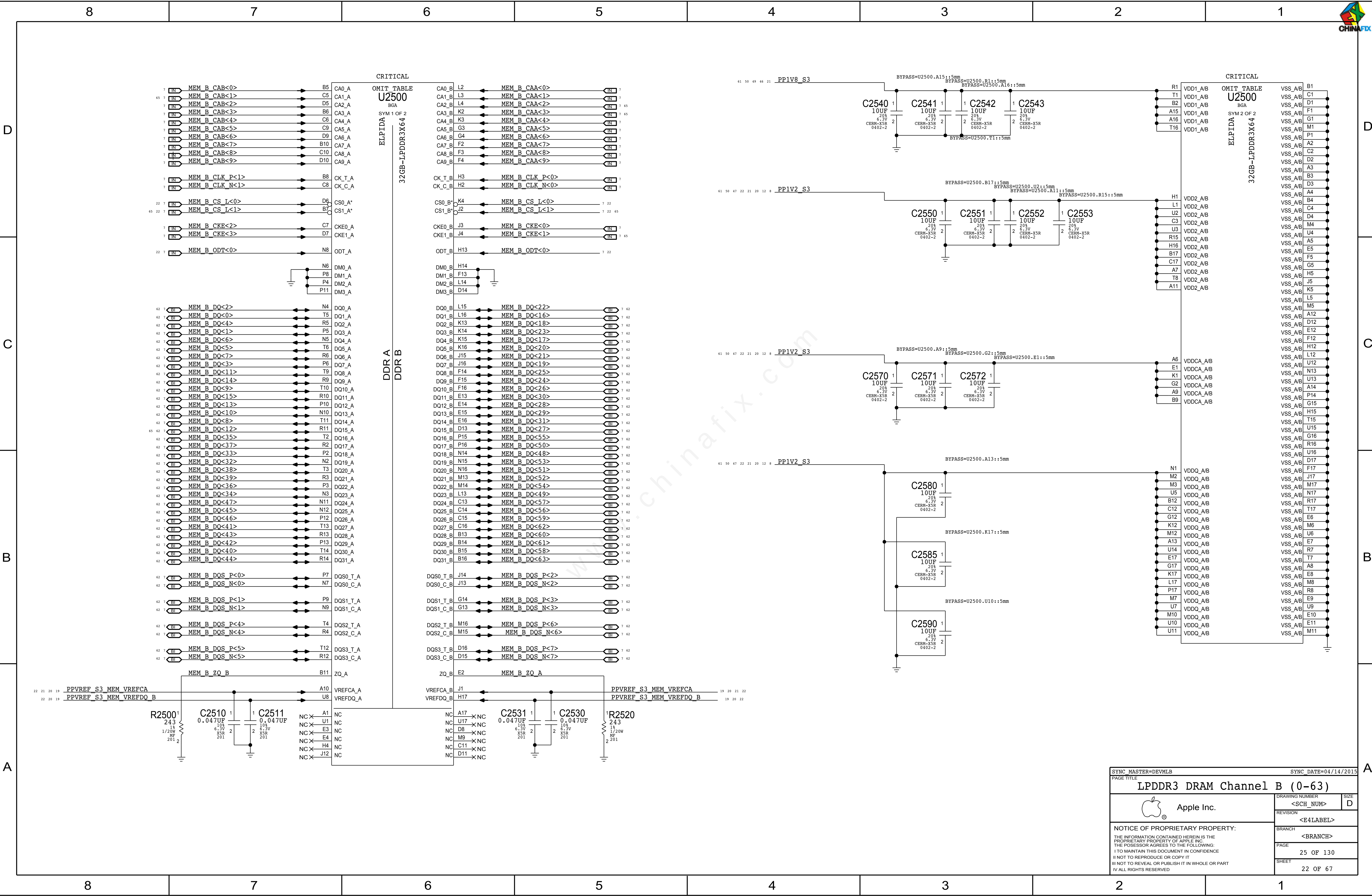
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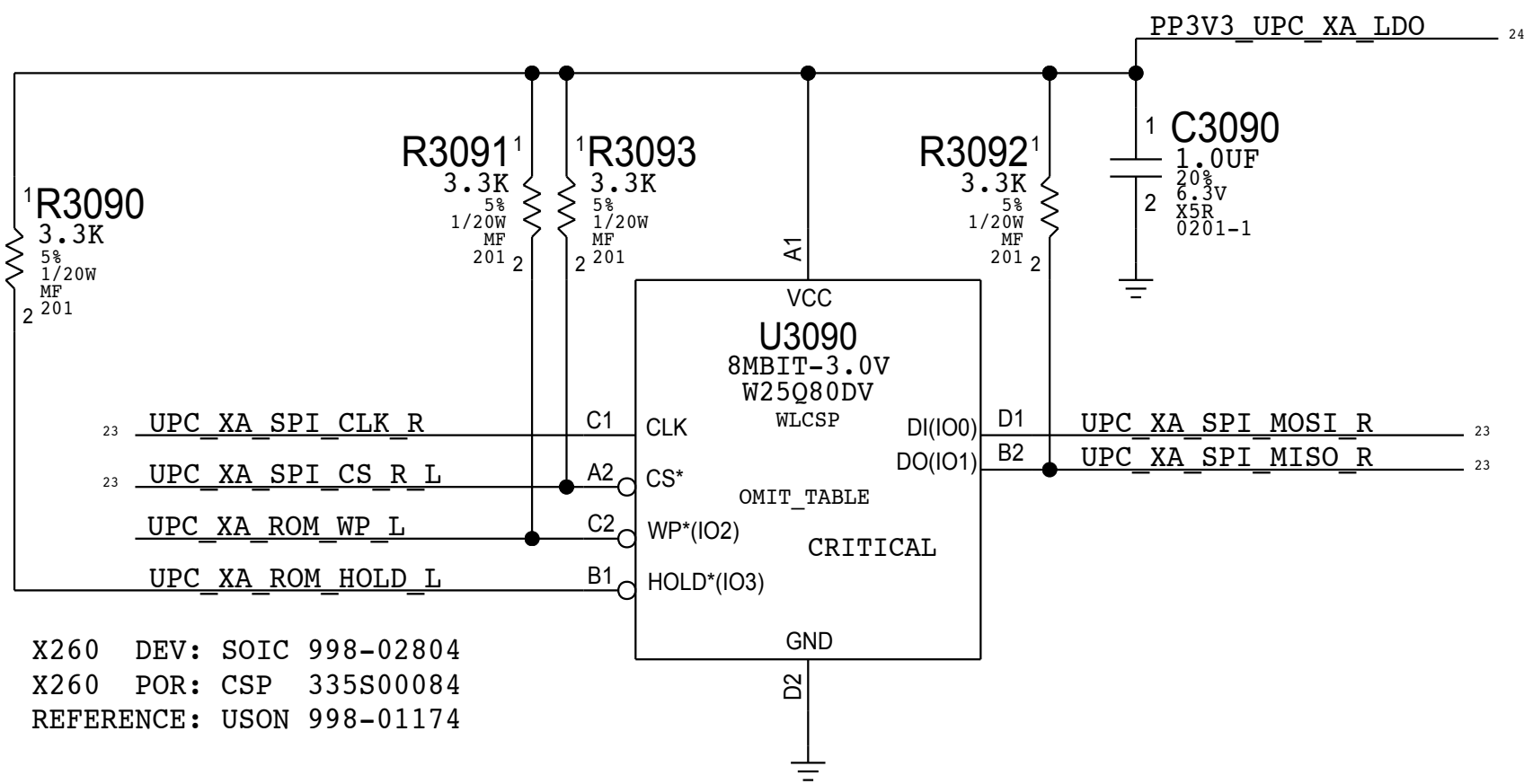
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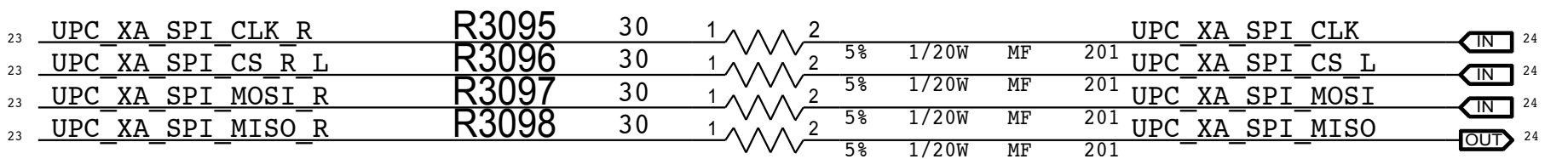




Port Controller ROM

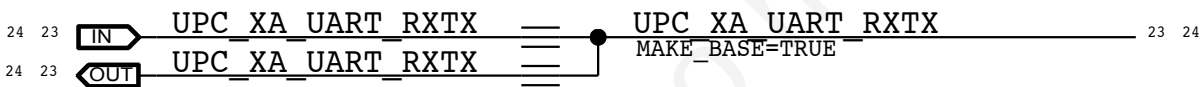


X260 DEV: SOIC 998-02804
X260 POR: CSP 335S00084
REFERENCE: USON 998-01174

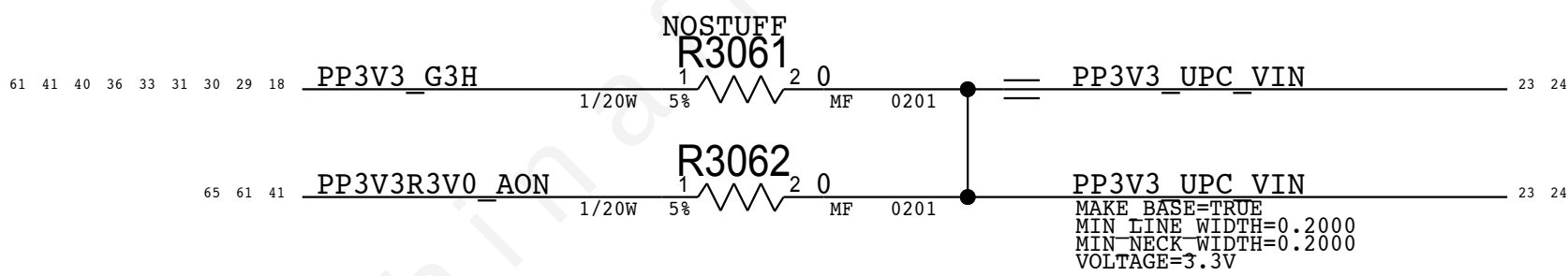


Place series R at T with dev debug header
Characterize series R on P0 Dev, hopefully remove for P0 POR

Single Port Support

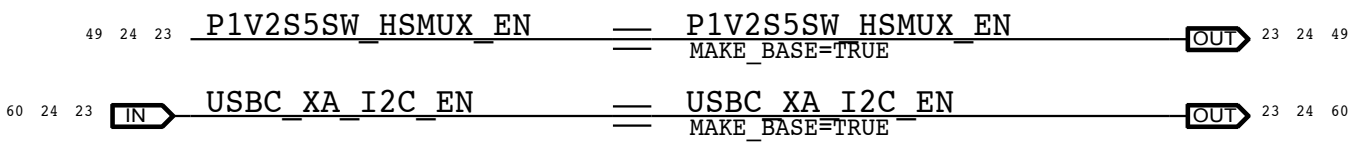


Bring Up Support

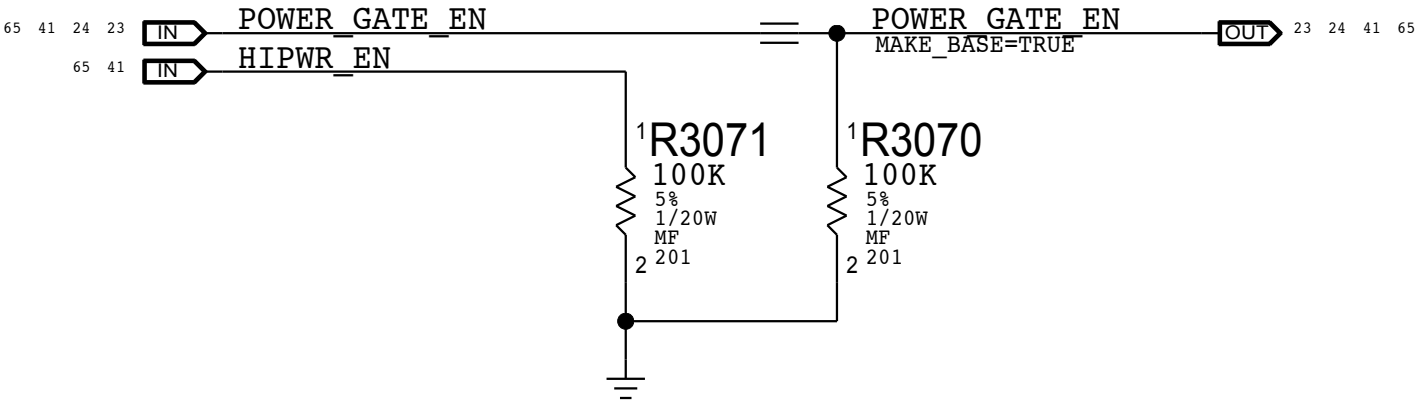


VIN_3V3 power currently required to drive POWER_GATE_EN high. Future OTP update may enable usage of 3V3_G3H instead.

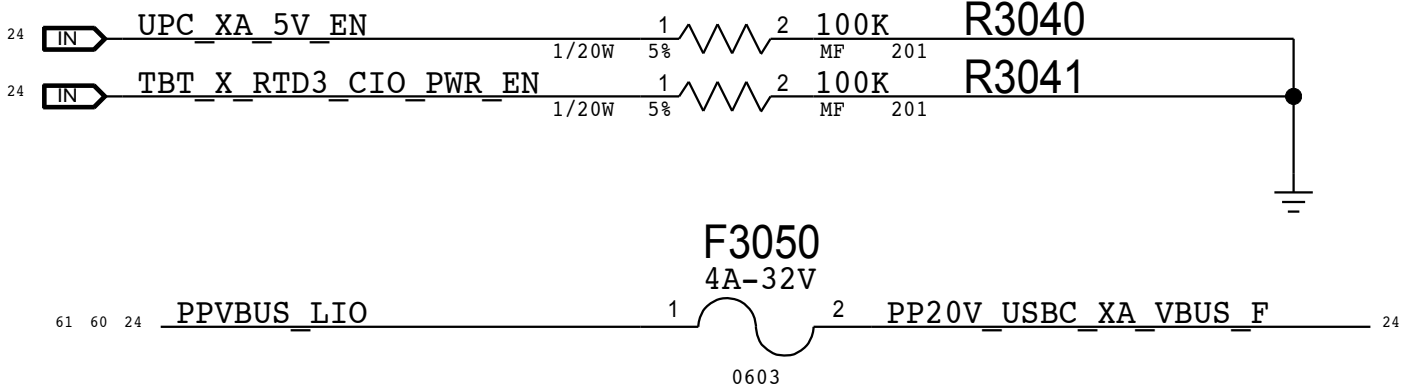
High Speed Mux Support



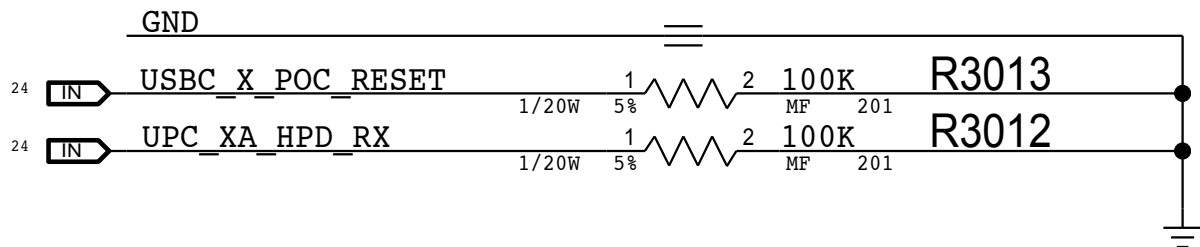
Bansuri Support



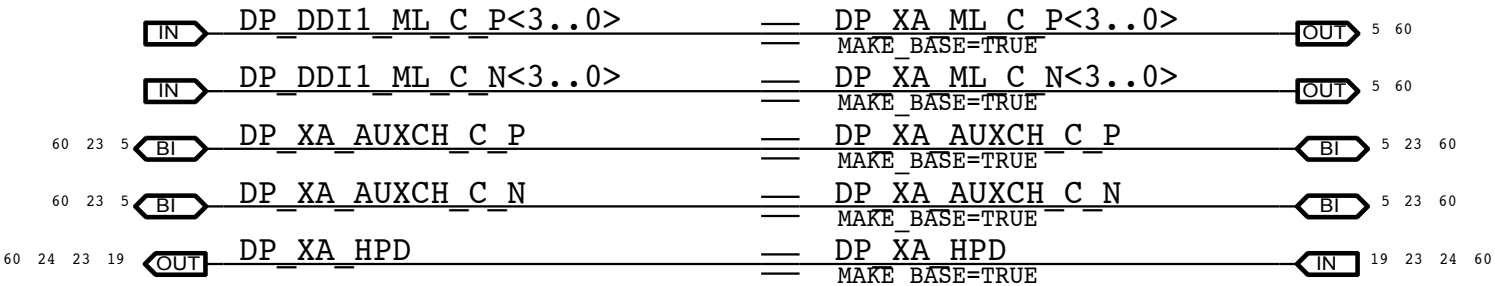
Other Support



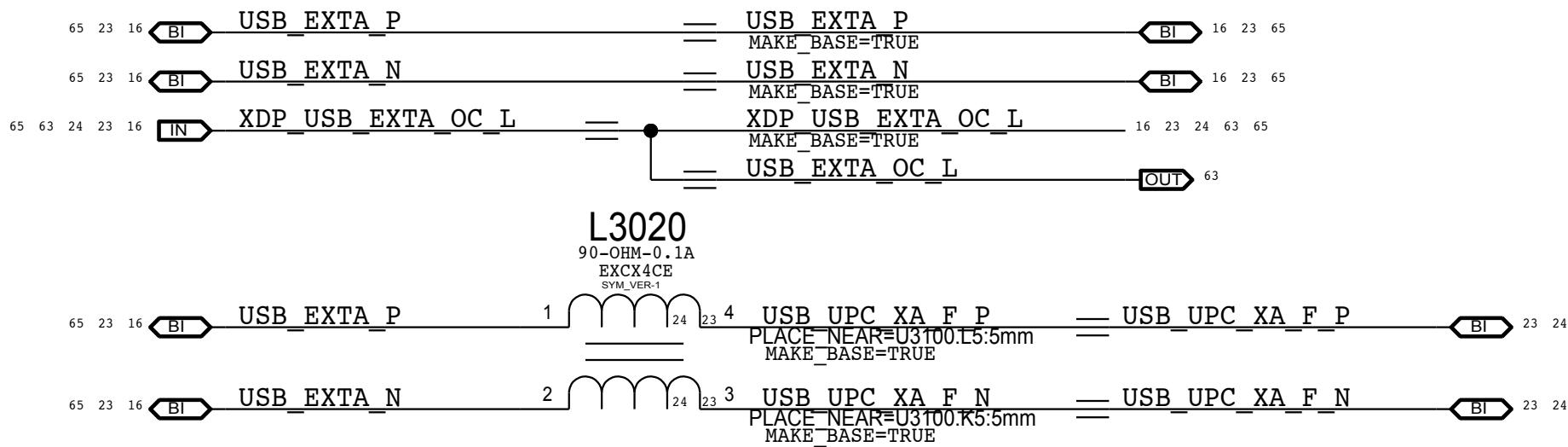
'Ridge-less Support



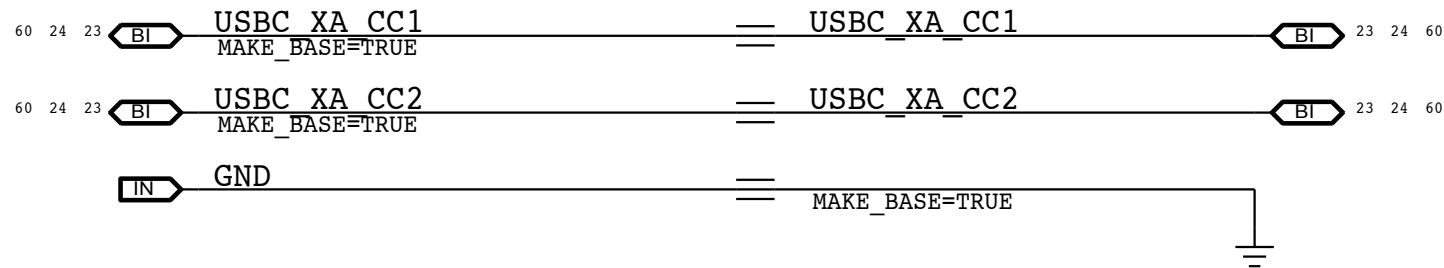
DP Support



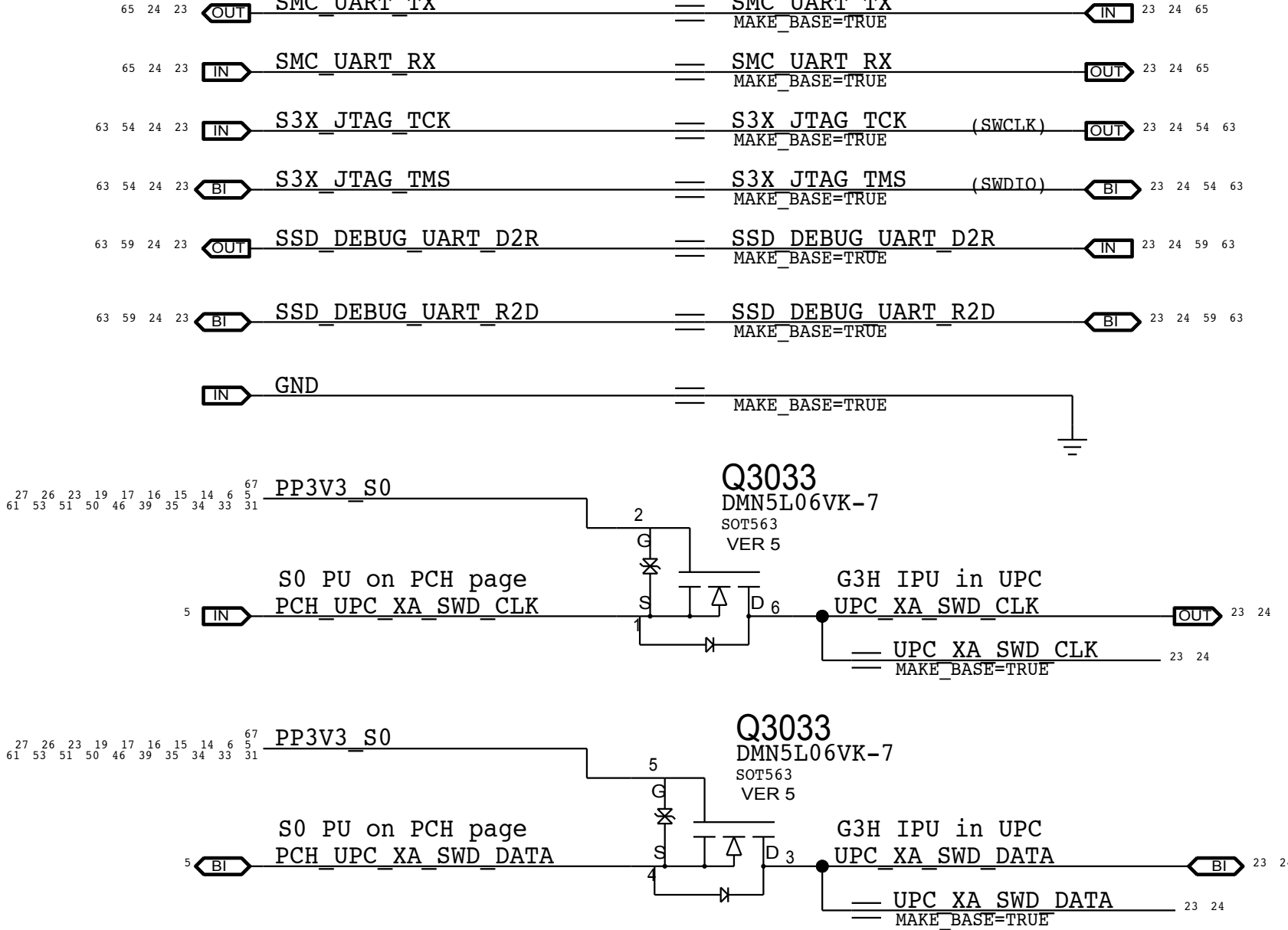
USB Support



Dead Battery Support

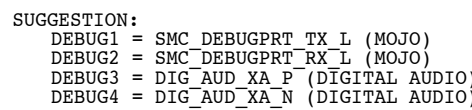


Debug Support



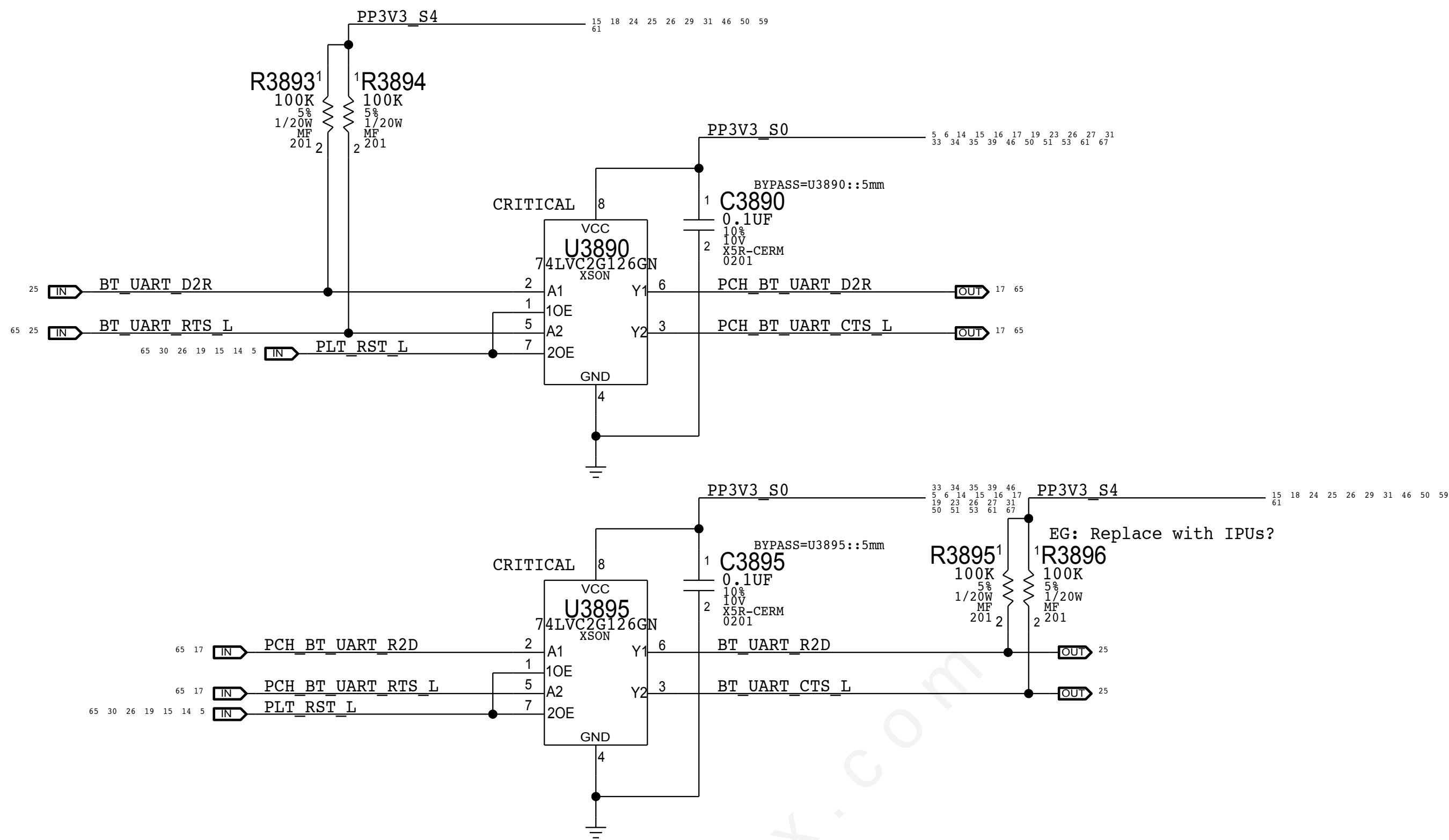
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USB-C Port Controller Support		DRAWING NUMBER		SIZE	
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BOM COST GROUP=USB-C



SYMC_MSTFTR=K240_EB3C PAGE TITLE	SYMC_MSTFTR=K240_EB3C DATE=04/04/2013										
USB-A PORT CONTROLLER A											
Apple Inc.	<table border="1" style="width:100%; border-collapse: collapse;"> <tr> <td style="width:80%; padding: 5px;">DRAWING NUMBER <SCH_NUM></td> <td style="width:20%; padding: 5px; text-align: center;">SIZE D</td> </tr> <tr> <td colspan="2" style="padding: 5px;">REVISION <B4 LABEL></td> </tr> <tr> <td colspan="2" style="padding: 5px;">BRANCH <BRANCH></td> </tr> <tr> <td colspan="2" style="padding: 5px;">PAGE 31 OF 130</td> </tr> <tr> <td colspan="2" style="padding: 5px;">SHEET 24 OF 67</td> </tr> </table>	DRAWING NUMBER <SCH_NUM>	SIZE D	REVISION <B4 LABEL>		BRANCH <BRANCH>		PAGE 31 OF 130		SHEET 24 OF 67	
DRAWING NUMBER <SCH_NUM>	SIZE D										
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BT UART Isolation





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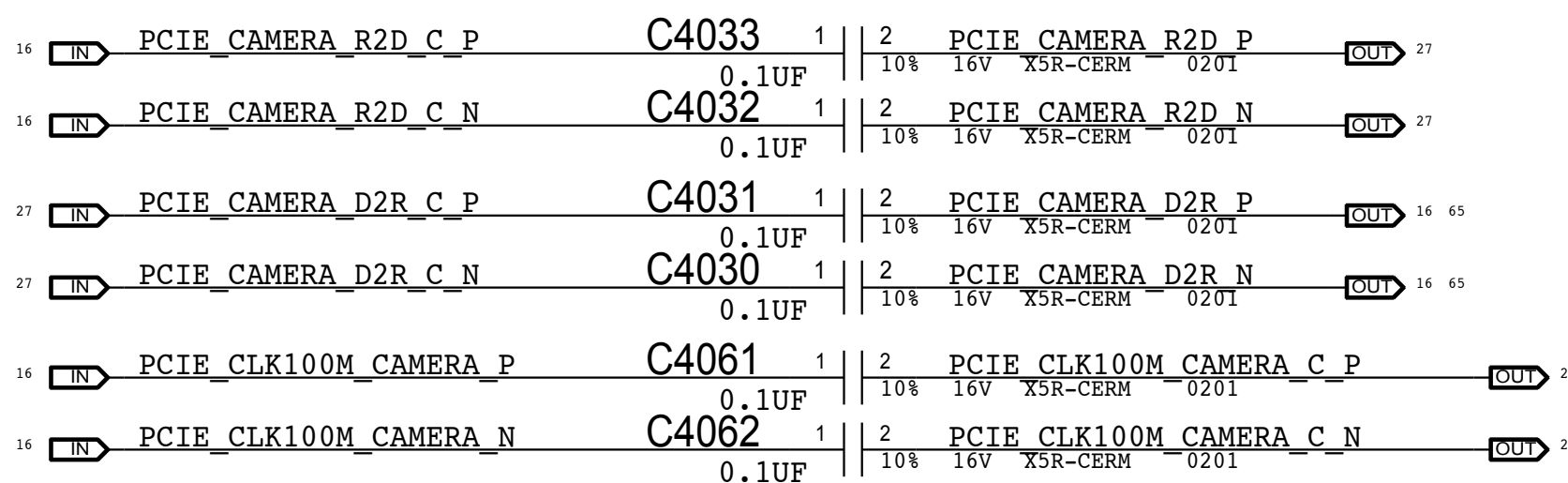
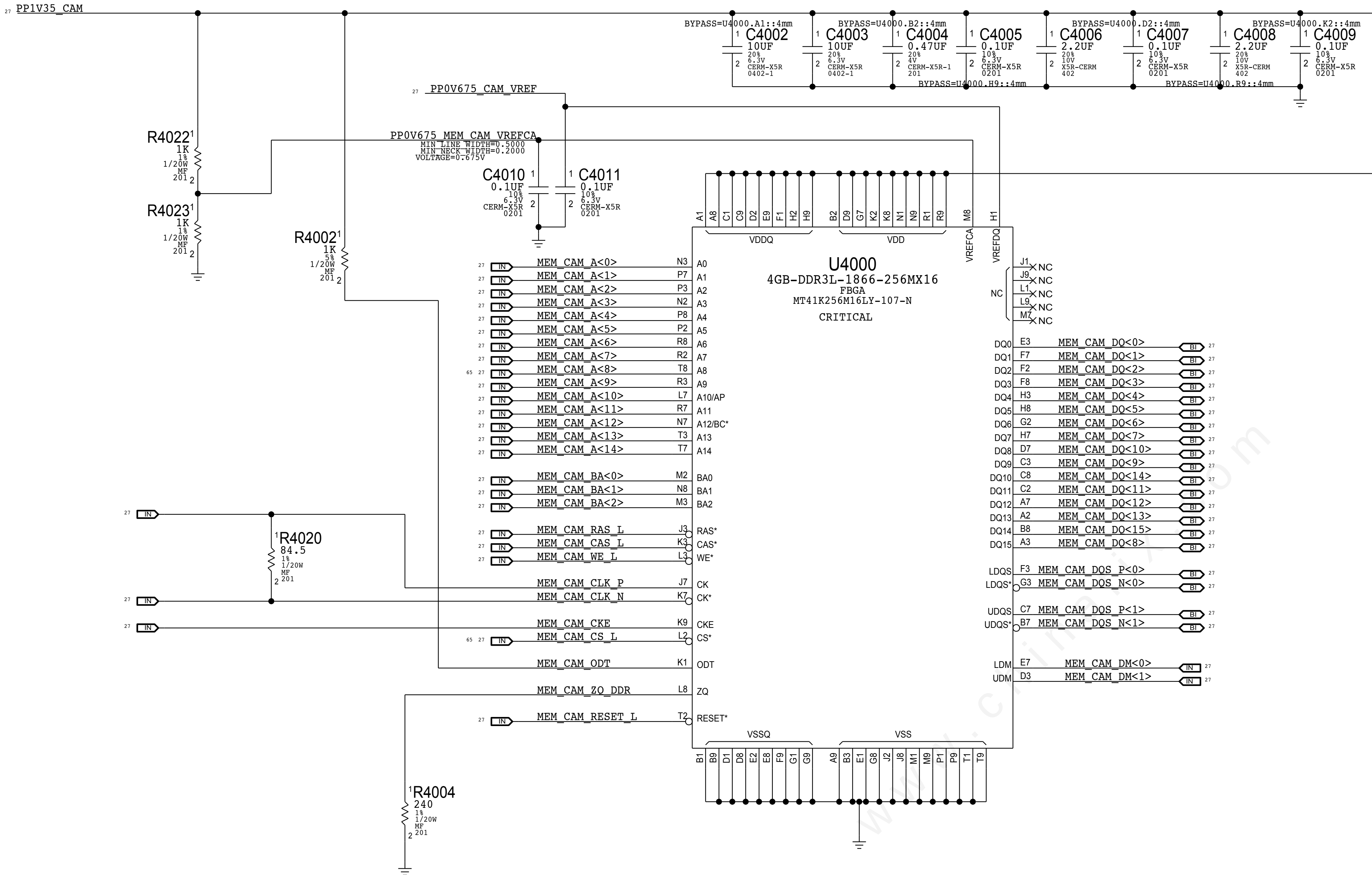
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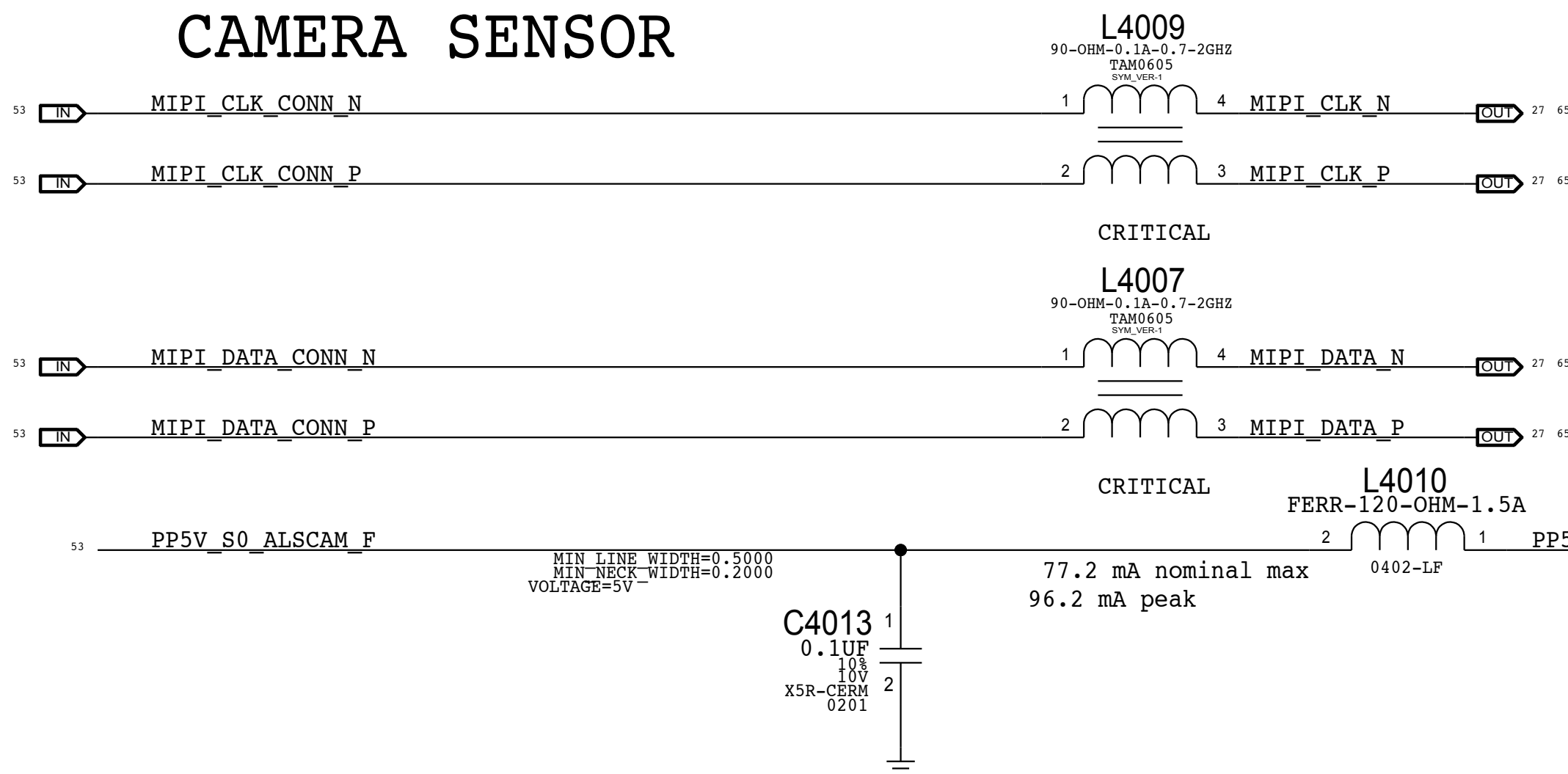
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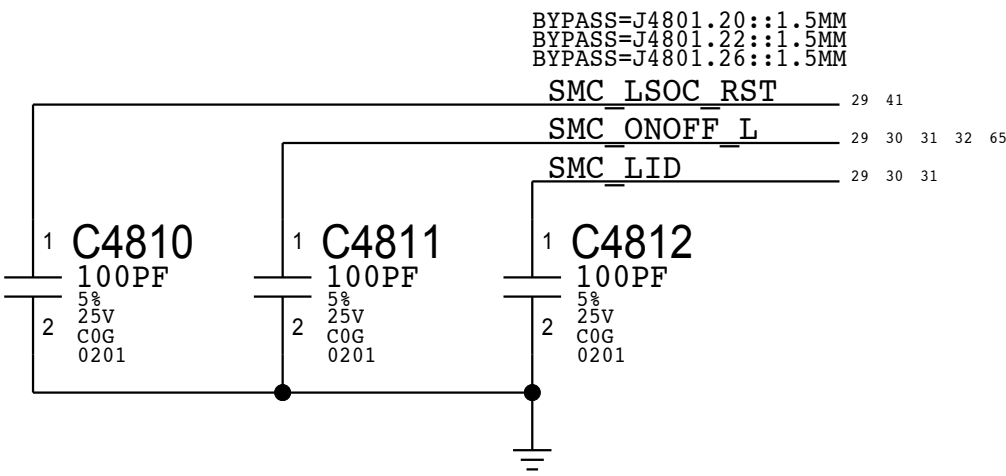
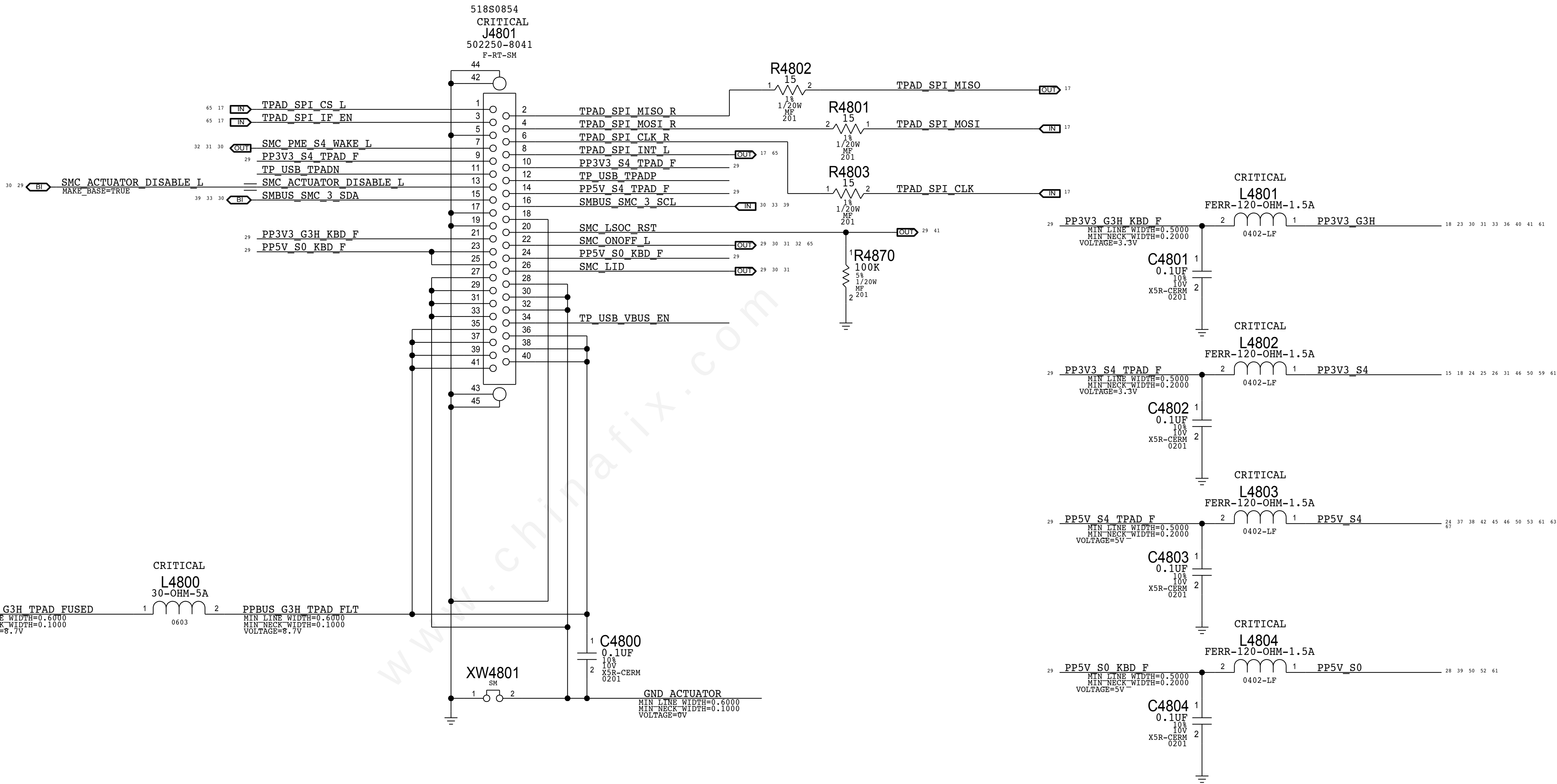
CAMERA SENSOR



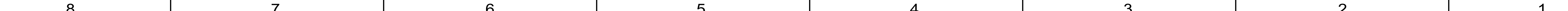
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	PAGE 40 OF 130		
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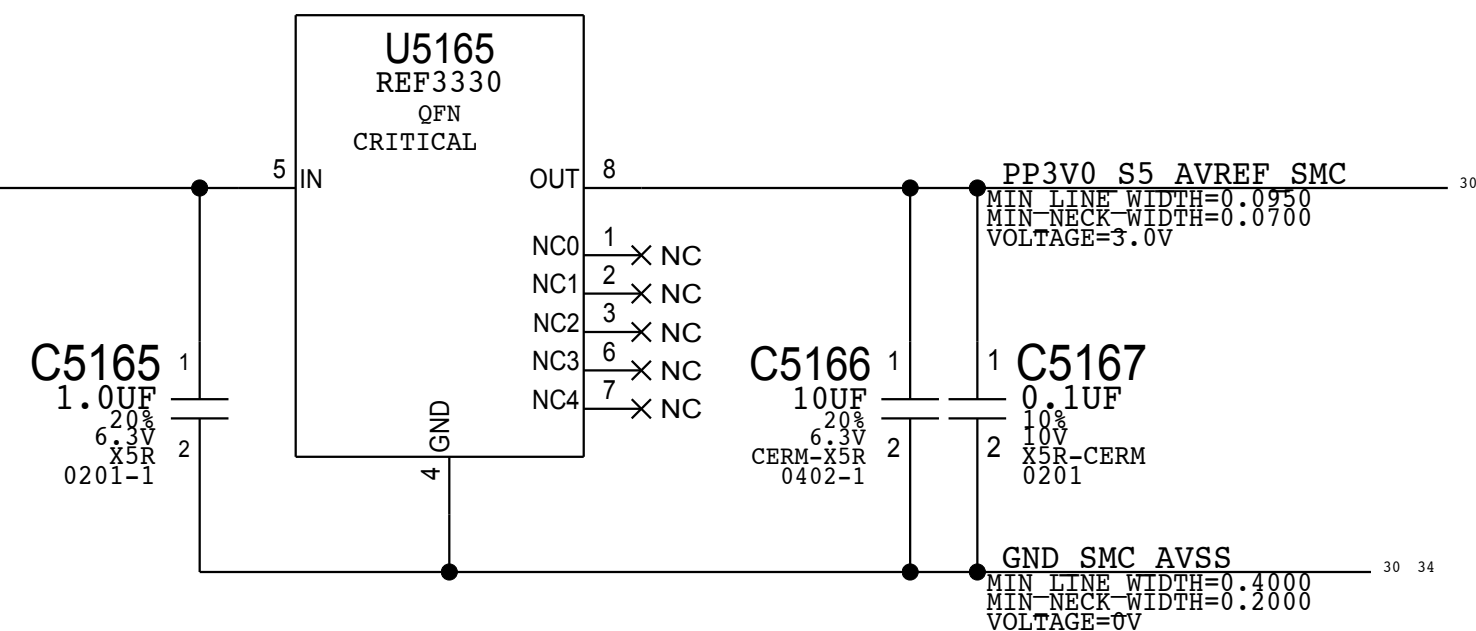
IPD ZIF CONNECTOR

Bottom side contacts used
Pinout reversed from flex

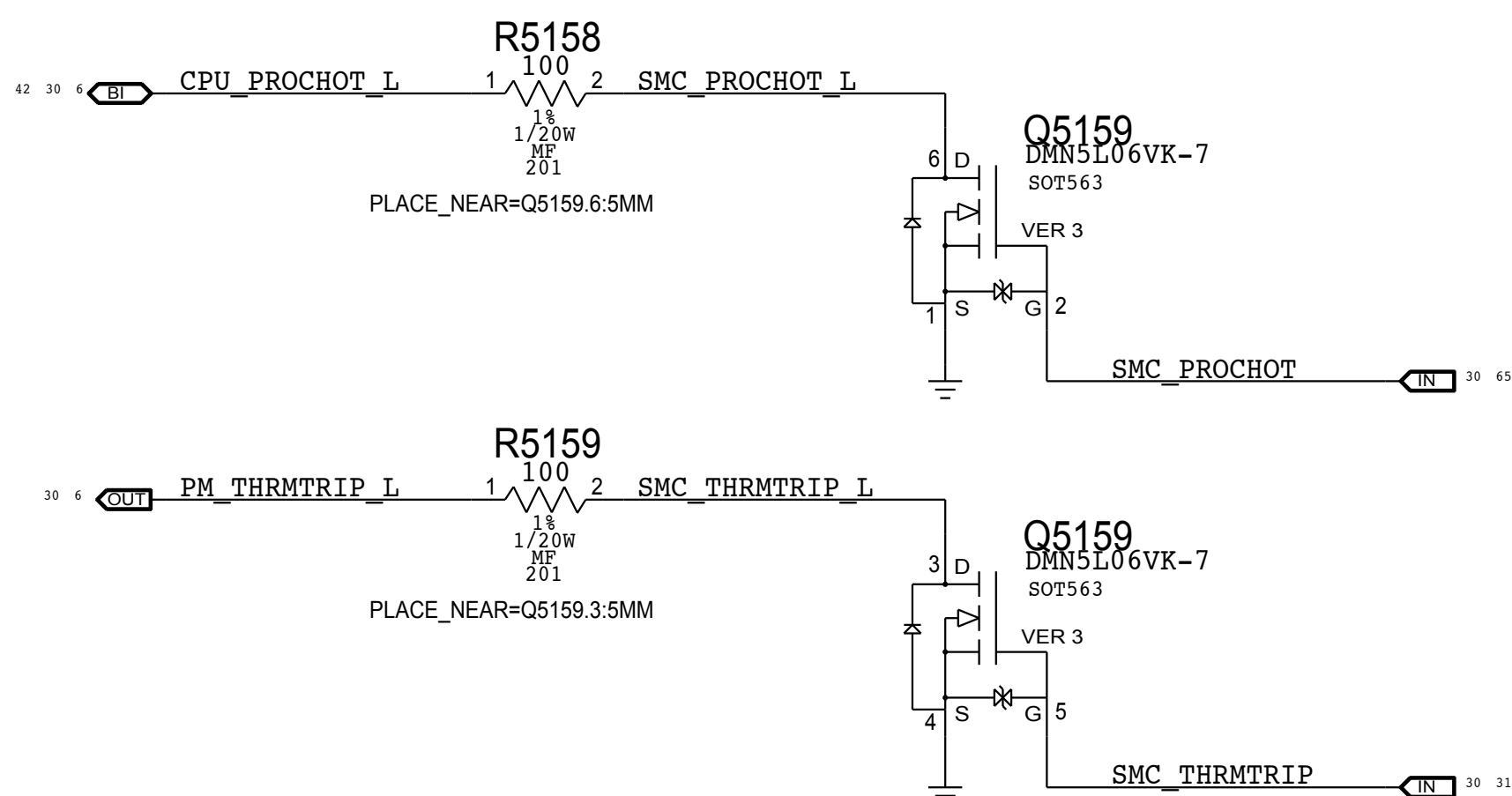


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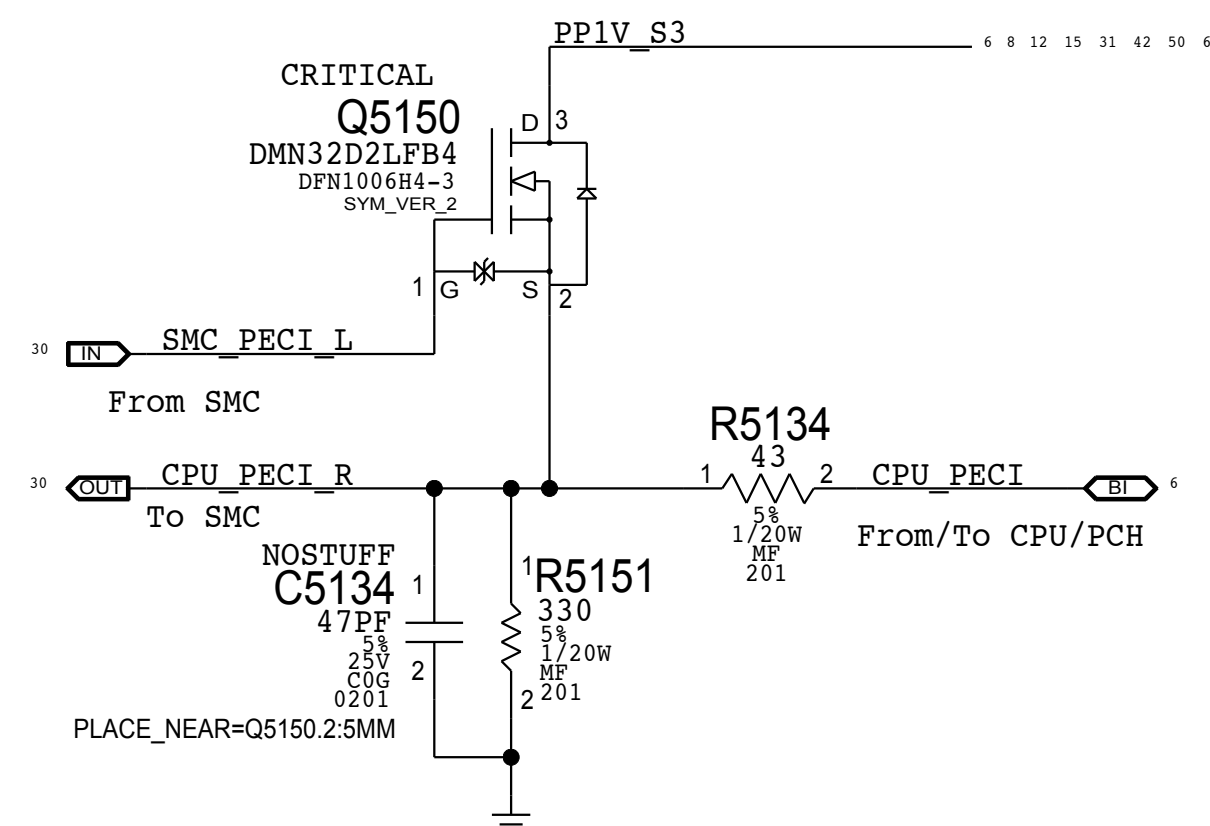




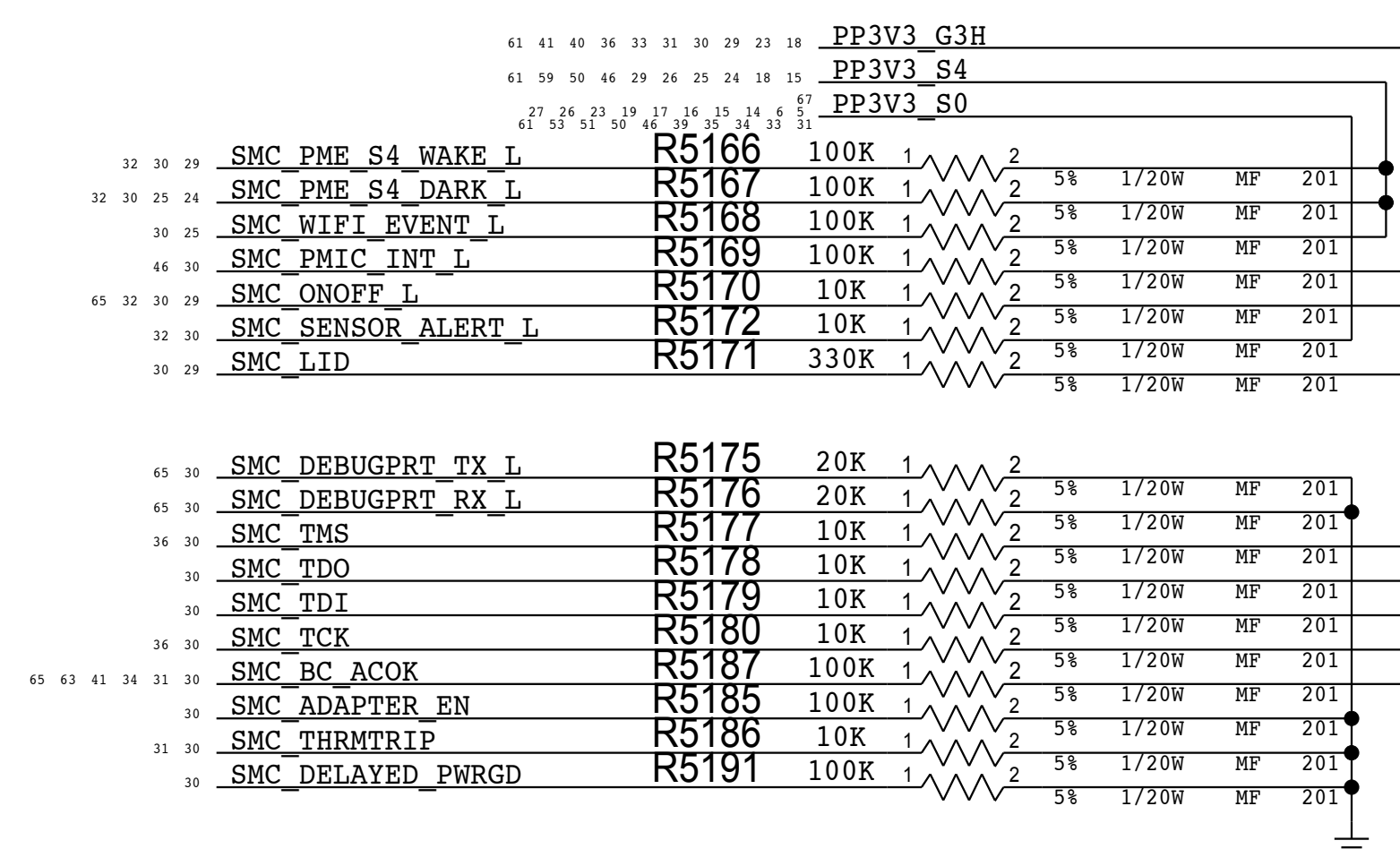
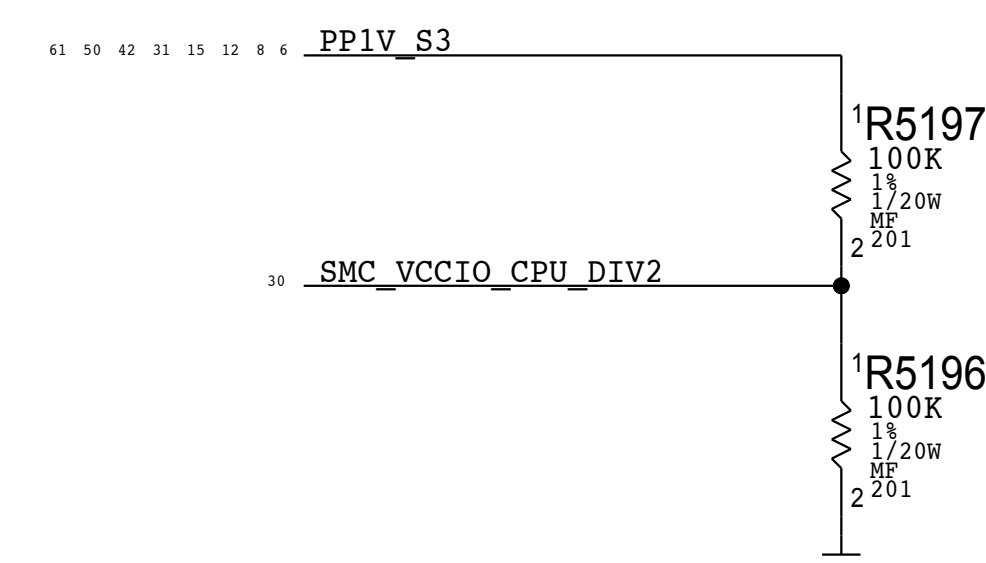
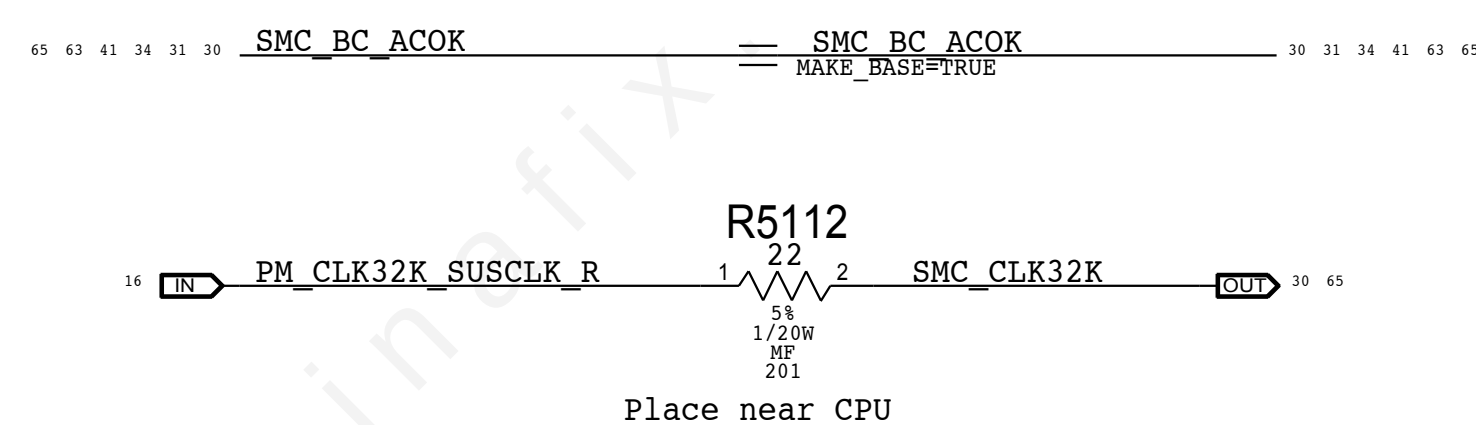
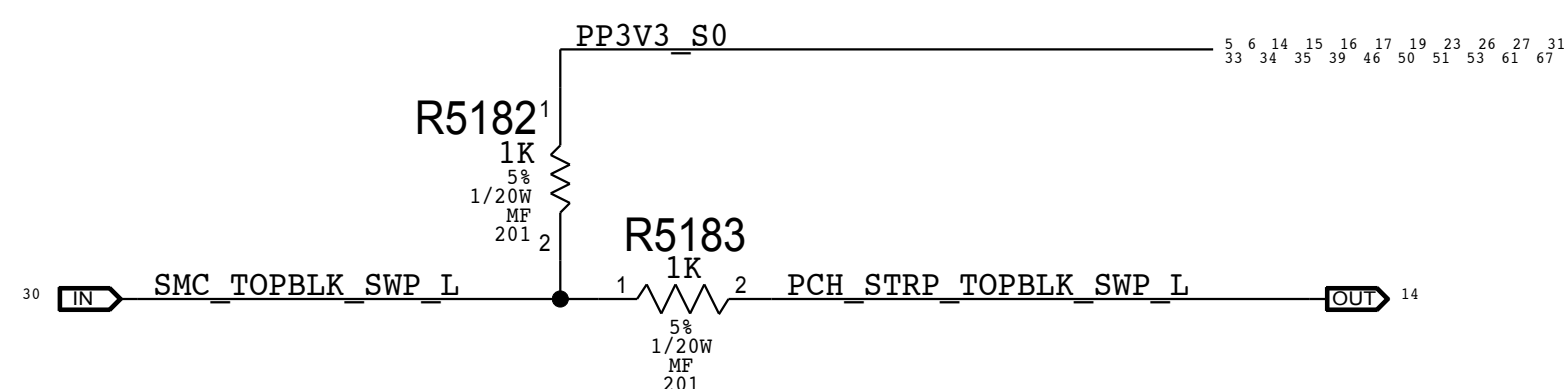
PROCHOT/THRMTRIP Support



PECI Support



Top-Block Swap



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		31 OF 67	



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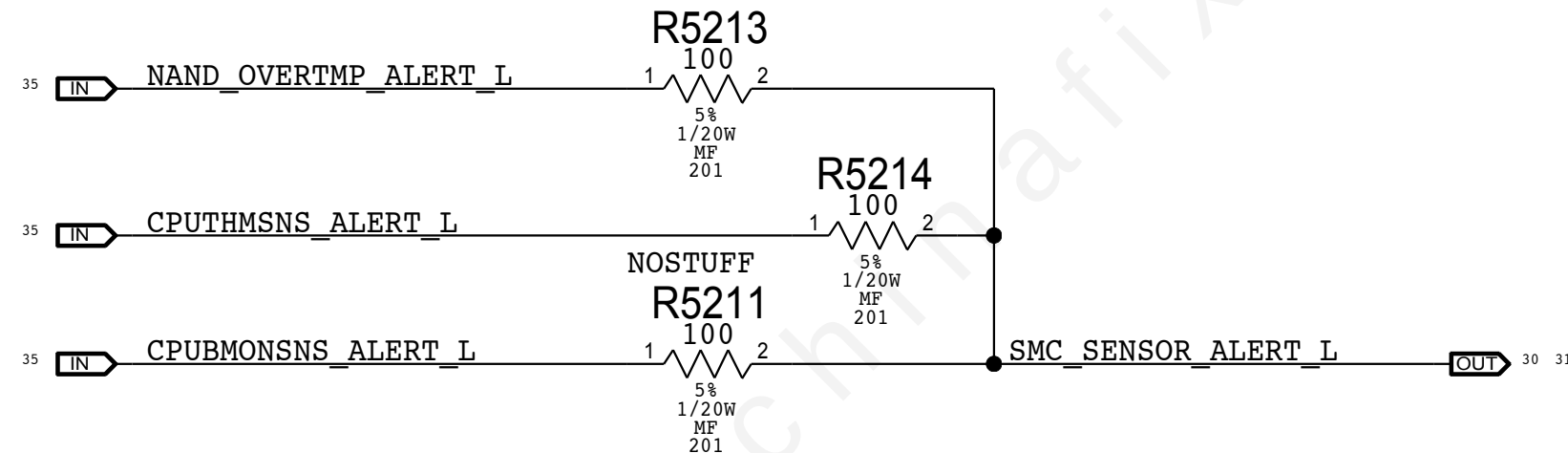
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52 OF 130			
SHEET			
32 OF 67			

34 32 30 SMC_DCIN_VSENSE == SMC_DCIN_VSENSE 30 32 34
MAKE_BASE=TRUE
34 32 30 SMC_DCIN_ISENSE == SMC_DCIN_ISENSE 30 32 34
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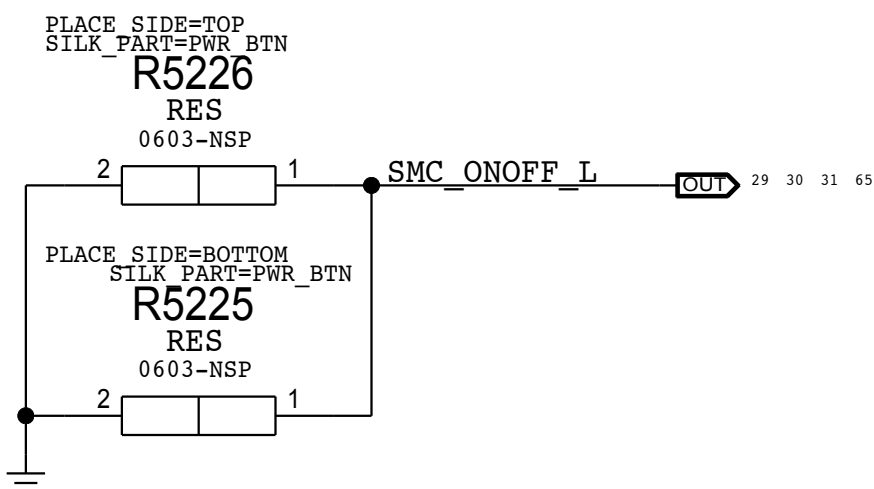
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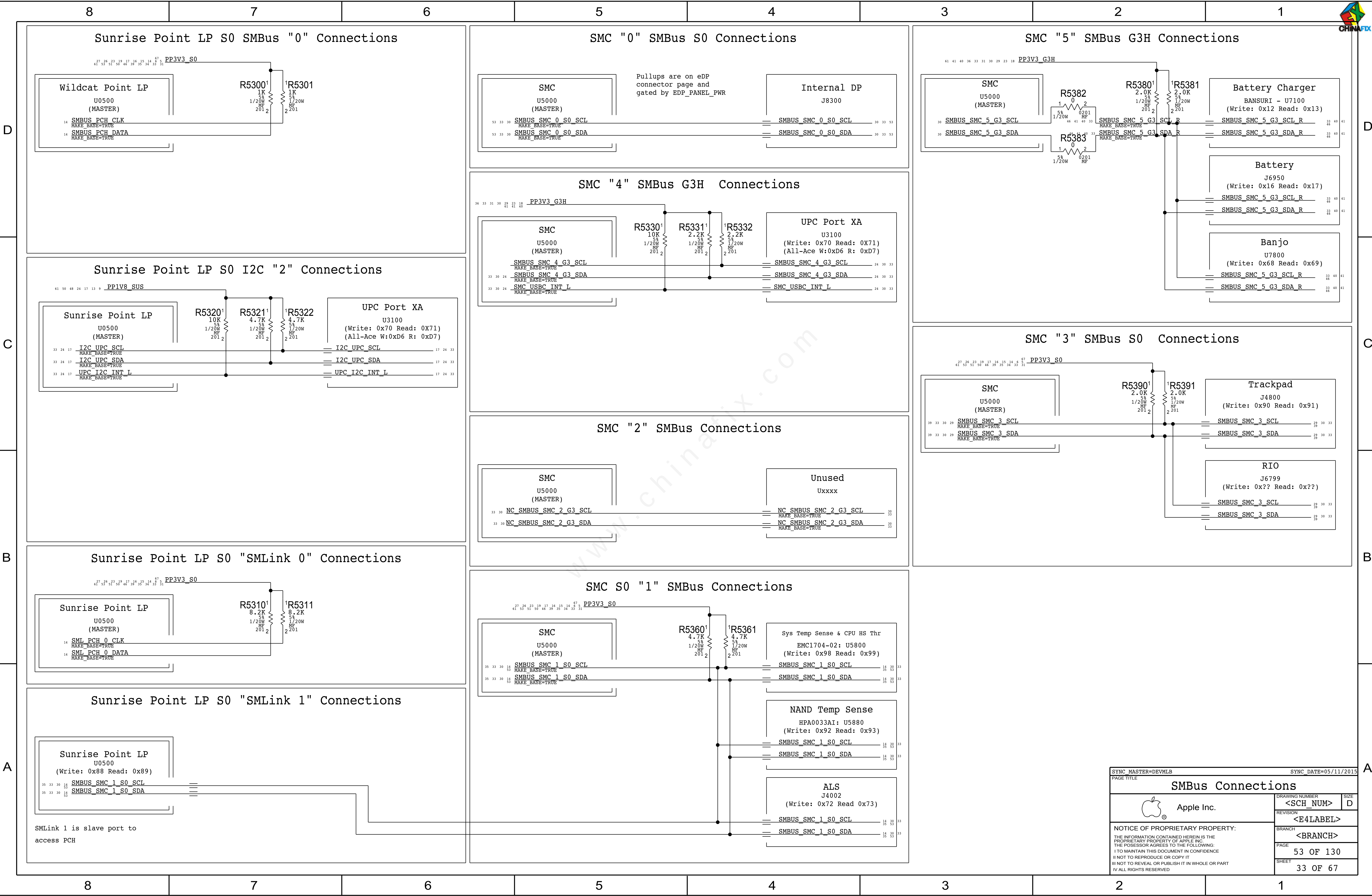


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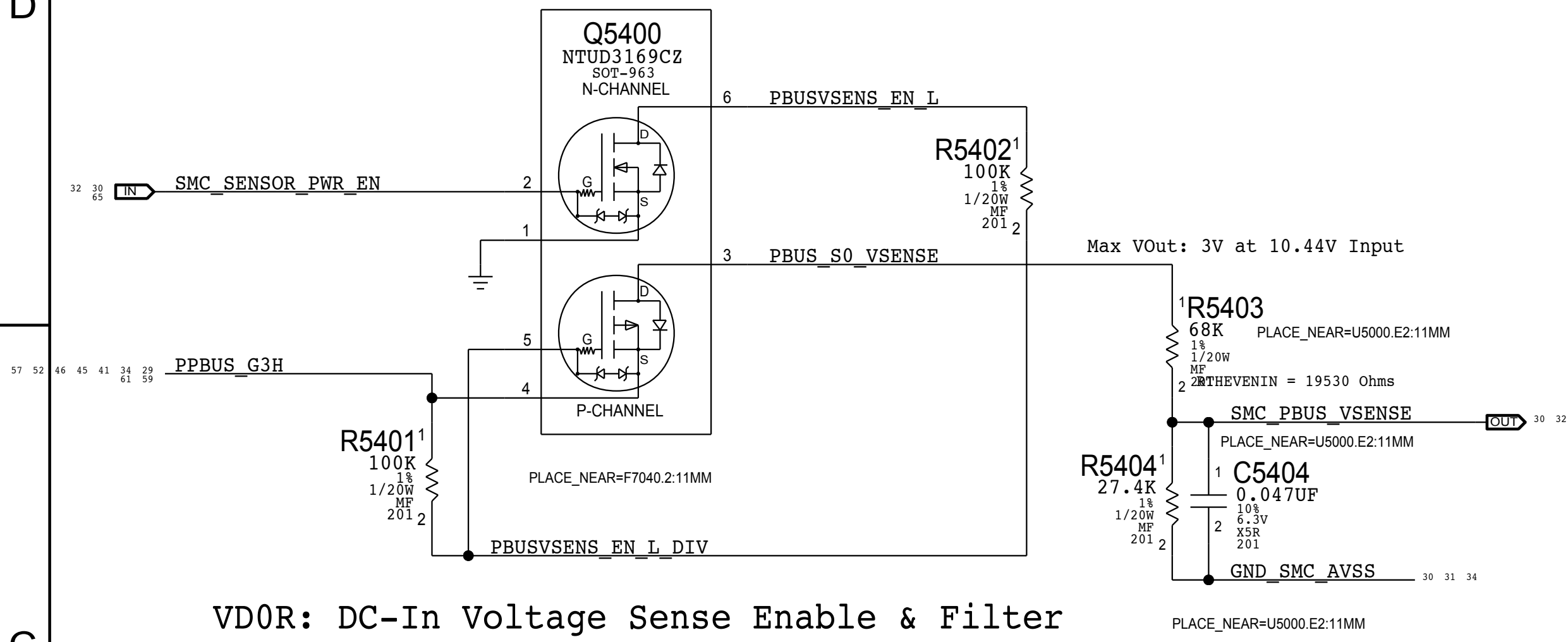
Debug Power "Buttons"



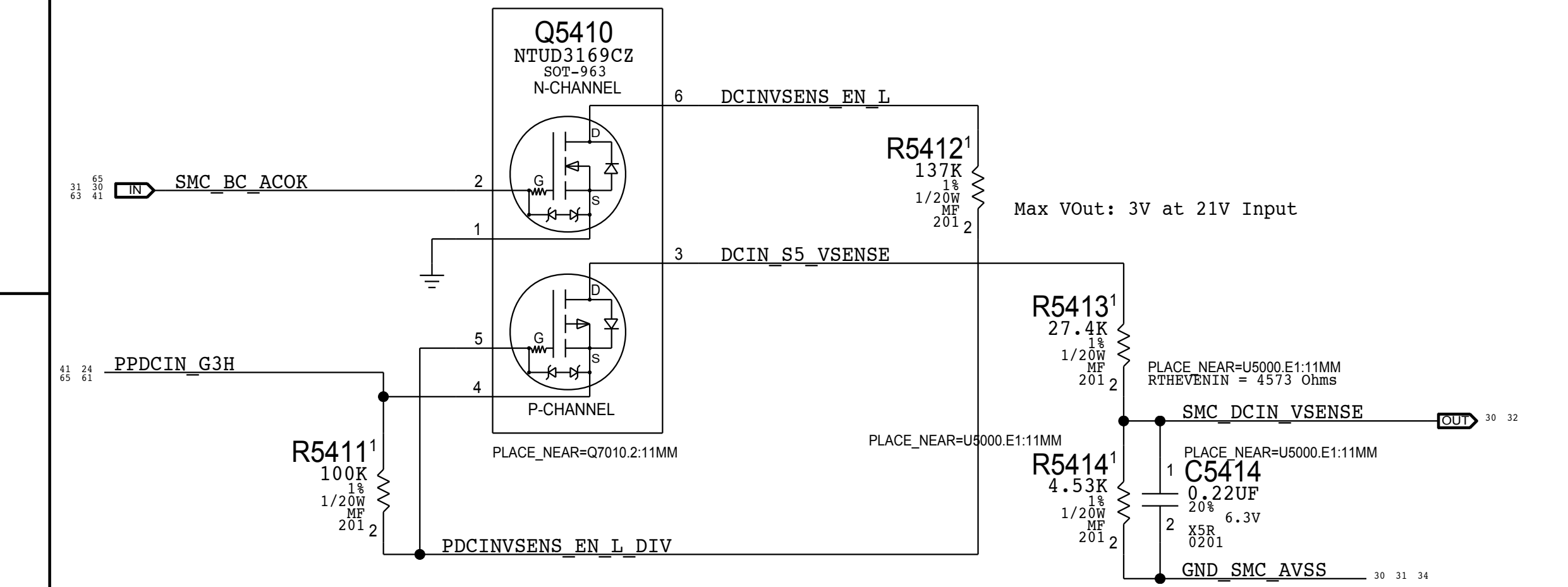


POR VOLTAGE / CURRENT SENSORS : TO BE USED IN PRODUCTION

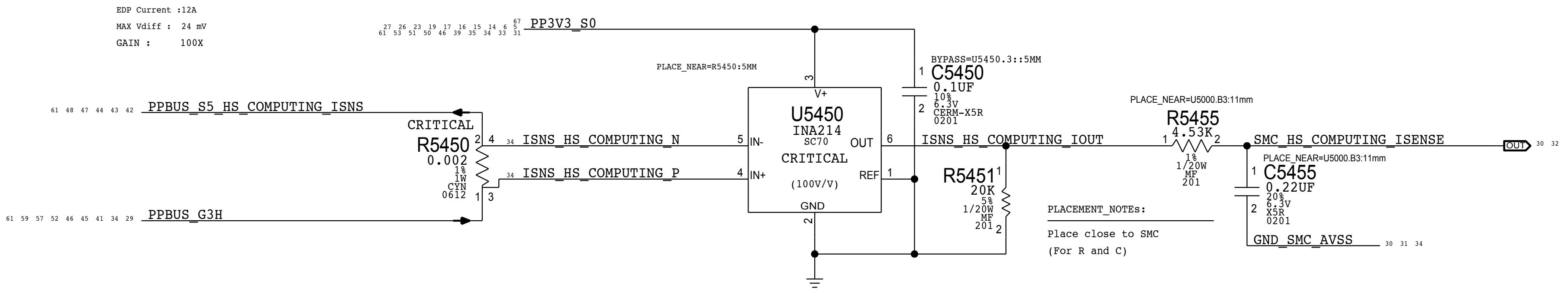
VP0R: PBUS Voltage Sense Enable & Filter



VD0R: DC-In Voltage Sense Enable & Filter

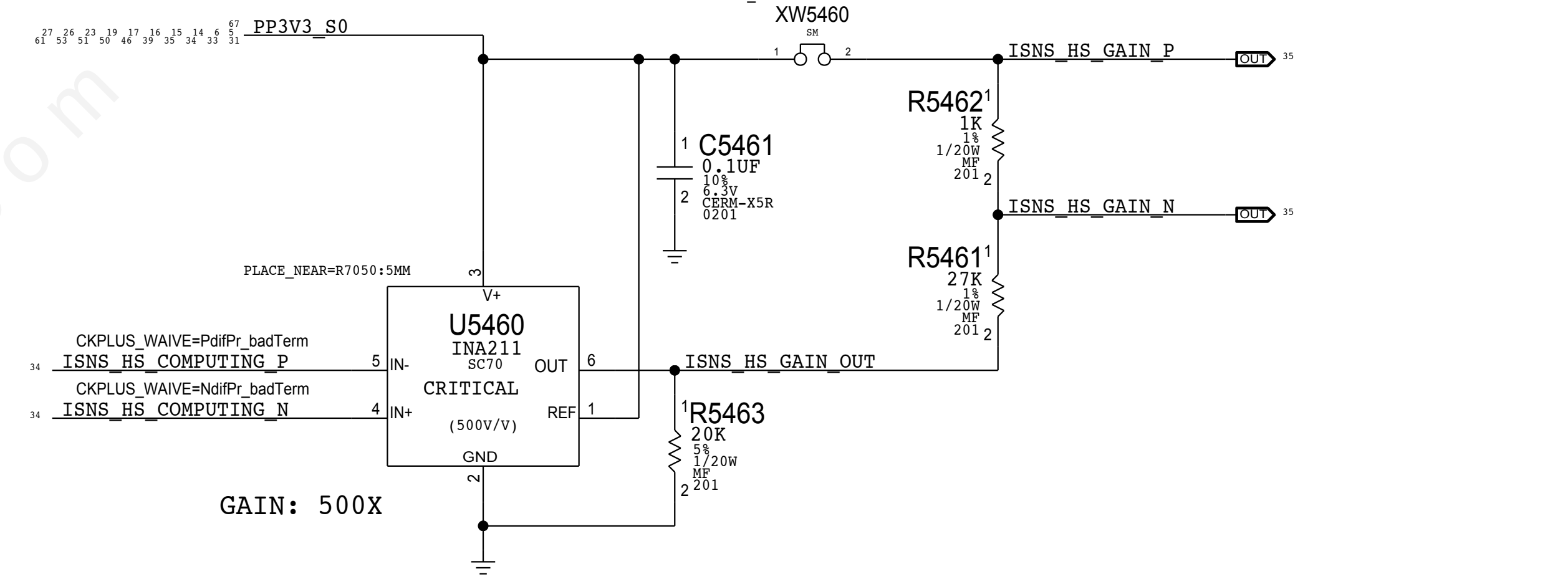


IC0R : COMPUTING High Side Current Sense

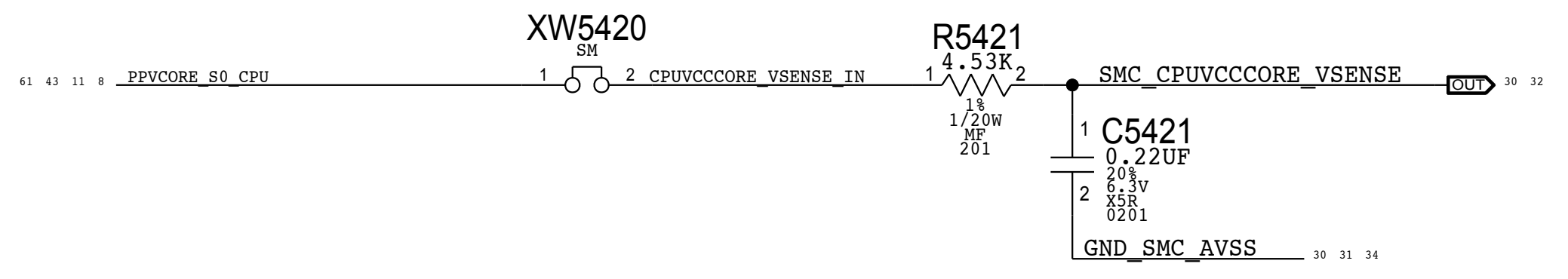


Need to set gains for ULX

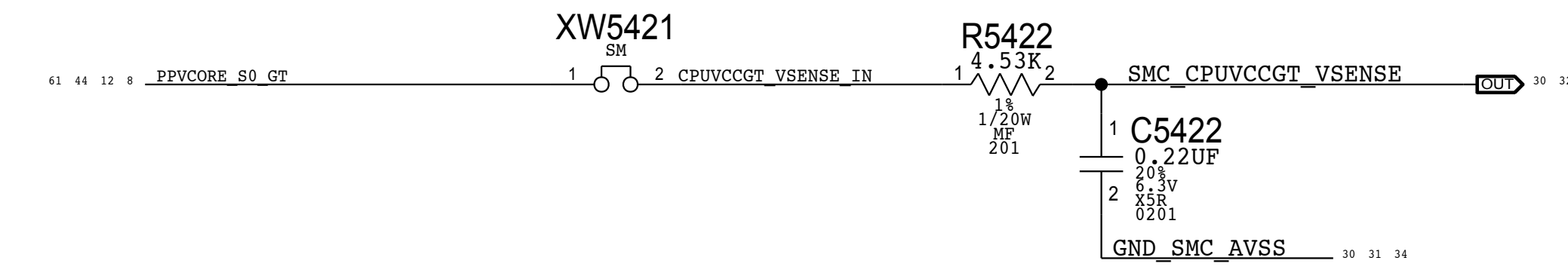
EMC1704 Computing High Side Gain Stage



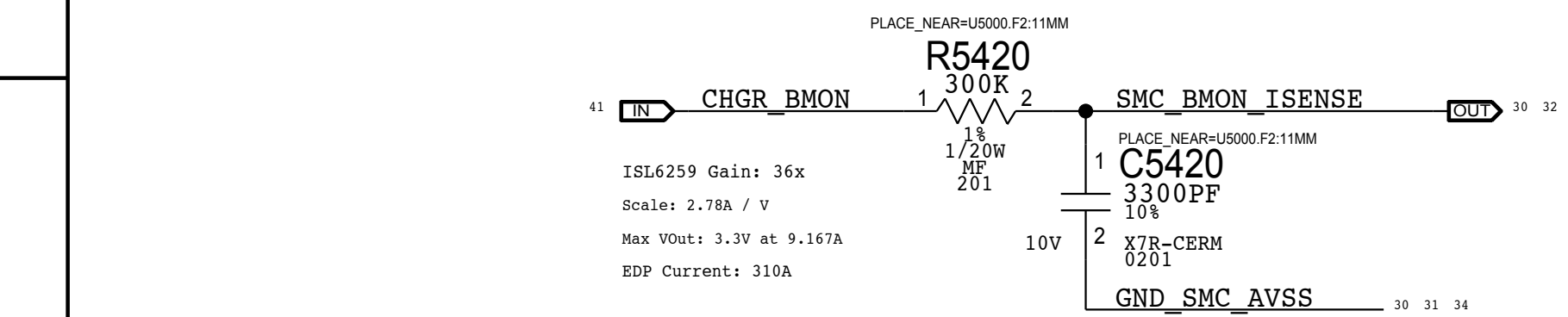
VC0C : CPU VCC CORE Voltage Sense



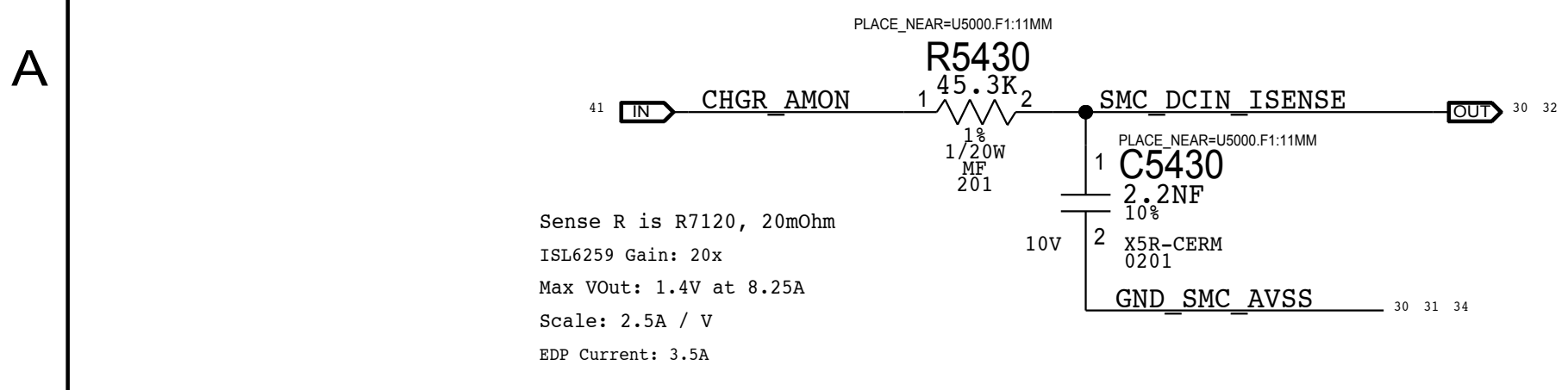
VC0G : CPU VCCGT Voltage Sense



CHARGER BMON High Side Current Sense

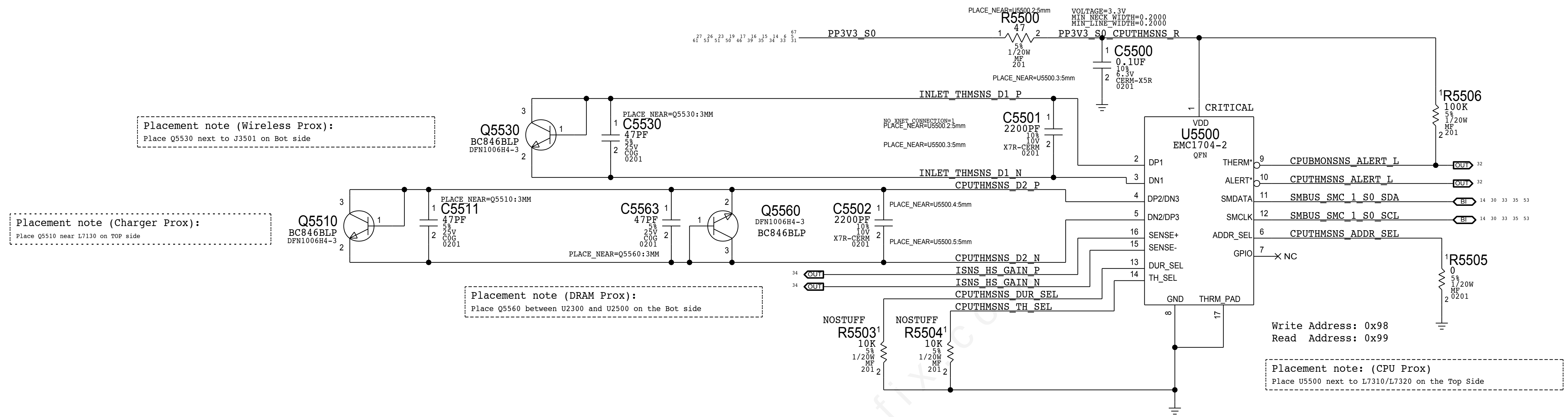


DC-IN (AMON) Current Sense

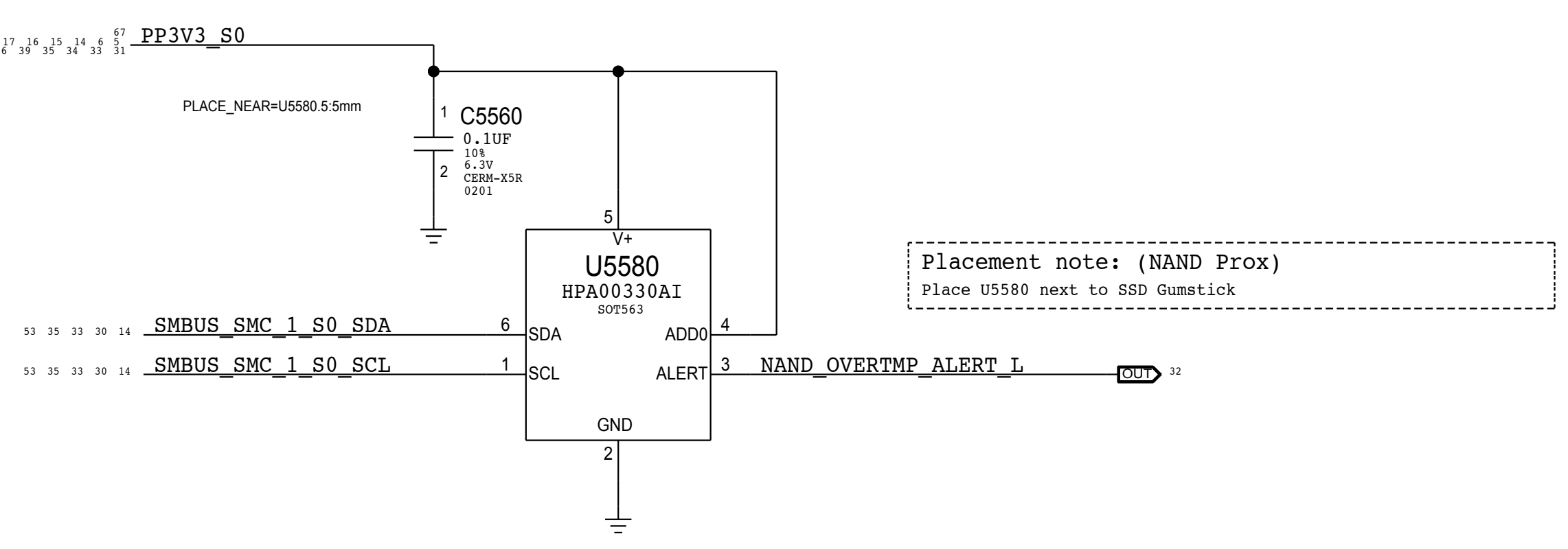


POR THERMAL SENSORS : TO BE USED IN PRODUCTION

CPU Proximity, Inlet ,DDR and BMON THR Sensor

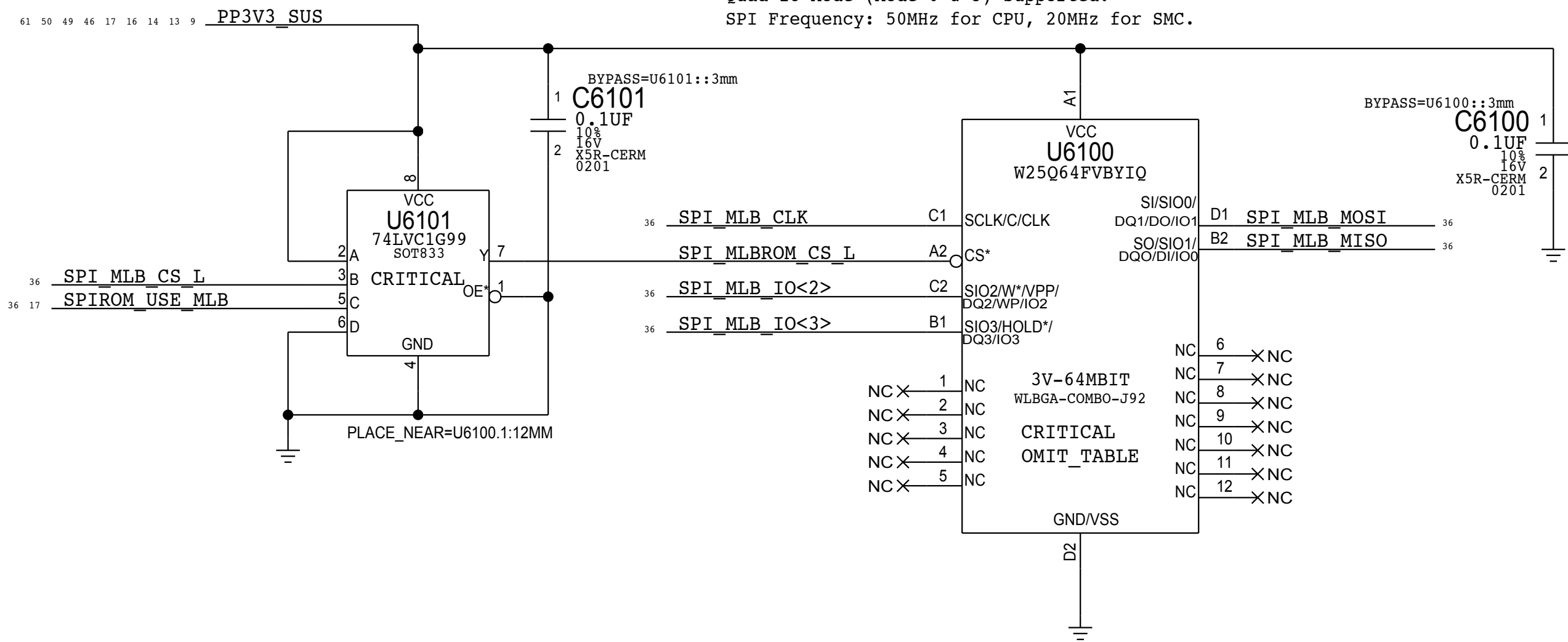


NAND Temp Sensor

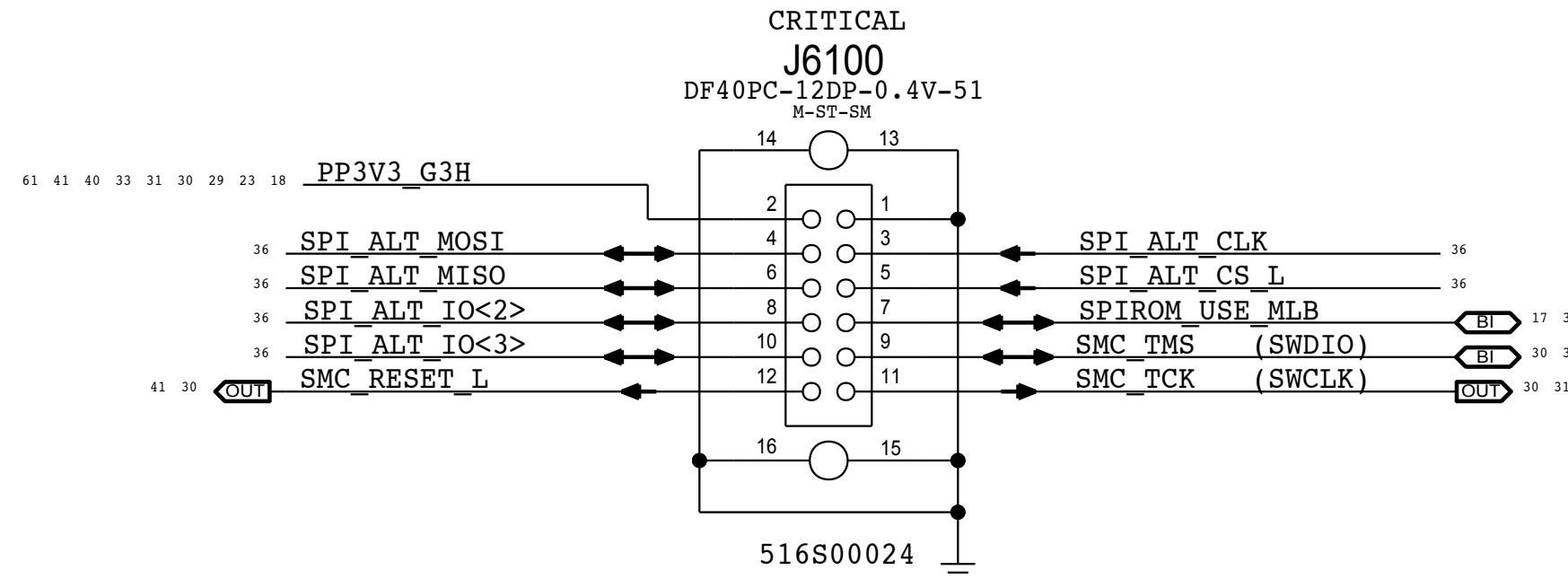


SPI ROM - Combo BGA Footprint (3 vendors)

Quad-IO Mode (Mode 0 & 3) supported.
SPI Frequency: 50MHz for CPU, 20MHz for SMC.

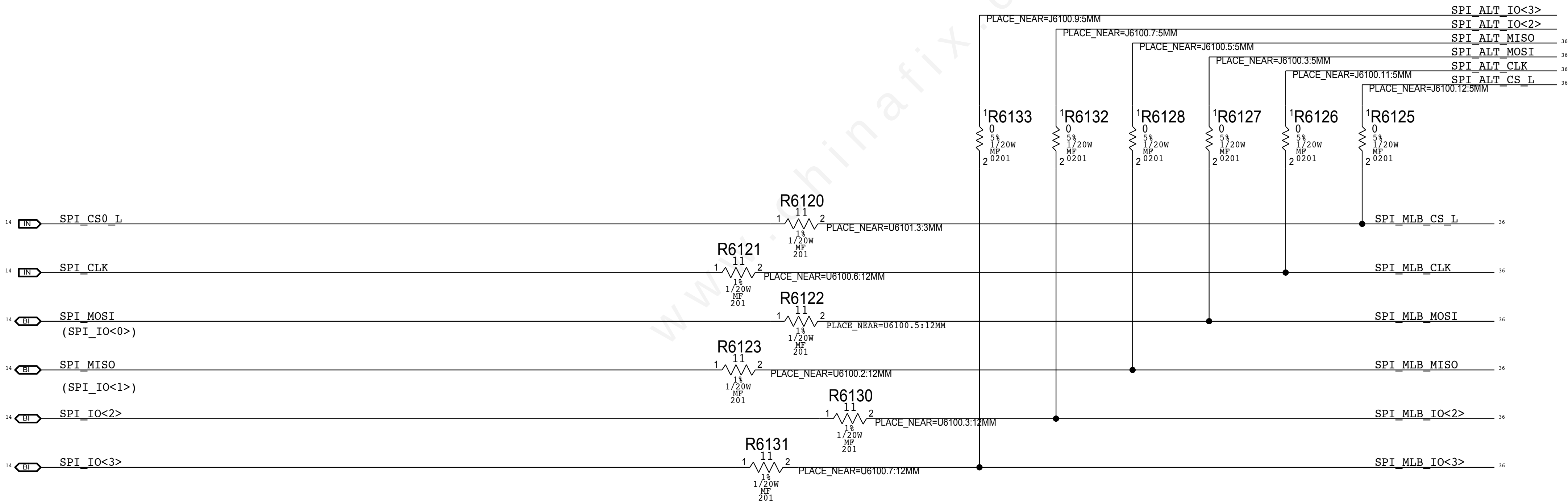


SPI+SWD SAM Connector



BootROM SPI Bus Series Termination

CPU Master

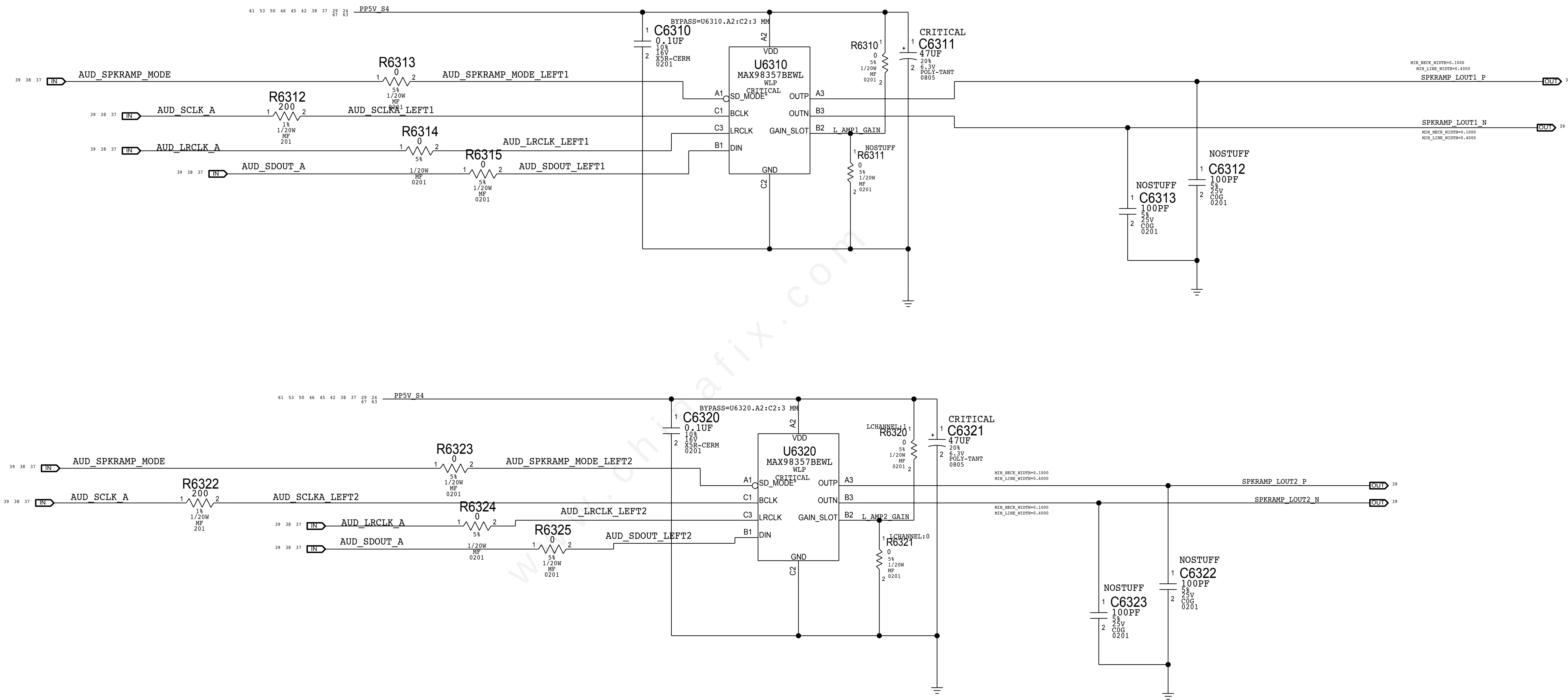


Sam Card ROM Slave

SPI ROM Slave

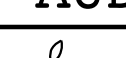
Left Speaker Amps

APPLE P/N 353S4265



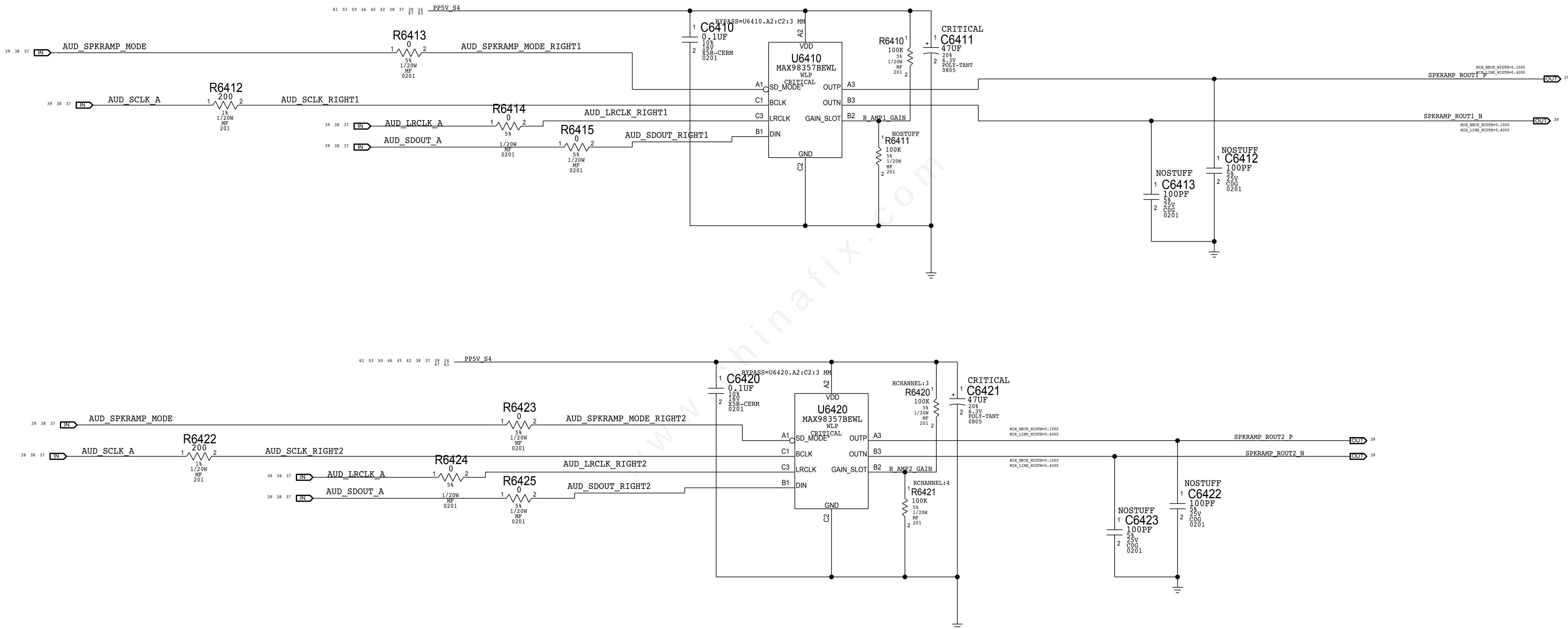
SPEAKER CONFIGS

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EQ:2CH	LCHANNEL:1,RCHANNEL:3
EQ:4CH	LCHANNEL:0,RCHANNEL:4

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		BRANCH	
		<BRANCH>	
		PAGE	63 OF 130
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Right Speaker Amps

APPLE P/N 353S4265



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PAGE TITLE			
AUDIO Right Speaker Amps			
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		PAGE	
		64 OF 130	
		SHEET	
		38 OF 67	

CODEC OUTPUT SIGNAL PATHS

FUNCTION	VOLUME	CONVERTER	PIN COMPLEX	MUTE CONTROL	DET ASSIGNMENT
HP/LINE OUT	0X02 (2)	0X02 (2)	0X09 (9,A)		0X09 (A)
SPEAKERS	0X04 (4)	0X04 (4)	0X0B (11)	GPIO_3	N/A

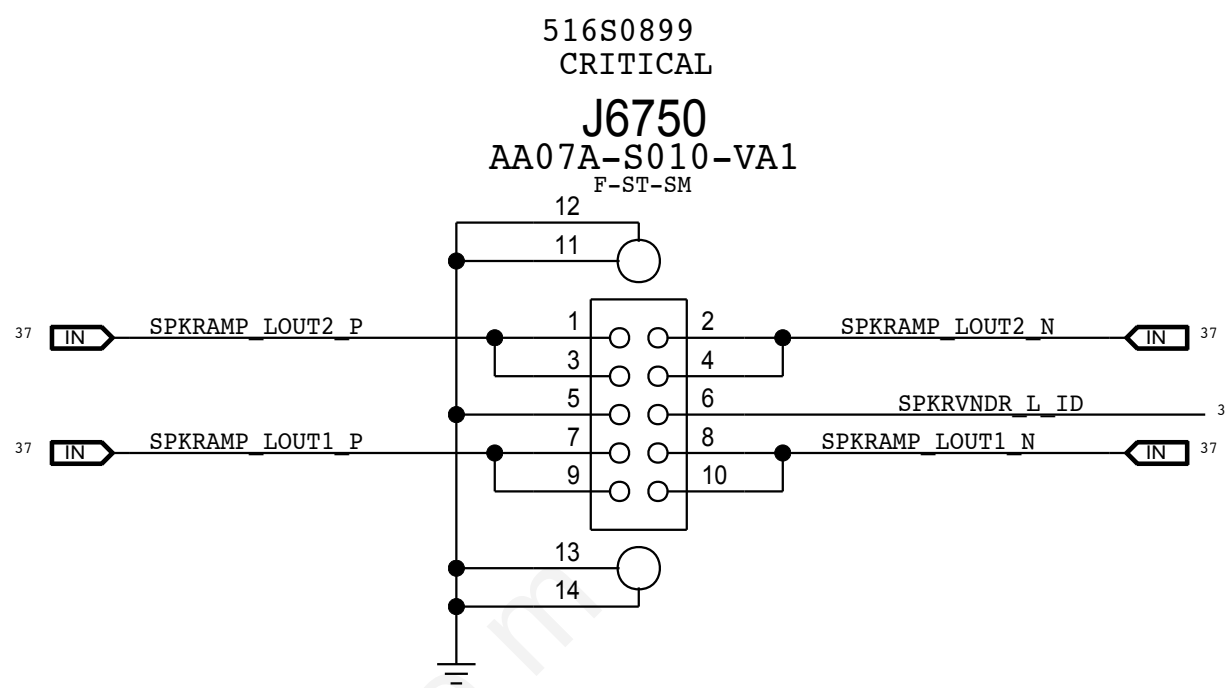
CODEC INPUT SIGNAL PATHS

FUNCTION	CONVERTER	PIN COMPLEX	VREF	DET ASSIGNMENT
BUILT-IN MIC	0X06 (6)	0X0D (13,B,RIGHT)	MIC_BIAS (80%)	N/A
HEADSET MIC	0X06 (6)	0X0D (13,V22,B,LEFT)	MIKEY	MIKEY

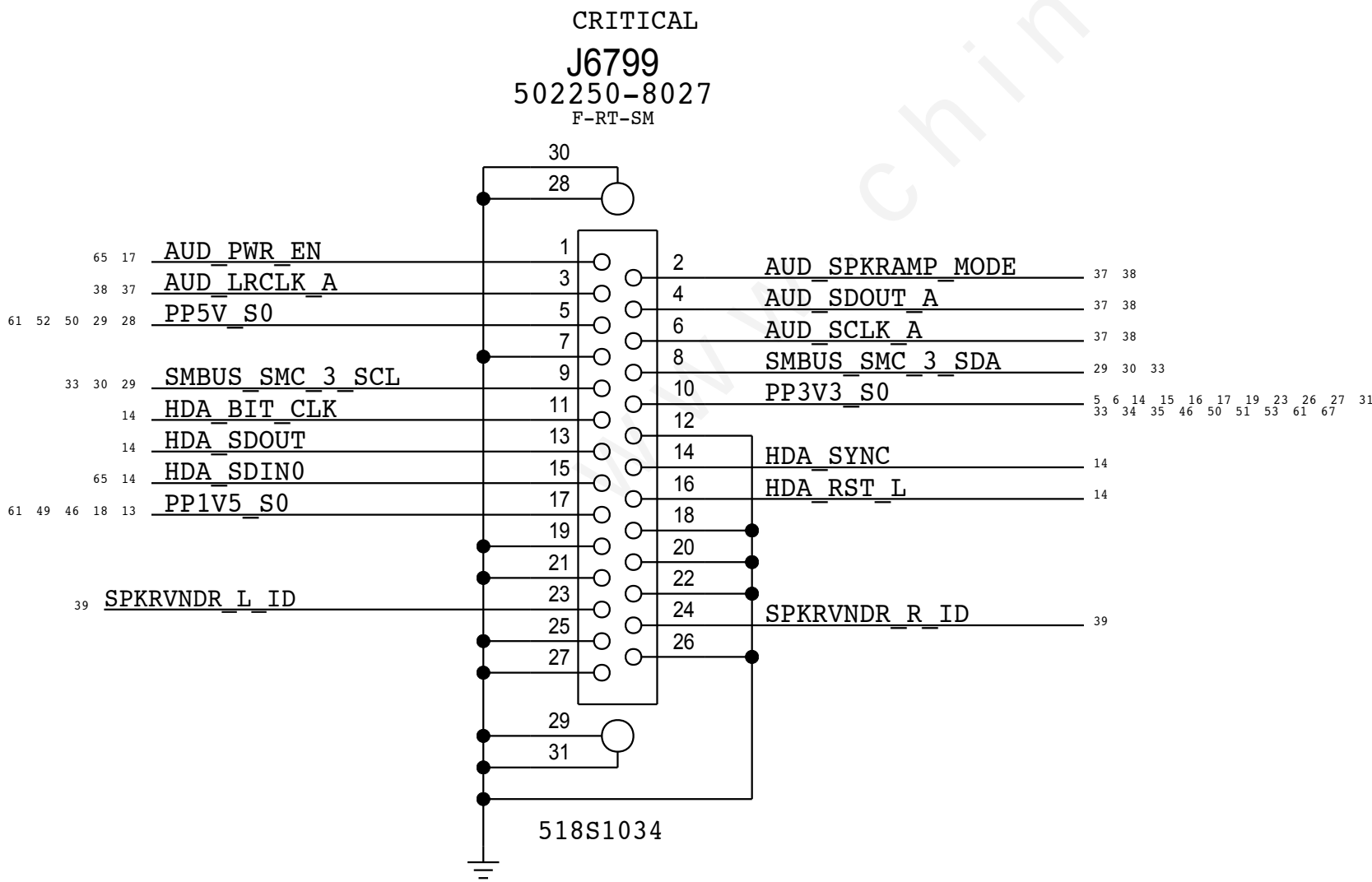
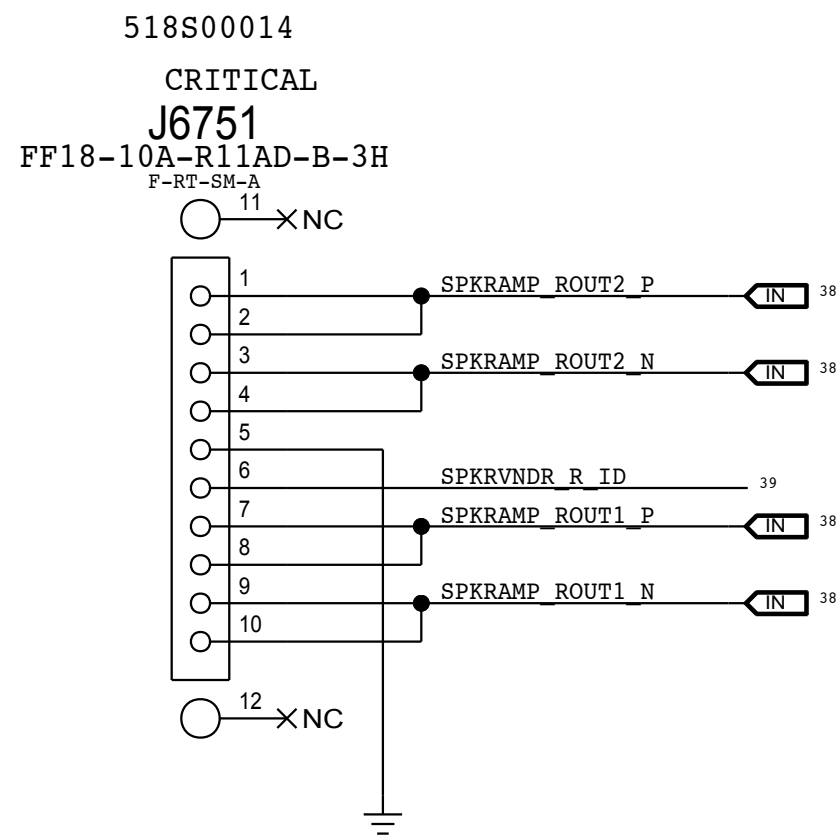
SOUTHBIDGE RESOURCE/PIN ALLOCATIONS

FUNCTION	NET NAME	SB GPIO/INT
PERIPHERAL/EXTRACTION DETECT	AUD_IP_PERIPHERAL_DETECT	GPIO 3
MIKEY INTERRUPT	AUD_I2C_INT_L	GPIO 5
MIKEY ENABLE	AUD_IPHS_SWITCH_EN	GPIO 16
MIKEY I2C BUS	I2C_MIKEY_SDA/SCL	PCH SMBUS 0

LEFT SPEAKER CONN

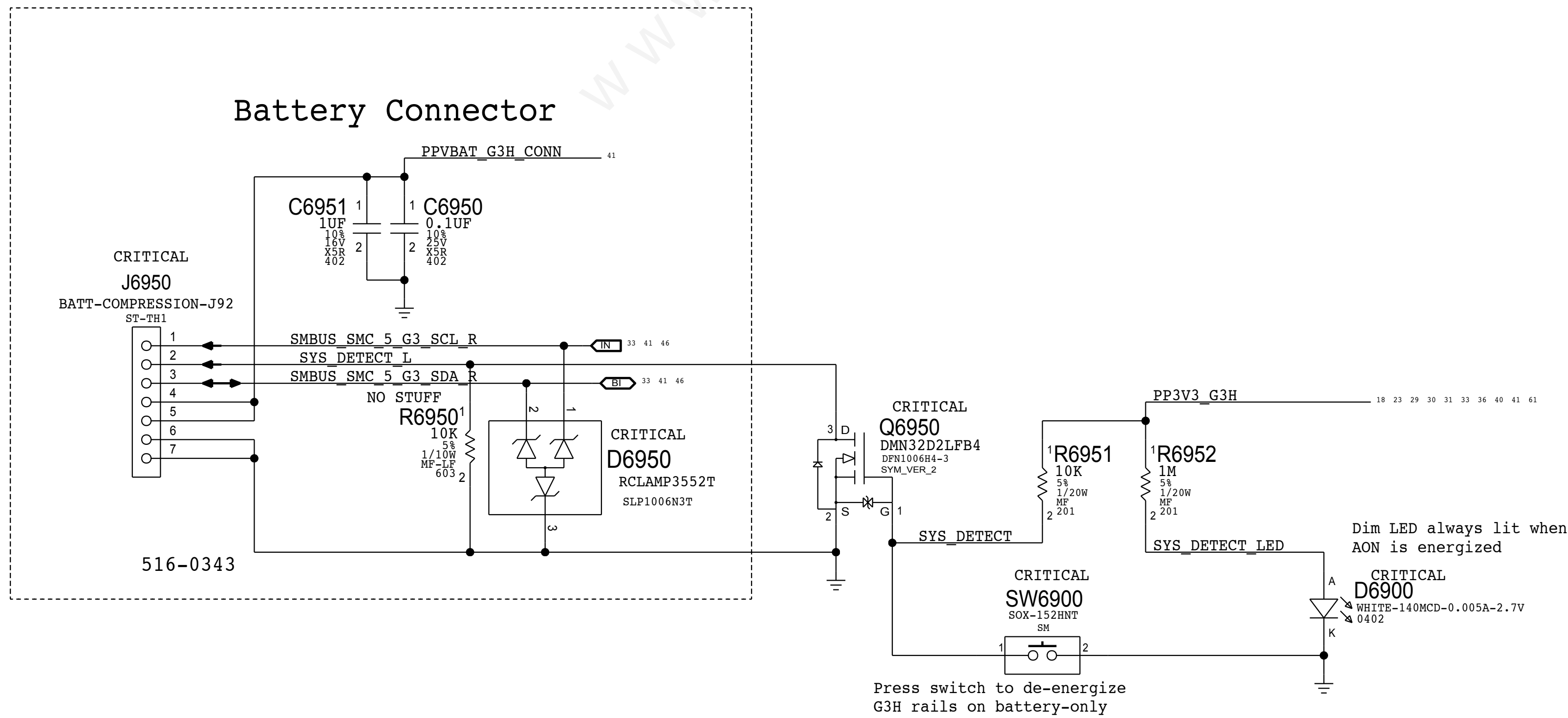
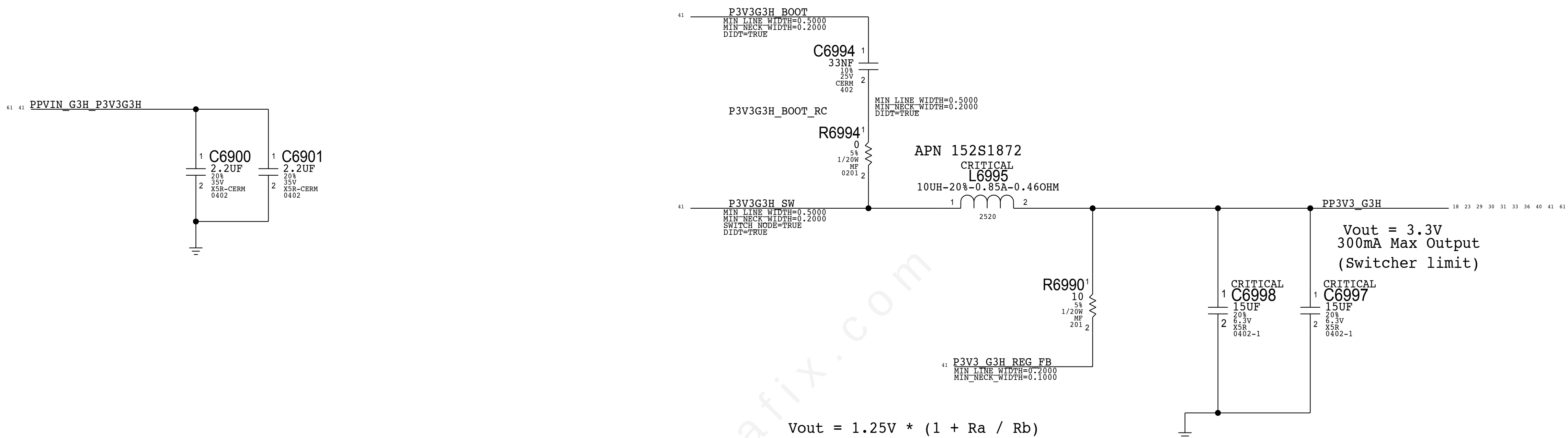


RIGHT SPEAKER CONN



RIO FLEX CONN

3.3V G3H VR - Bansuri



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Battery Connector & 3.3V G3Hot			
 Apple Inc.	DRAWING NUMBER <SCH_NUM>		SIZE D
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		SHEET	40 OF 67

Charger PMIC - Bansuri

D

C

B

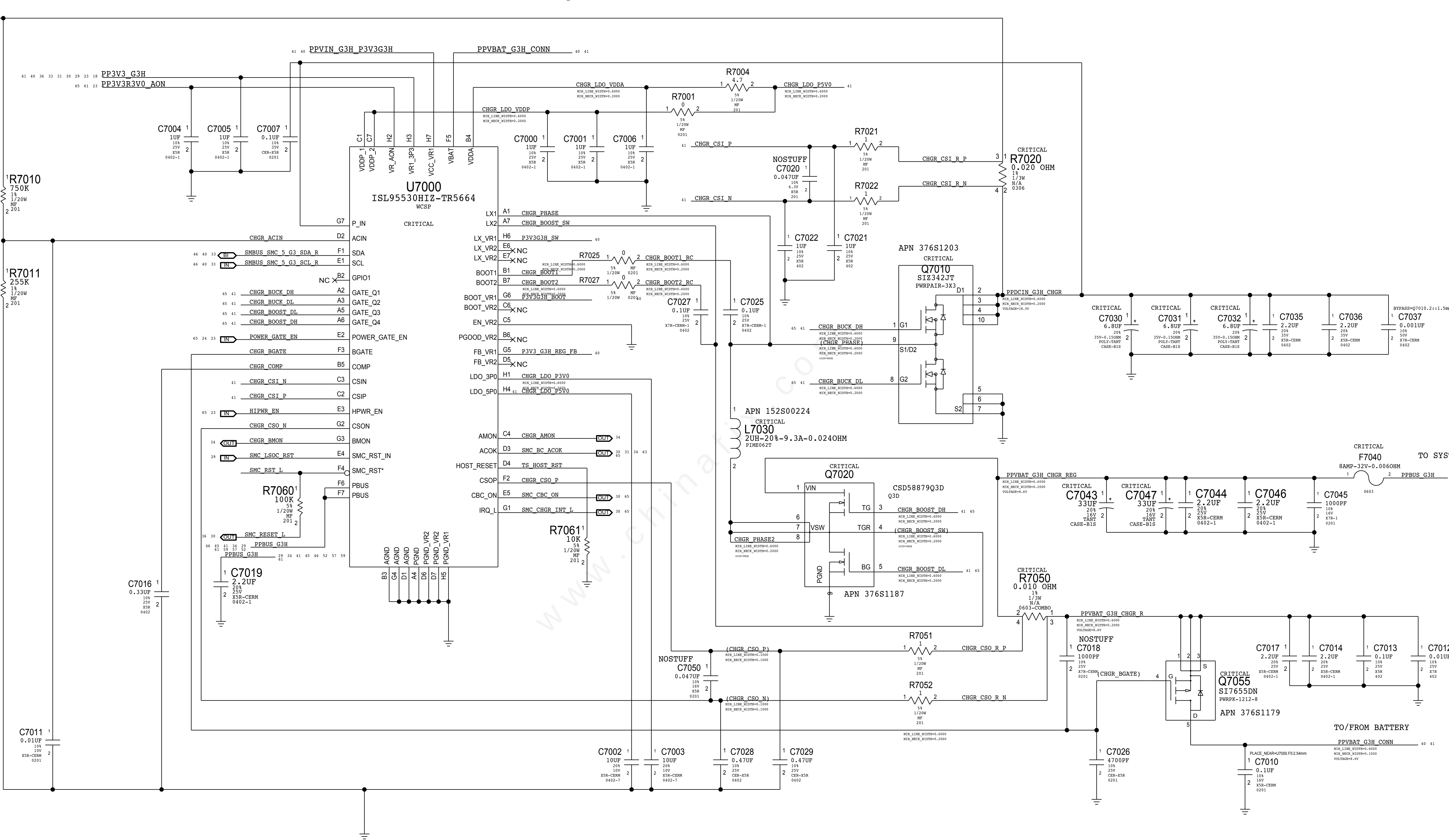
A

D

C

B

A



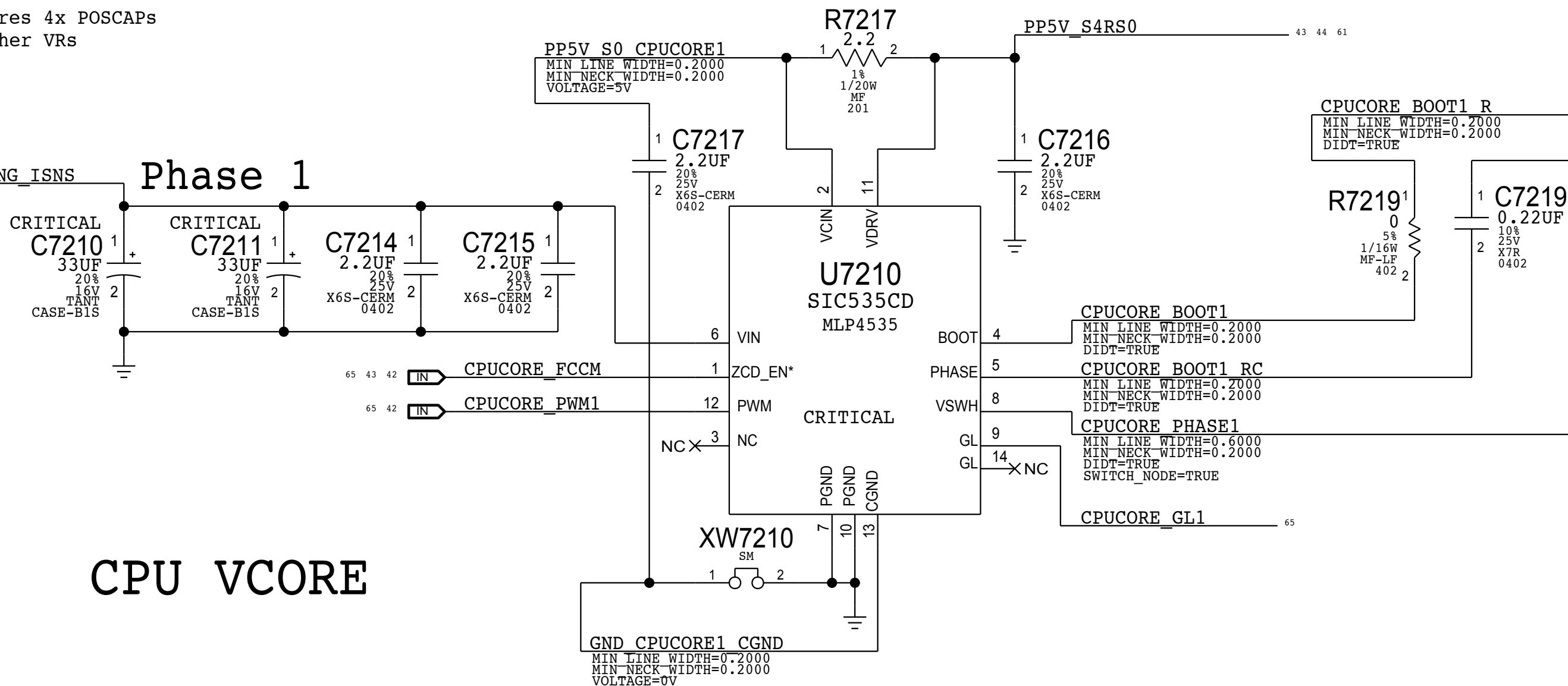
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PBus Supply & Battery Charger			
 Apple Inc.		DRAWING NUMBER	SIZE
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		<E4LABEL>	
		BRANCH	
		<BRANCH>	
		PAGE	70 OF 130
		SHEET	41 OF 67



Note:Vcore design requires 4x POSCAPs
3x shared from other VRs

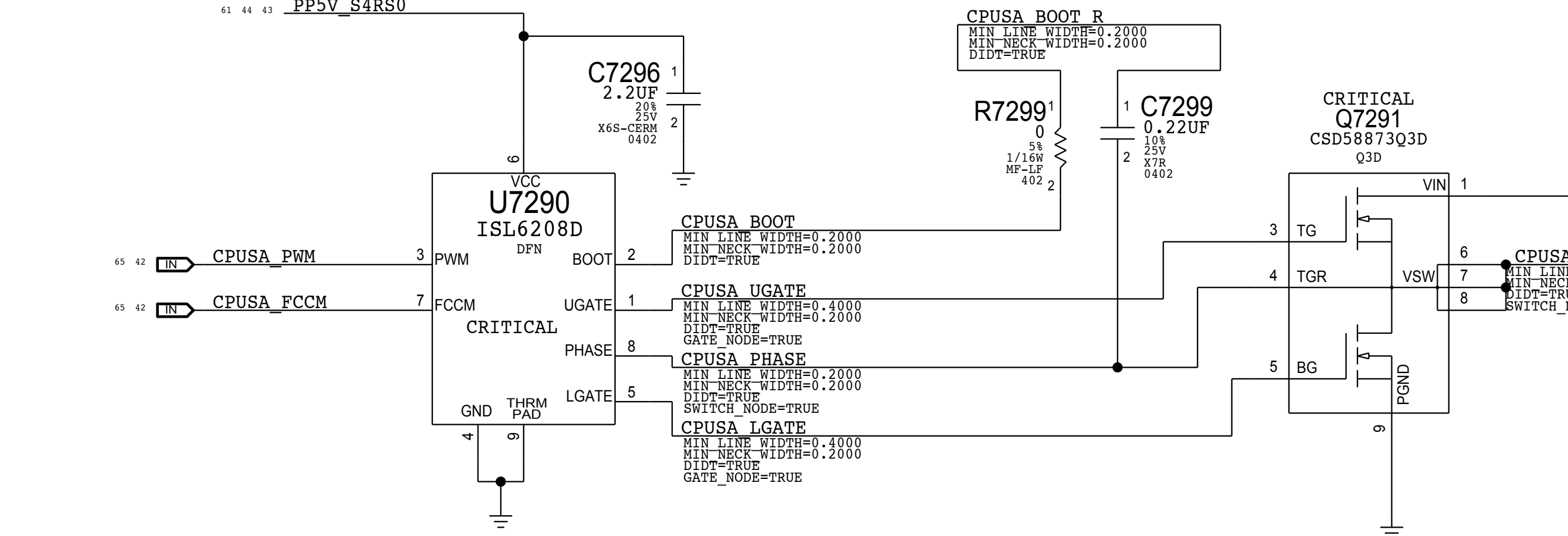
CPU VCORE

Phase 1



CPU VCCSA

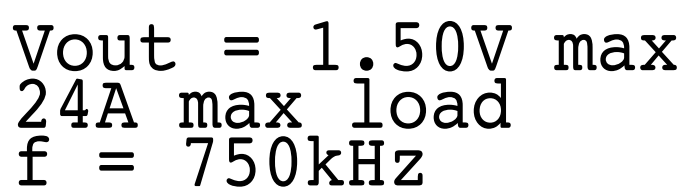
Phase 2



Vout = 1.50V max
24A max load
f = 750kHz

Vout = 1.05V max
4.1A max load
f = 750kHz

SYNC_MASTER=DEVMLB		SYNC_DATE=05/19/2015	
PAGE TITLE		CPU VCore/VccSA Power Stage	
 Apple Inc.	DRAWING NUMBER	<SCH_NUM>	
	REVISION	<E4LABEL>	
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		PAGE	72 OF 130
		SHEET	43 OF 67



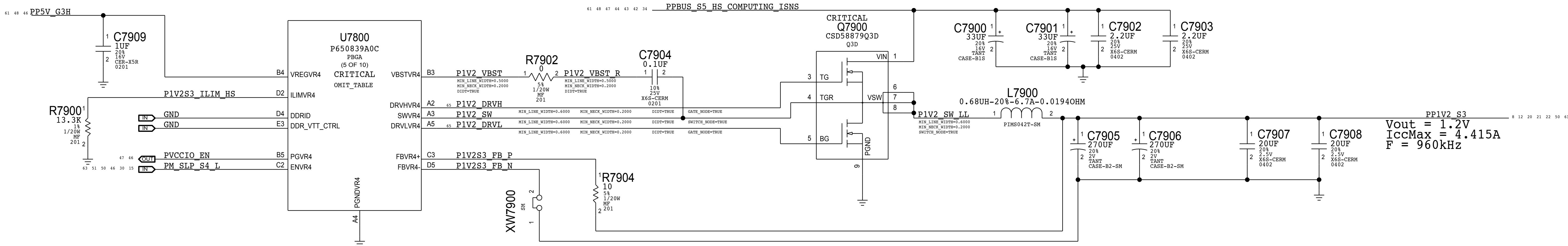


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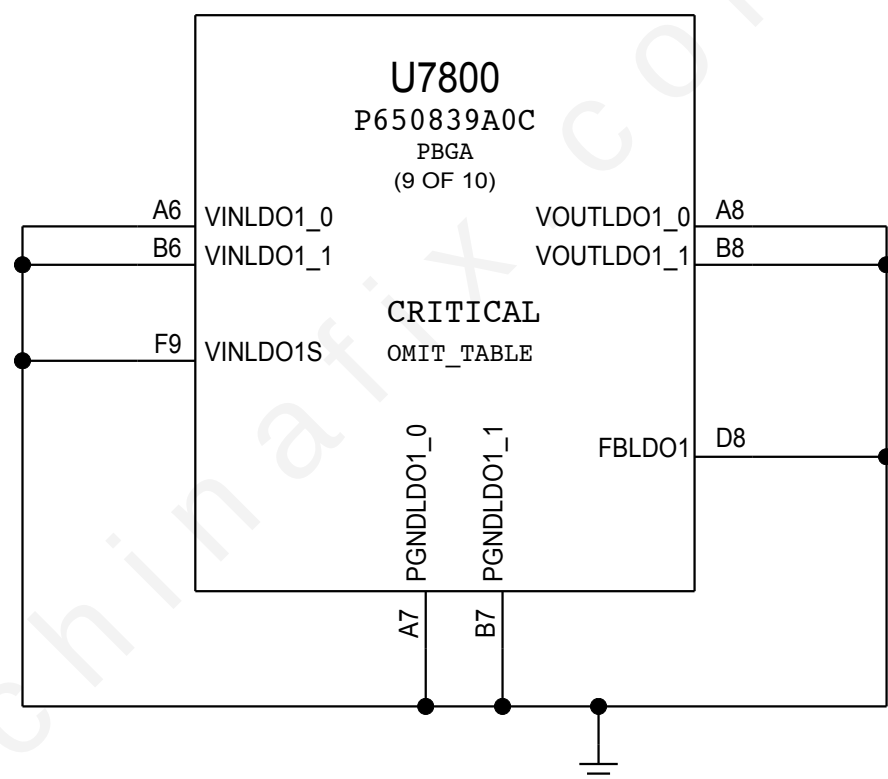


	A
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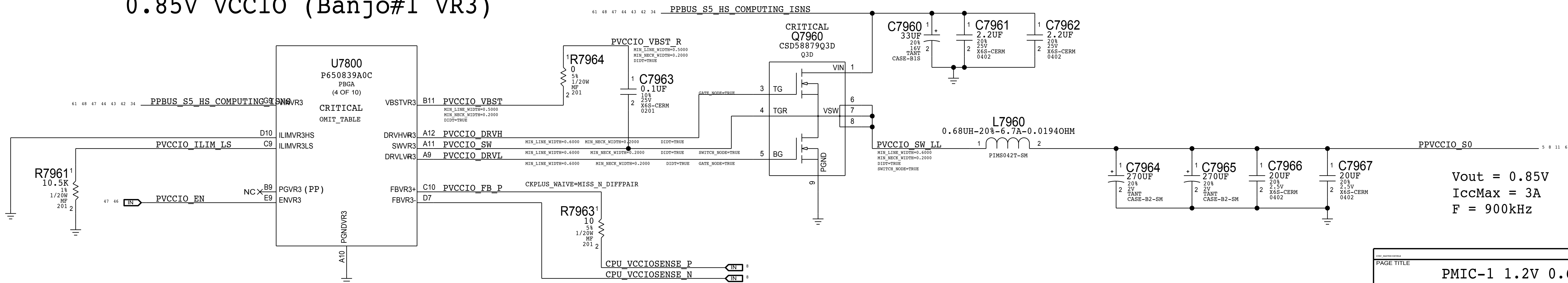
1.2V VDDQ (Banjo#1 VR4)



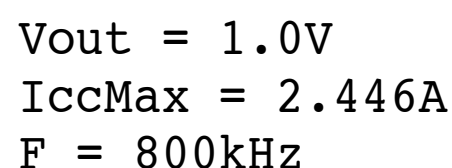
0.6V VTT (Banjo#1 LDO1)



0.85V VCCIO (Banjo#1 VR3)



PAGE TITLE			
PMIC-1 1.2V 0.6V VCCIO			
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		PAGE	79 OF 130
		SHEET	47 OF 67
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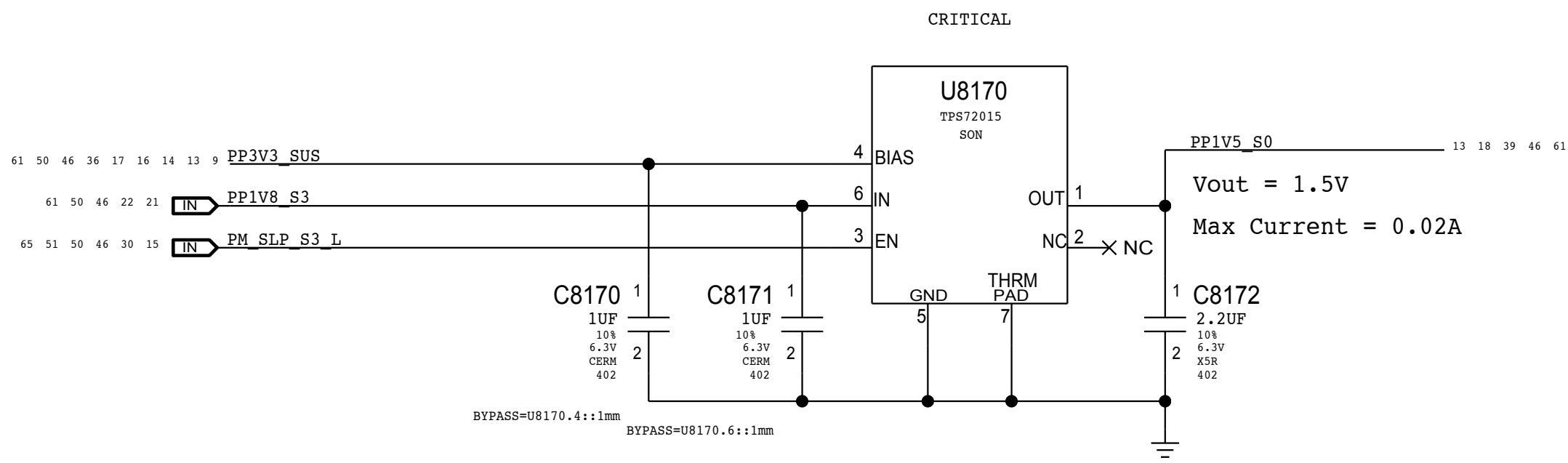
Vout = 1.8V
IccMax = 0.314A
F = 2MHz

Vout = Based on VID
IccMax = 1.1A
F = 800kHz

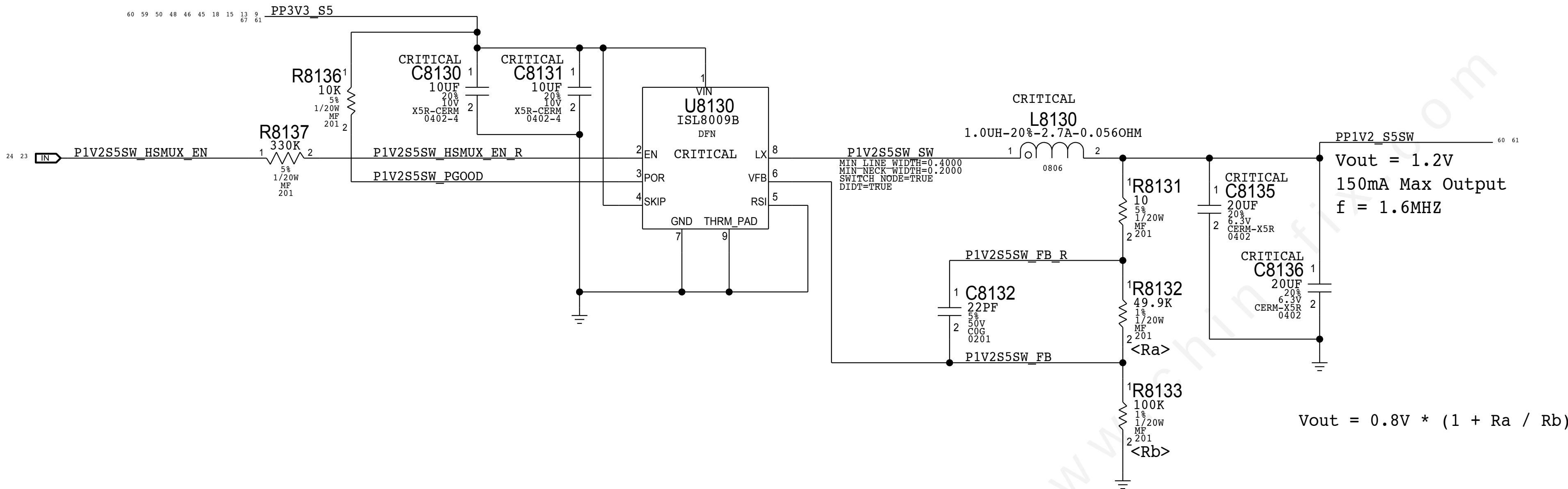
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		SHEET		48 OF 67	



1.5V S0 LDO



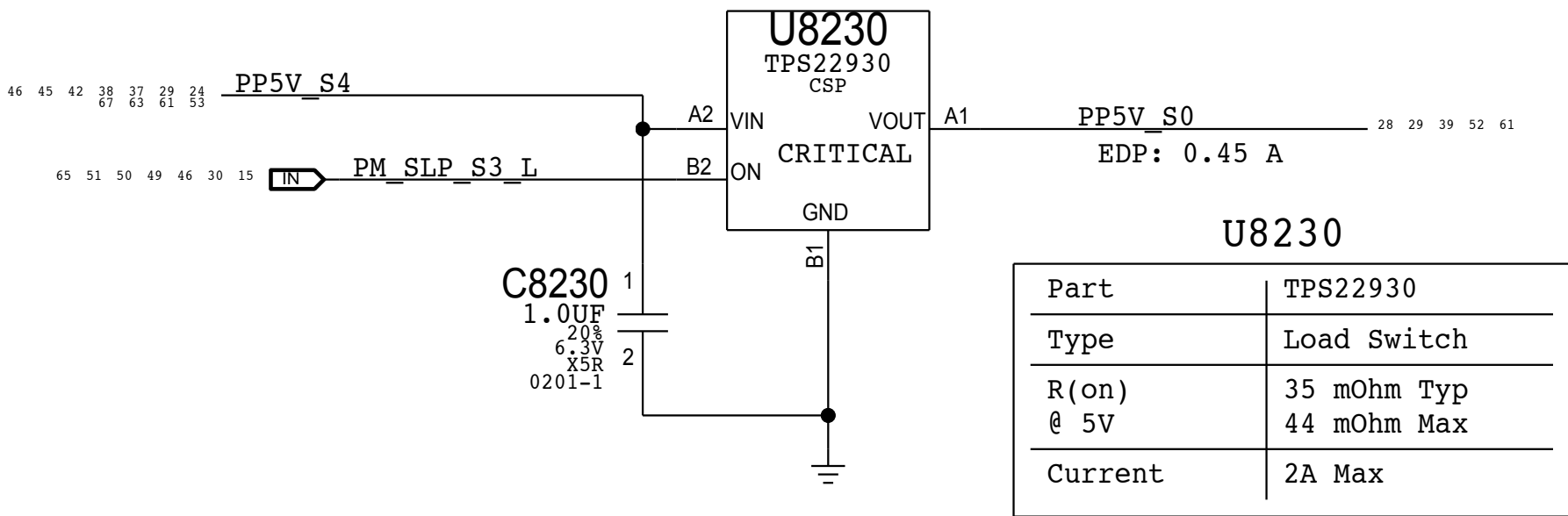
1.2V S5 USB-C HS Mux



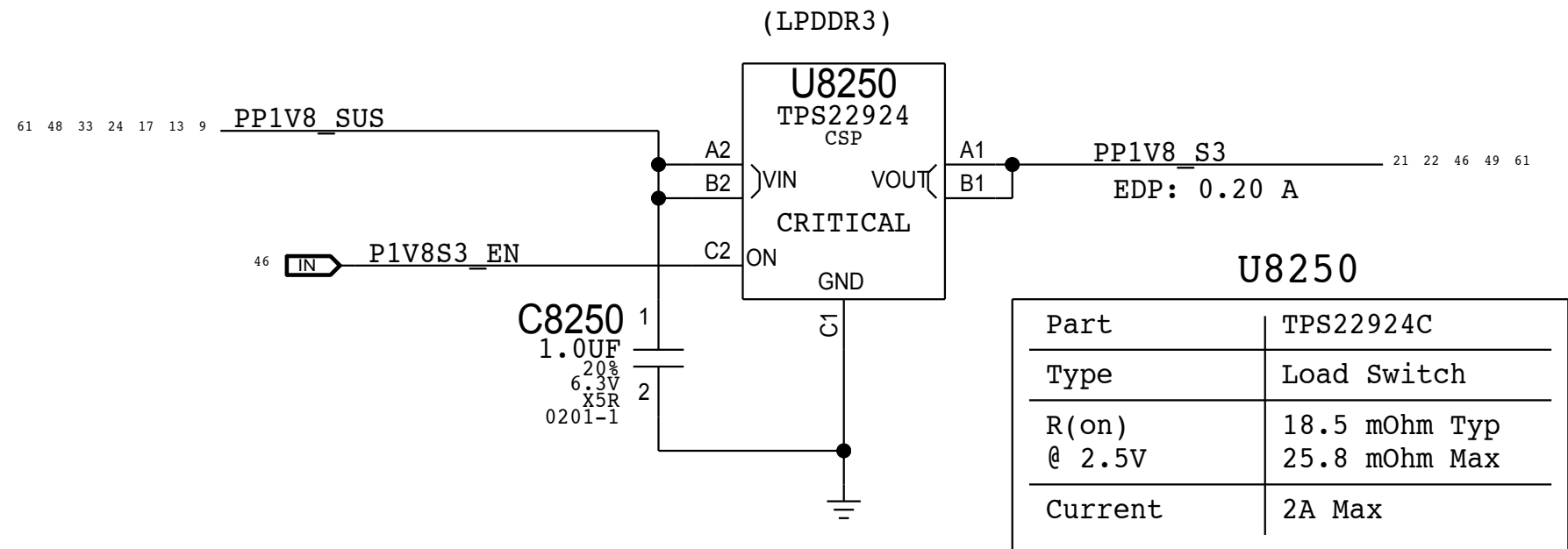
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Misc Power Supplies			
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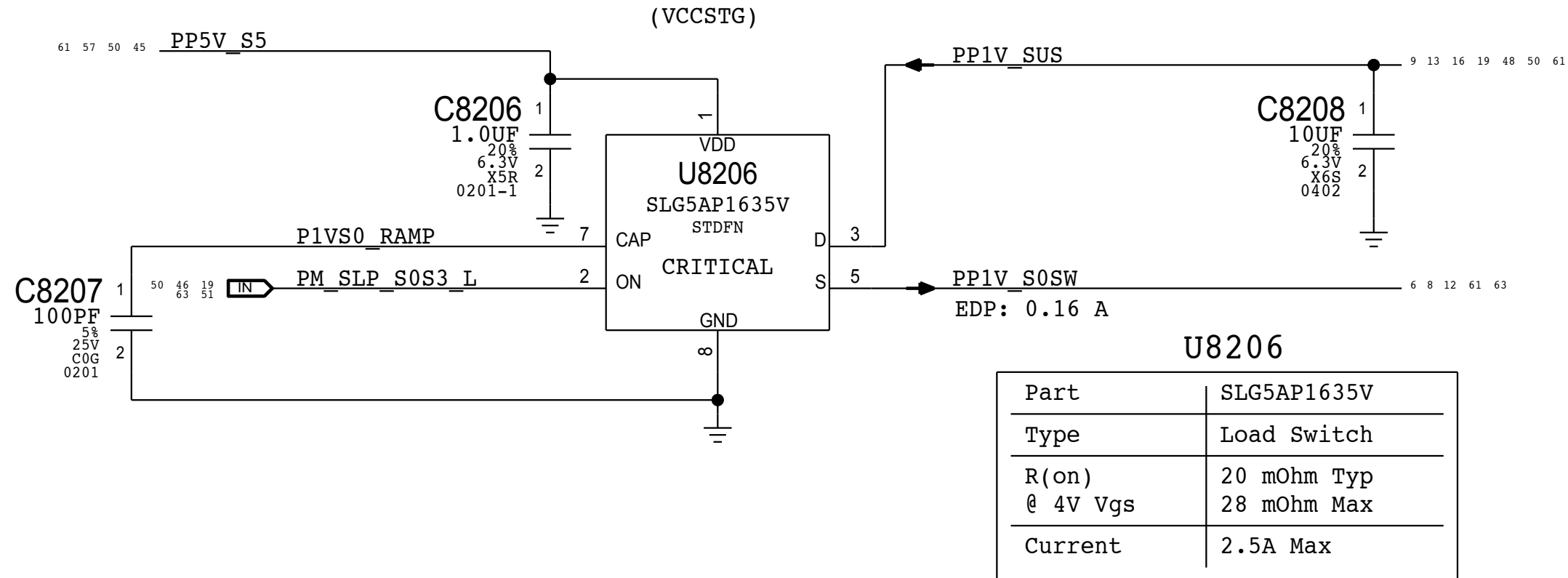
5V S0 Switch



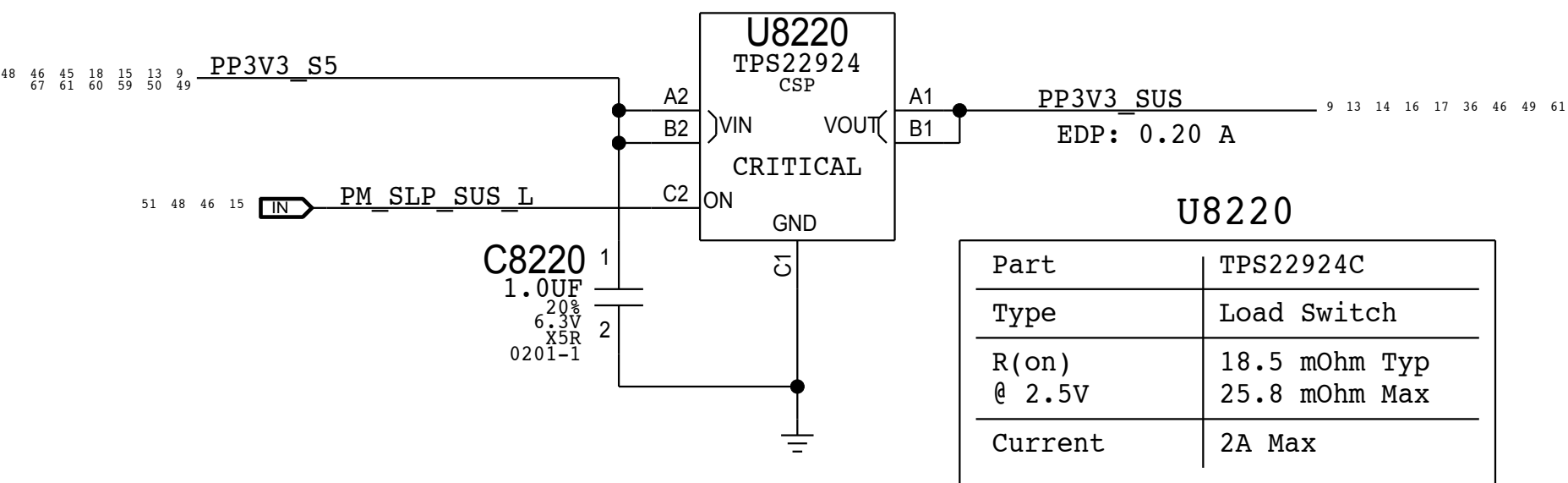
1.8V S3 Switch



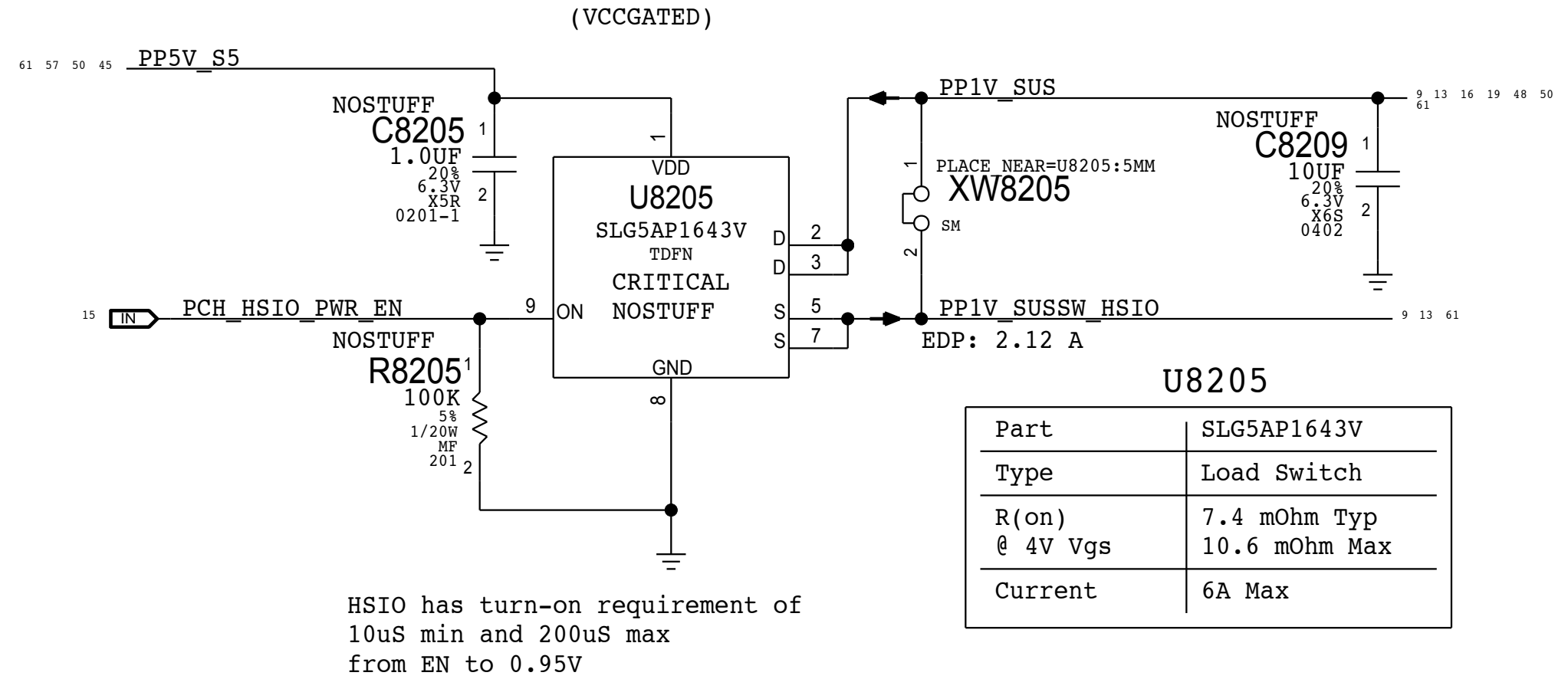
1.0V S0SW Switch



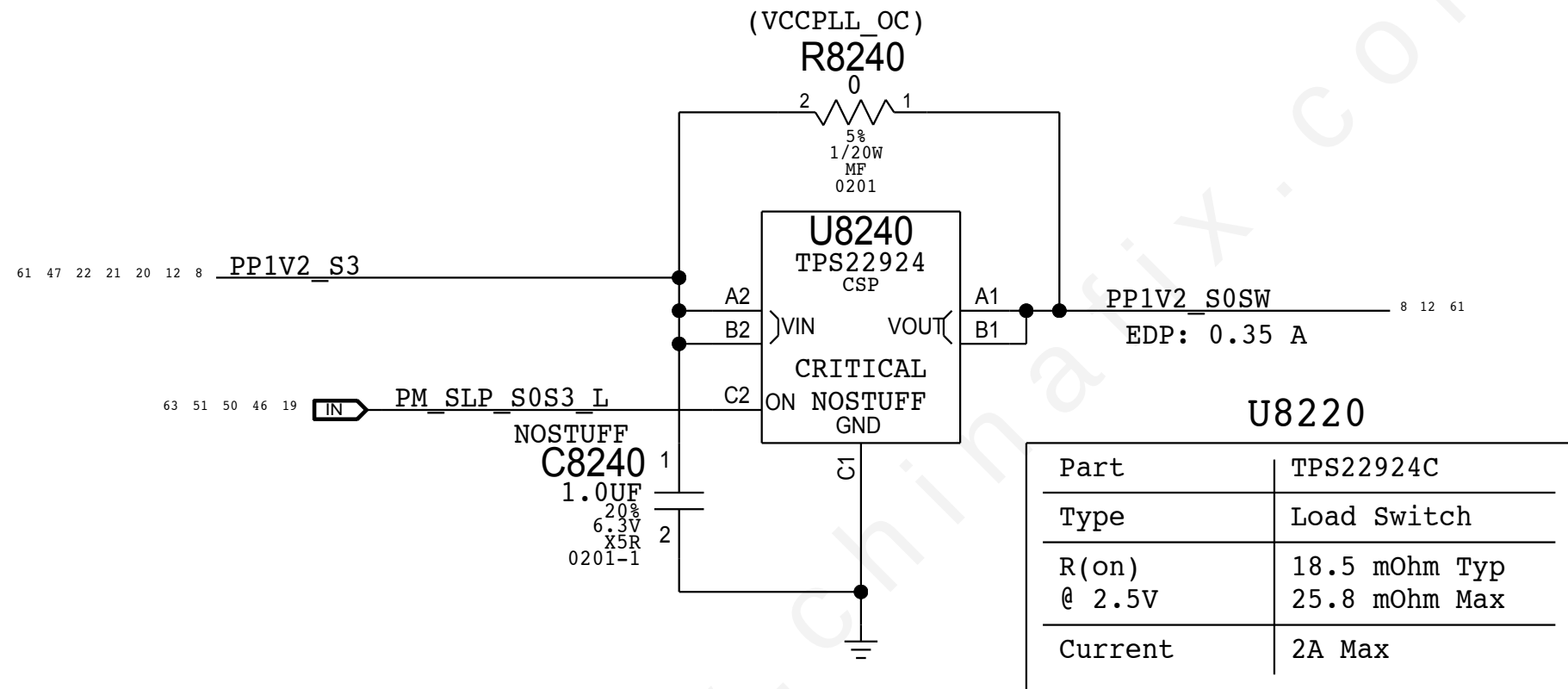
3.3V SUS Switch



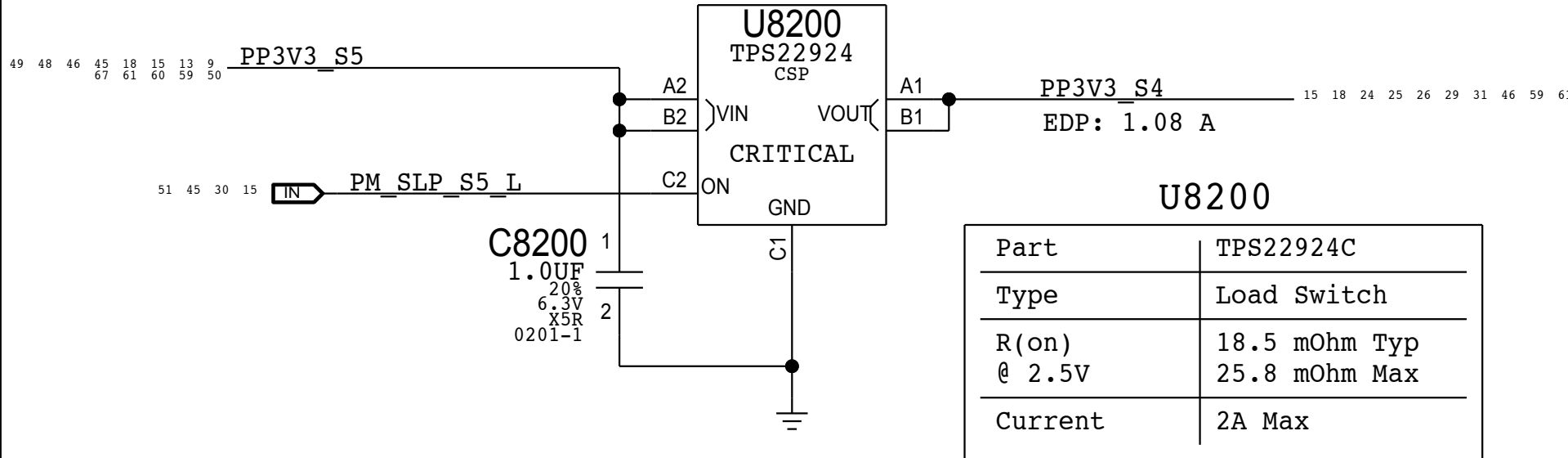
1.0V SUSSW HSIO Switch



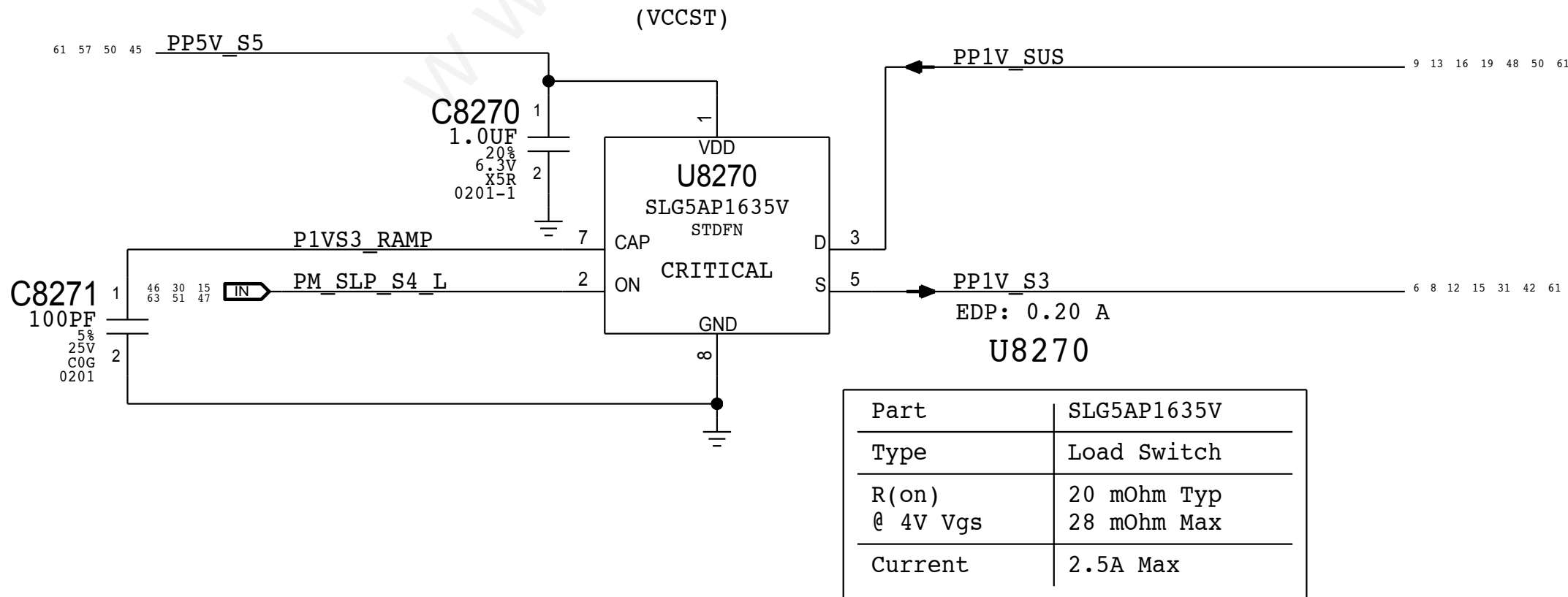
1.2V S0SW Switch



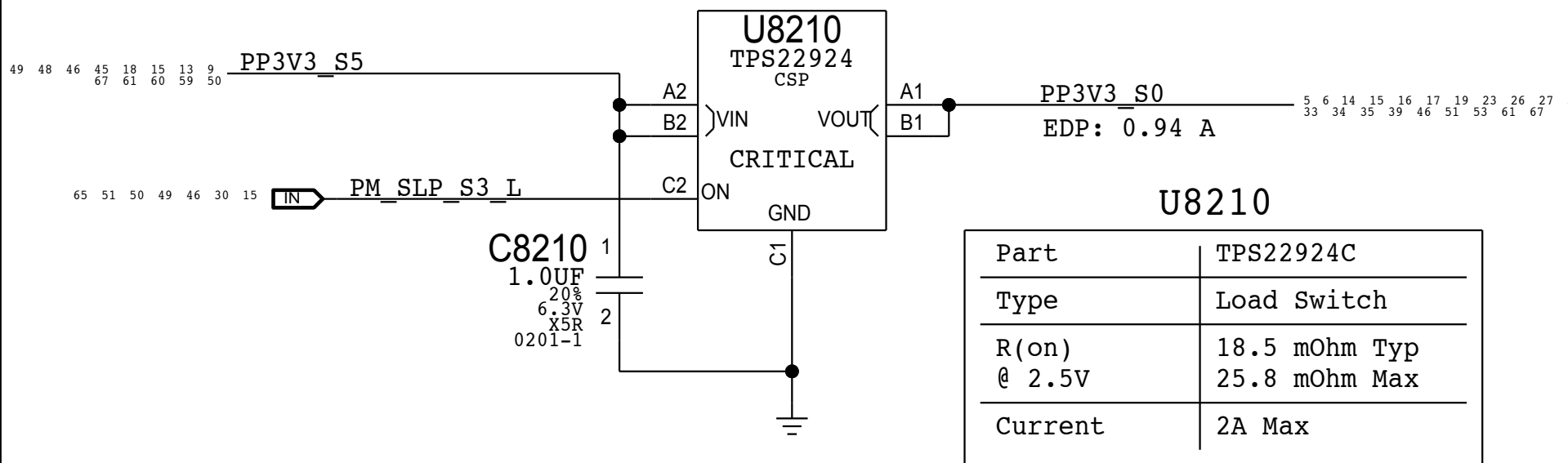
3.3V S4 Switch



1.0V S3 Switch

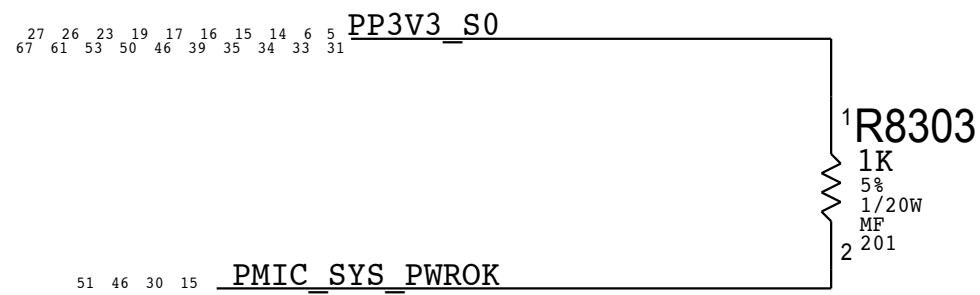
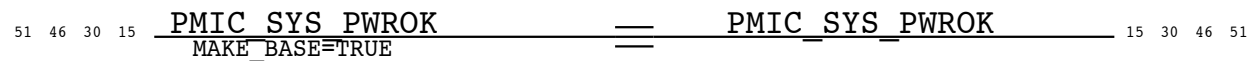
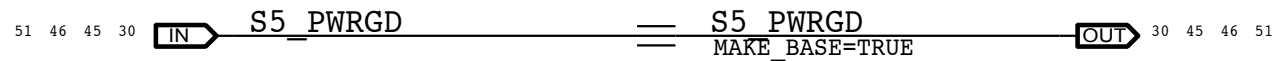


3.3V S0 Switch

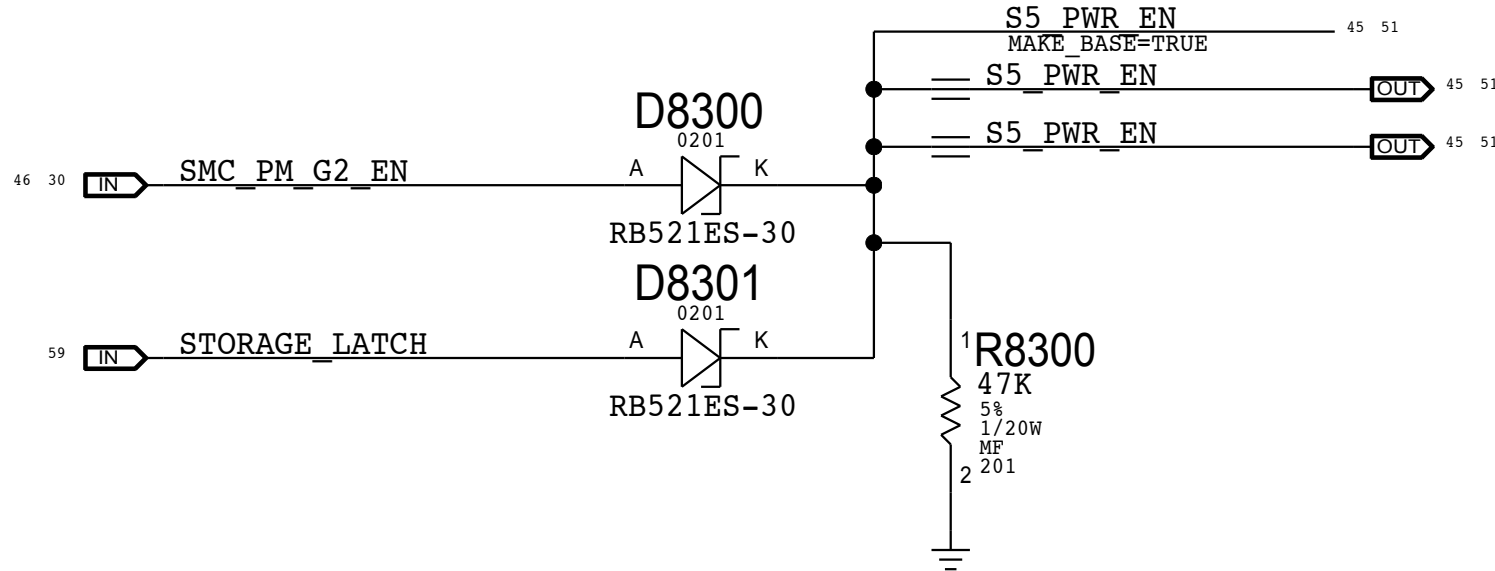


Mobile System Power State Table						
State	SMC_ADAPTER_EN	SMC_PM_G2_ENABLE	PM_SUS_EN	PM_SLP_S5_L	PM_SLP_S4_L	PM_SLP_S3_L
Run (S0)	X	1	1	1	1	1
Sleep (S3AC)	1	1	1	1	1	0
Sleep (S3)	0	1	1	1	1	0
Deep Sleep (S4)	0	1	0	1	0	0
Deep Sleep (S5)	0	1	0	0	0	0
Battery Off (G3BotAC)	toggle 3Hz	0	0	0	0	0
Battery Off (G3Rot)	1	0	0	0	0	0

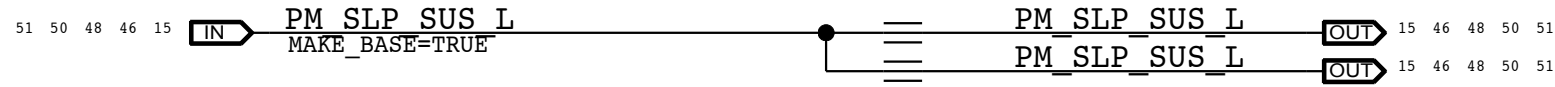
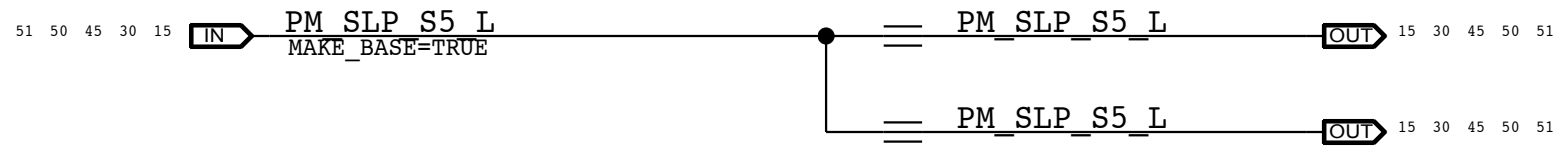
S5 Power Good (3.3V DSW PGOOD based)



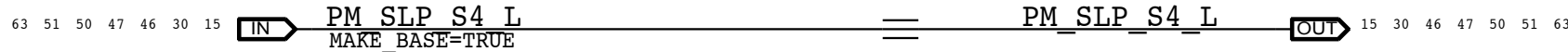
S5 Enables



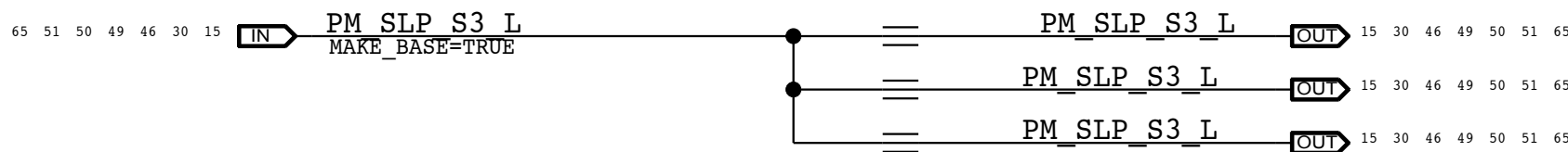
SUS & S4 Enables



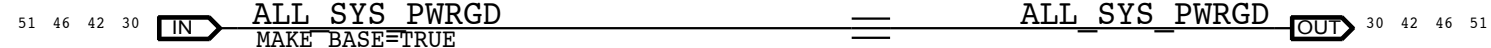
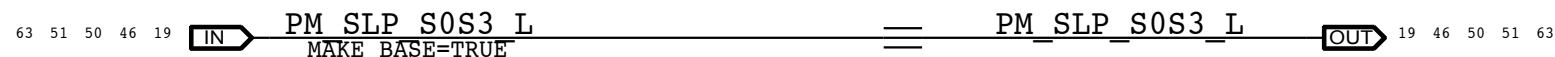
S3 Enables



S0 Enables



S0I Enables



D

D

C

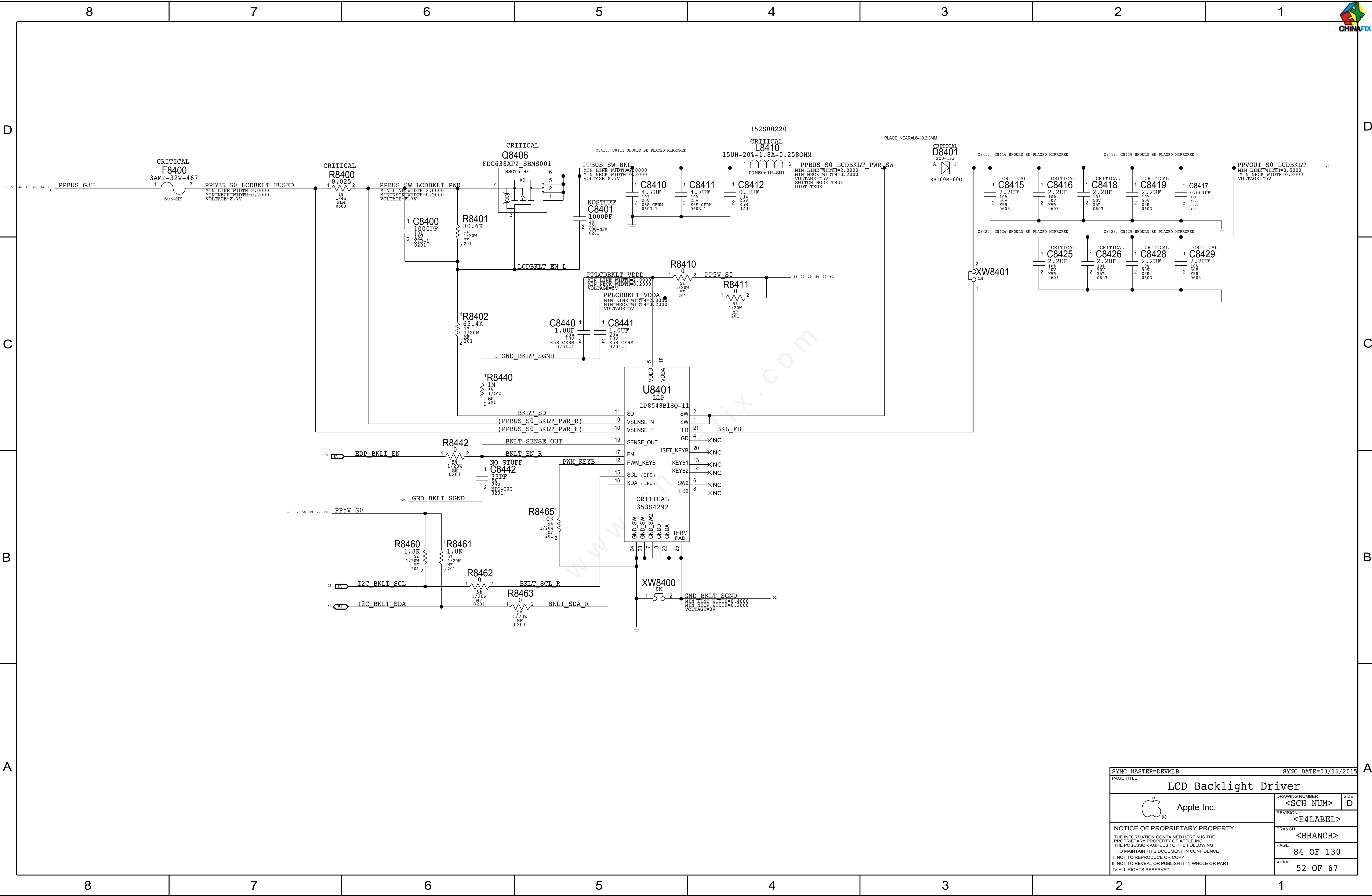
C

B

B

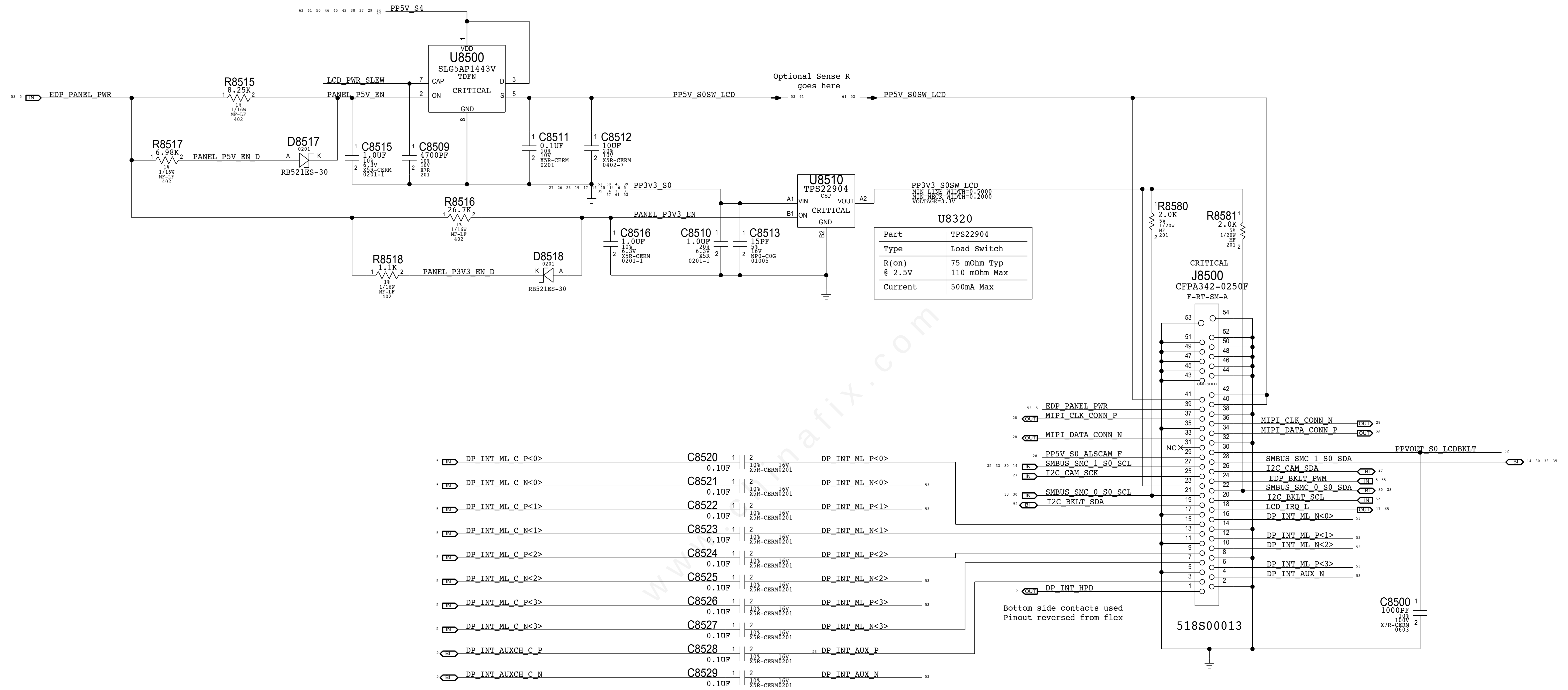
A

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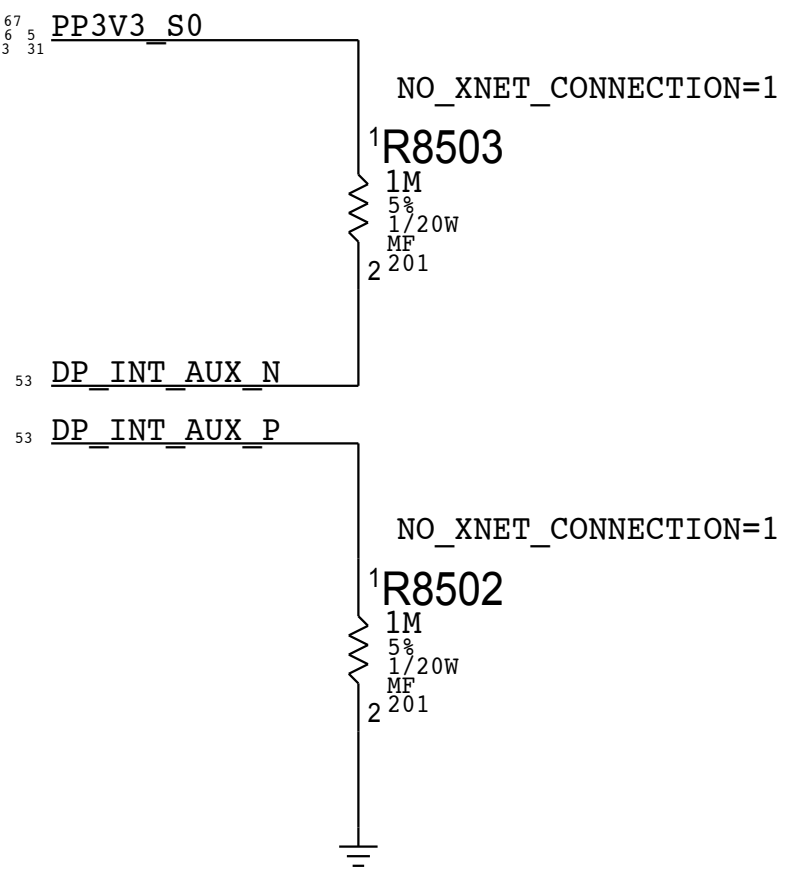


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		PAGE	84 OF 130
		SHEET	52 OF 67

LCD PANEL INTERFACE (eDP) + Camera (MIPI)



LCD Panel HPD & AUX strapping



1

1



OMIT_TABLE			
U8600			
S3X			
BGA			
SYM 2 OF 7			
NC X H2	AN14_I00	AN15_I00	R2 X NC
NC X S3	AN14_I01	AN15_I01	P3 X NC
NC X S2	AN14_I02	AN15_I02	P2 X NC
NC X G1	AN14_I03	AN15_I03	P1 X NC
NC X D2	AN14_I04	AN15_I04	L7 X NC
NC X D1	AN14_I05	AN15_I05	L1 X NC
NC X C2	AN14_I06	AN15_I06	K2 X NC
NC X C1	AN14_I07	AN15_I07	K1 X NC
NC X G6	AN14_NCE(0)	AN15_NCE(0)	B3 X NC
NC X E7	AN14_NCE(1)	AN15_NCE(1)	D4 X NC
NC X E6	AN14_NCE(2)	AN15_NCE(2)	E4 X NC
NC X G7	AN14_NCE(3)	AN15_NCE(3)	B4 X NC
NC X G8	AN14_NCE(4)	AN15_NCE(4)	A4 X NC
NC X A6	AN14_NCE(5)	AN15_NCE(5)	D5 X NC
NC X H9	AN14_NCE(6)	AN15_NCE(6)	B5 X NC
NC X G8	AN14_NCE(7)	AN15_NCE(7)	A5 X NC
NC X F6	AN14_ALE	AN15_ALE	F4 X NC
NC X C3	AN14_CLE	AN15_CLE	A3 X NC
NC X D6	AN14_NWE	AN15_NWE	K3 X NC
NC X B6	AN14_RNB	AN15_RNB	C4 X NC
NC X C6	AN14_PPM_IN	AN15_PPM_IN	G5 X NC
NC X E2	AN14_PPM_OUT	AN15_PPM_OUT	E5 X NC
NC X F1	AN14_NRE_P	AN15_NRE_P	M2 X NC
NC X F1	AN14_NRE_N	AN15_NRE_N	M1 X NC
NC X F2	AN14_DQS_P	AN15_DQS_P	N2 X NC
NC X F1	AN14_DQS_N	AN15_DQS_N	N1 X NC
NC X H1	AN14_VREF	AN15_VREF	R1 X NC
NC X C2	AN16_I00	AN17_I00	T2 X NC
NC X D1	AN16_I01	AN17_I01	U1 X NC
NC X D2	AN16_I02	AN17_I02	U2 X NC
NC X D3	AN16_I03	AN17_I03	U3 X NC
NC X G1	AN16_I04	AN17_I04	Y1 X NC
NC X G2	AN16_I05	AN17_I05	Y2 X NC
NC X H1	AN16_I06	AN17_I06	AA1 X NC
NC X H2	AN16_I07	AN17_I07	AA2 X NC
NC X A11	AN16_NCE(0)	AN17_NCE(0)	AK5 X NC
NC X J11	AN16_NCE(1)	AN17_NCE(1)	AK7 X NC
NC X H10	AN16_NCE(2)	AN17_NCE(2)	AK1 X NC
NC X H9	AN16_NCE(3)	AN17_NCE(3)	AK6 X NC
NC X H12	AN16_NCE(4)	AN17_NCE(4)	AK6 X NC
NC X H11	AN16_NCE(5)	AN17_NCE(5)	AK7 X NC
NC X G9	AN16_NCE(6)	AN17_NCE(6)	AG7 X NC
NC X G11	AN16_NCE(7)	AN17_NCE(7)	AG5 X NC
NC X J9	AN16_ALE	AN17_ALE	AG8 X NC
NC X H9	AN16_CLE	AN17_CLE	AK6 X NC
NC X H3	AN16_NWE	AN17_NWE	AA3 X NC
NC X G12	AN16_RNB	AN17_RNB	AF6 X NC
NC X G10	AN16_PPM_IN	AN17_PPM_IN	AG8 X NC
NC X F10	AN16_PPM_OUT	AN17_PPM_OUT	AK6 X NC
NC X F1	AN16_NRE_P	AN17_NRE_P	W1 X NC
NC X F2	AN16_NRE_N	AN17_NRE_N	W2 X NC
NC X E1	AN16_DQS_P	AN17_DQS_P	V1 X NC
NC X E2	AN16_DQS_N	AN17_DQS_N	V2 X NC
NC X C1	AN16_VREF	AN17_VREF	T1 X NC

OMIT_TABLE			
U8600			
S3X			
BGA			
SYM 7 OF 7			
G23	VSS	VSS	P13
G24	VSS	VSS	P15
G25	VSS	VSS	P17
G26	VSS	VSS	P19
G27	VSS	VSS	P21
H3	VSS	VSS	P23
H4	VSS	VSS	P27
H6	VSS	VSS	R3
H7	VSS	VSS	R5
H8	VSS	VSS	R8
H11	VSS	VSS	R10
H12	VSS	VSS	R12
H13	VSS	VSS	R14
H14	VSS	VSS	R16
H19	VSS	VSS	R18
H21	VSS	VSS	R20
H23	VSS	VSS	R22
H25	VSS	VSS	R26
H27	VSS	VSS	R28
H28	VSS	VSS	T3
J5	VSS	VSS	T4
J8	VSS	VSS	T7
J10	VSS	VSS	T9
J18	VSS	VSS	T11
J22	VSS	VSS	T13
J24	VSS	VSS	T15
J26	VSS	VSS	T17
K4	VSS	VSS	T19
K9	VSS	VSS	T21
K11	VSS	VSS	T23
K21	VSS	VSS	T24
K27	VSS	VSS	T27
L3	VSS	VSS	T28
L5	VSS	VSS	U5
L8	VSS	VSS	U8
L10	VSS	VSS	U10
L12	VSS	VSS	U12
L14	VSS	VSS	U14
L16	VSS	VSS	U16
L18	VSS	VSS	U18
L20	VSS	VSS	U20
L22	VSS	VSS	U22
L26	VSS	VSS	U26
L28	VSS	VSS	V3
M3	VSS	VSS	V4
M4	VSS	VSS	V9
M6	VSS	VSS	V11
M7	VSS	VSS	V13
M9	VSS	VSS	V19
M11	VSS	VSS	V21
M13	VSS	VSS	V23
M15	VSS	VSS	V27
M17	VSS	VSS	V28
M19	VSS	VSS	W3
M21	VSS	VSS	W5
M23	VSS	VSS	W8
M24	VSS	VSS	W10
M25	VSS	VSS	W12
M27	VSS	VSS	W14
M28	VSS	VSS	W16
N3	VSS	VSS	W20
N5	VSS	VSS	W22
N8	VSS	VSS	W26
N10	VSS	VSS	W28
N12	VSS	VSS	Y3
N14	VSS	VSS	Y4
N16	VSS	VSS	Y9
N18	VSS	VSS	Y11
N20	VSS	VSS	Y13
N22	VSS	VSS	Y15
N26	VSS	VSS	Y17
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P9	VSS	VSS	Y28
P11	VSS	VSS	V15

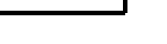
VSS_S TP_S3X_VSS_S

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U8600			
S3X			
BGA			
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U24	VSS_PLL5	VSS	AG21
K24	VSS_PLL6	VSS	AG23
N24	VSS_PLL7	VSS	AG25
K7	VSS_PLL8	VSS	AG27
N7	VSS_PLL9	VSS	AG28
Y7	VSS_PLL10	VSS	AH4
U7	VSS_PLL11	VSS	AH14
V17	VSS_PLL_C	VSS	AH27
A1	VSS	VSS	AJ1
A2	VSS	VSS	AJ2
A9	VSS	VSS	AJ4
A20	VSS	VSS	AJ6
A29	VSS	VSS	AJ10
A30	VSS	VSS	AJ12
AA5	VSS	VSS	AJ13
AA12	VSS	VSS	AJ14
AA14	VSS	VSS	AJ17
AA16	VSS	VSS	AJ19
AA18	VSS	VSS	AJ21
AA20	VSS	VSS	AJ25
AA22	VSS	VSS	AJ27
AA24	VSS	VSS	AJ29
AA26	VSS	VSS	AJ30
AB5	VSS	VSS	AK1
AB8	VSS	VSS	AK2
AB10	VSS	VSS	AK29
AB14	VSS	VSS	AK30
AB17	VSS	VSS	B1
AB18	VSS	VSS	B2
AB21	VSS	VSS	B9
AB23	VSS	VSS	B10
AB26	VSS	VSS	B11
AC3	VSS	VSS	B12
AC4	VSS	VSS	B13
AC5	VSS	VSS	B14
AC7	VSS	VSS	B15
AC9	VSS	VSS	B16
AC11	VSS	VSS	B17
AC13	VSS	VSS	B18
AC15	VSS	VSS	B27
AC17	VSS	VSS	B28
AC18	VSS	VSS	B29
AC19	VSS	VSS	B30
AC20	VSS	VSS	C5
AC22	VSS	VSS	C7
AC24	VSS	VSS	C9
AC26	VSS	VSS	C18
AC27	VSS	VSS	C22
AC28	VSS	VSS	C23
AD4	VSS	VSS	C24
AD5	VSS	VSS	C25
AD6	VSS	VSS	C26
AD7	VSS	VSS	C27
AD8	VSS	VSS	D3
AD9	VSS	VSS	D9
AD14	VSS	VSS	D10
AD16	VSS	VSS	D11
AD18	VSS	VSS	D12
AD19	VSS	VSS	D13
AD24	VSS	VSS	D14
AD25	VSS	VSS	D15
AD26	VSS	VSS	D16
AD27	VSS	VSS	D17
AE3	VSS	VSS	D18
AE4	VSS	VSS	D27
AE5	VSS	VSS	D28
AE6	VSS	VSS	E3
AE7	VSS	VSS	E13
AE8	VSS	VSS	E14
AE9	VSS	VSS	E15
AE10	VSS	VSS	E16
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AE19	VSS	VSS	E28
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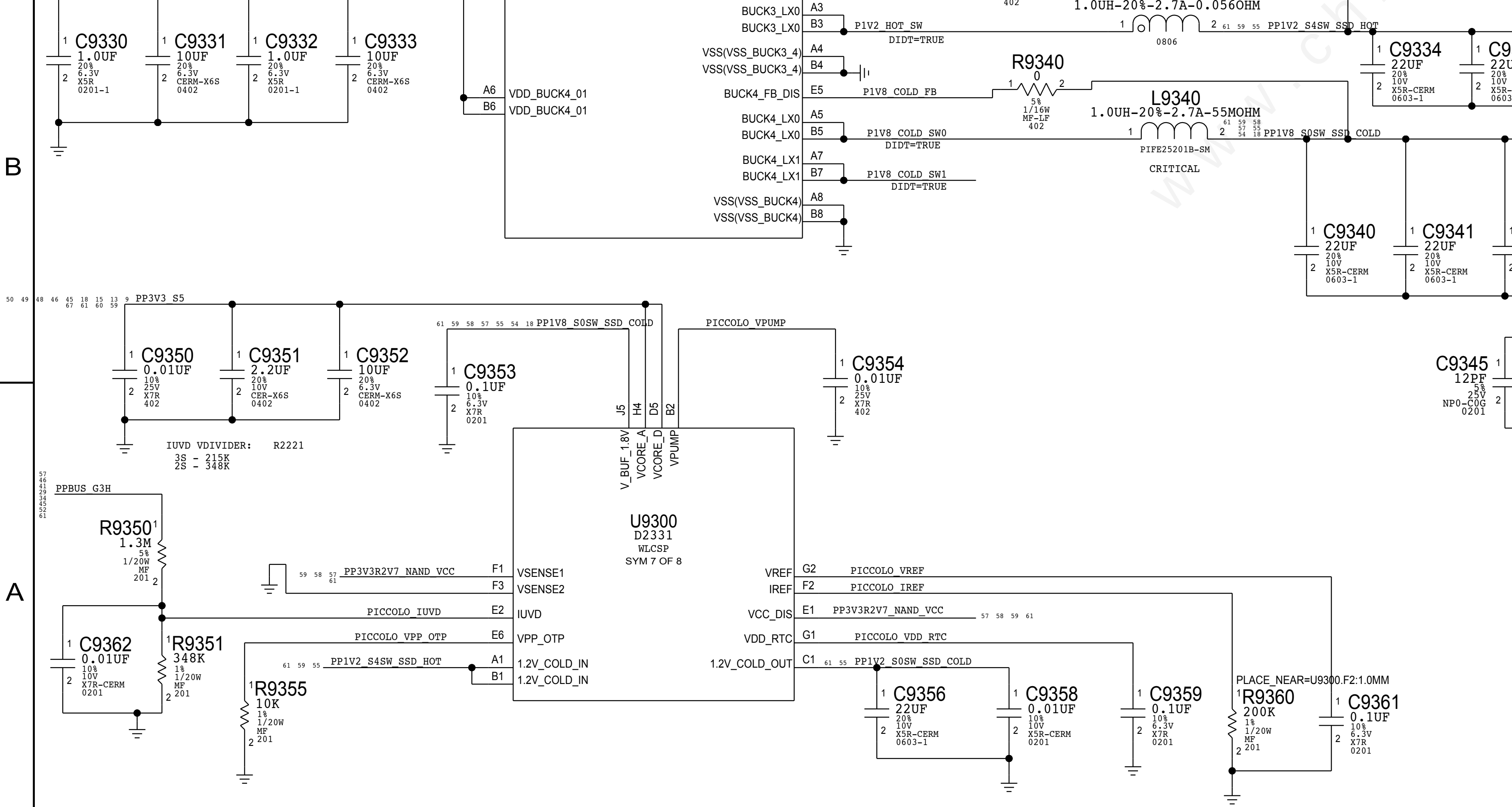
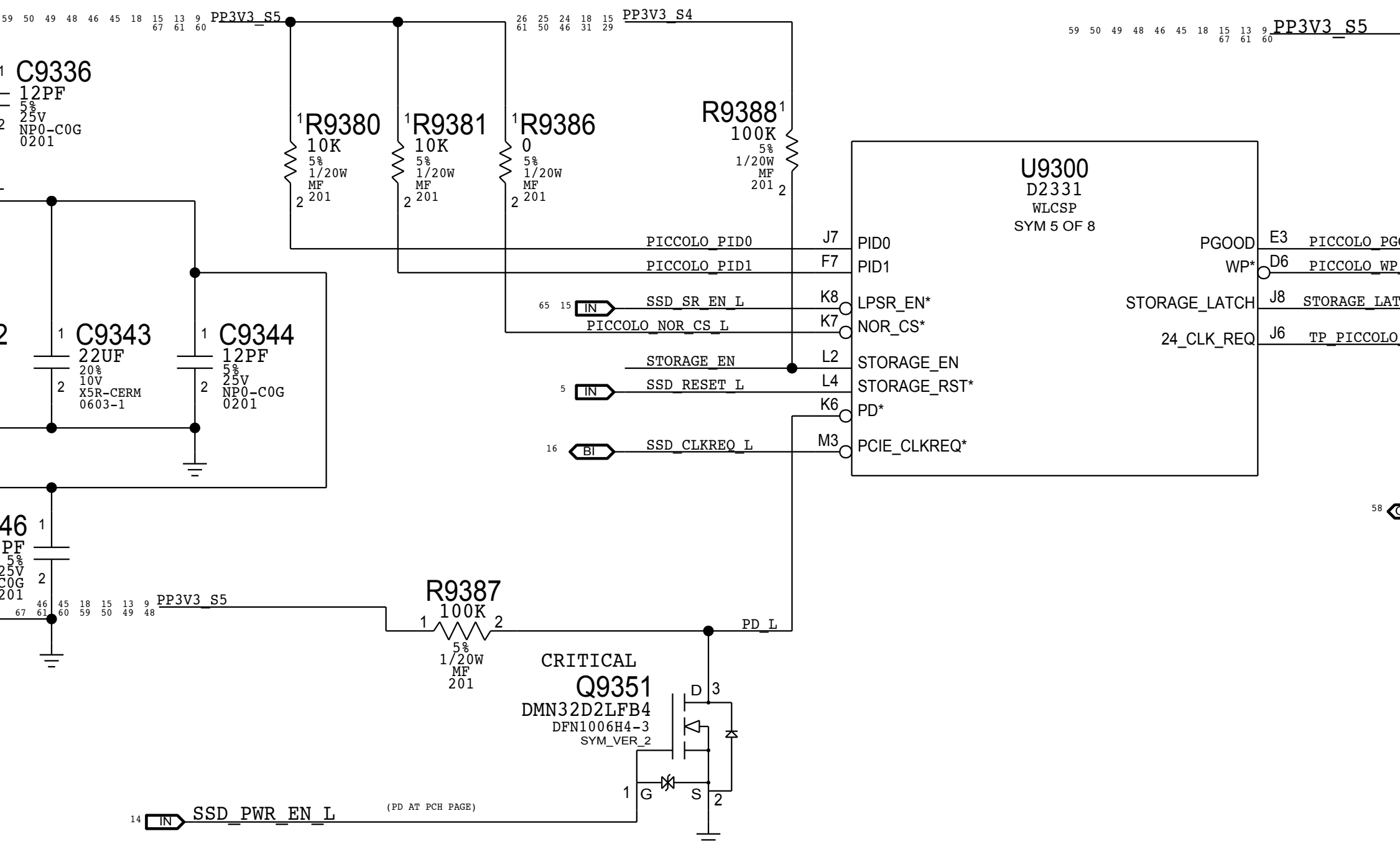
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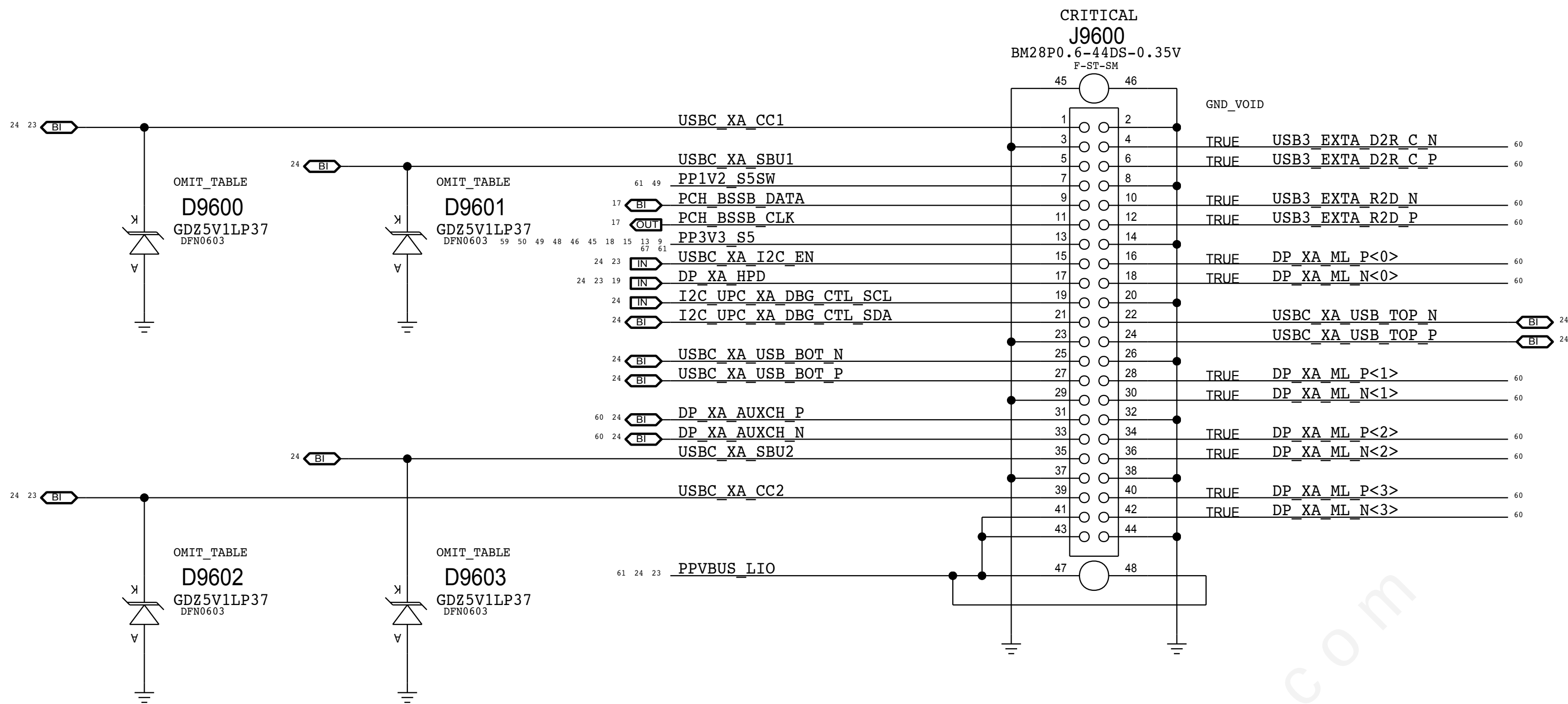






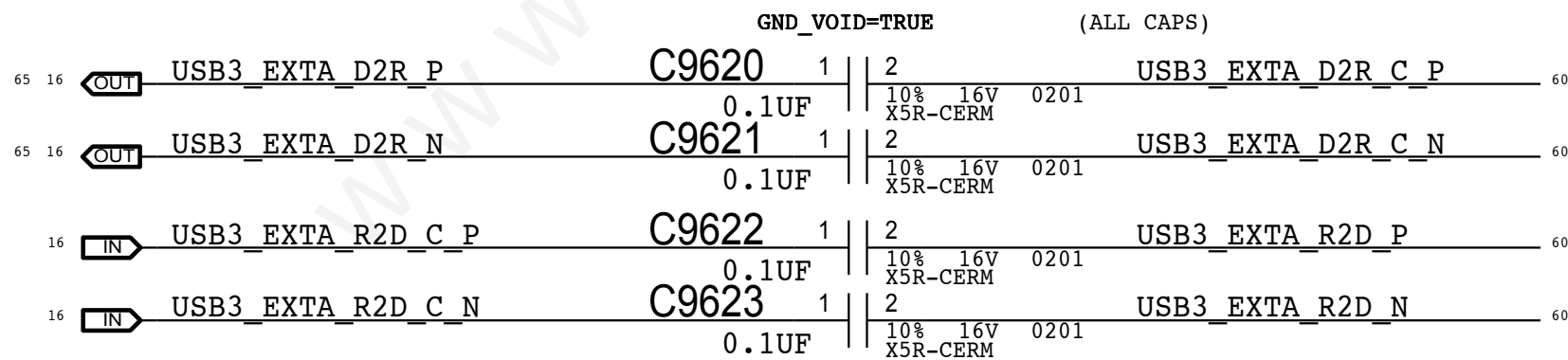


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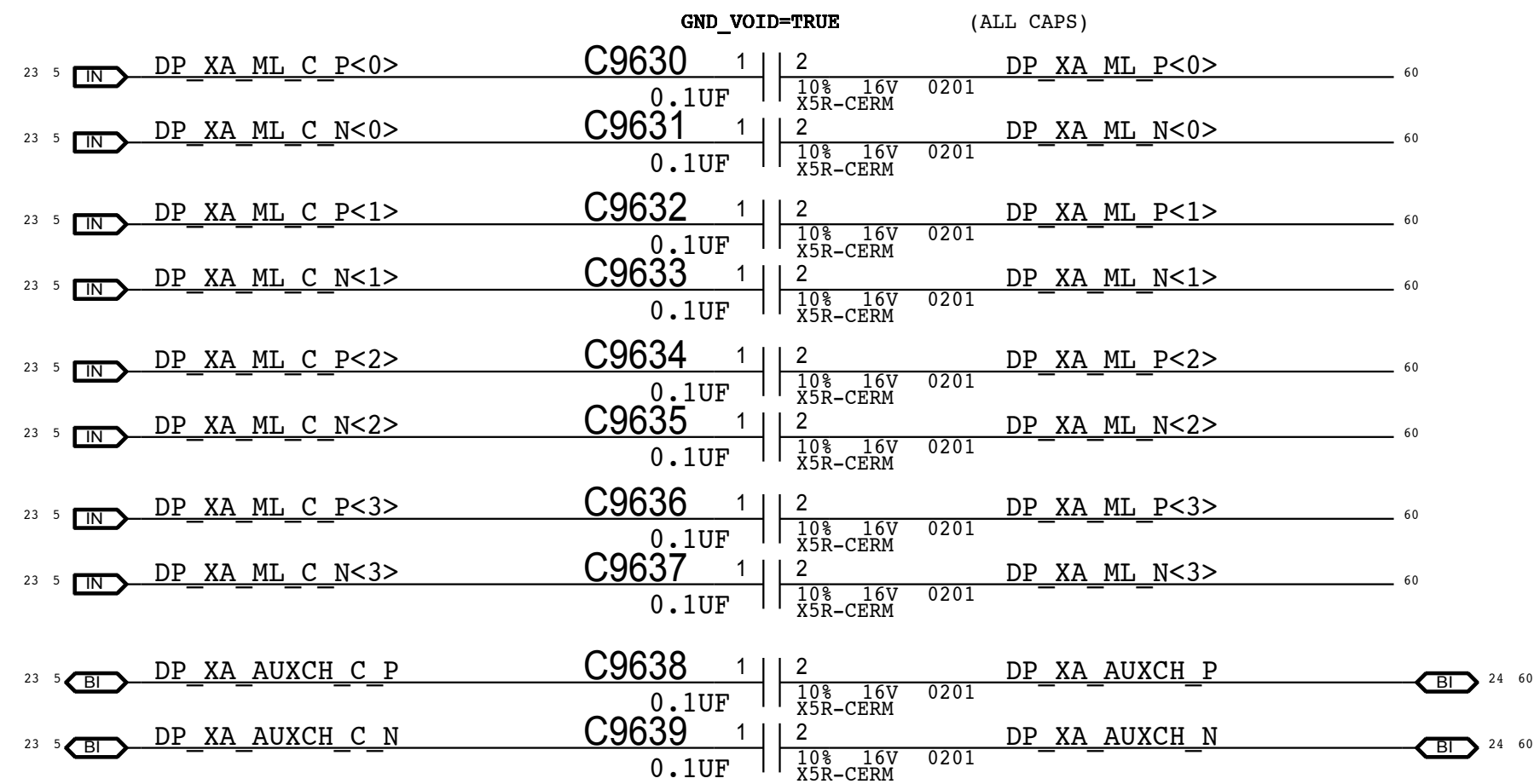


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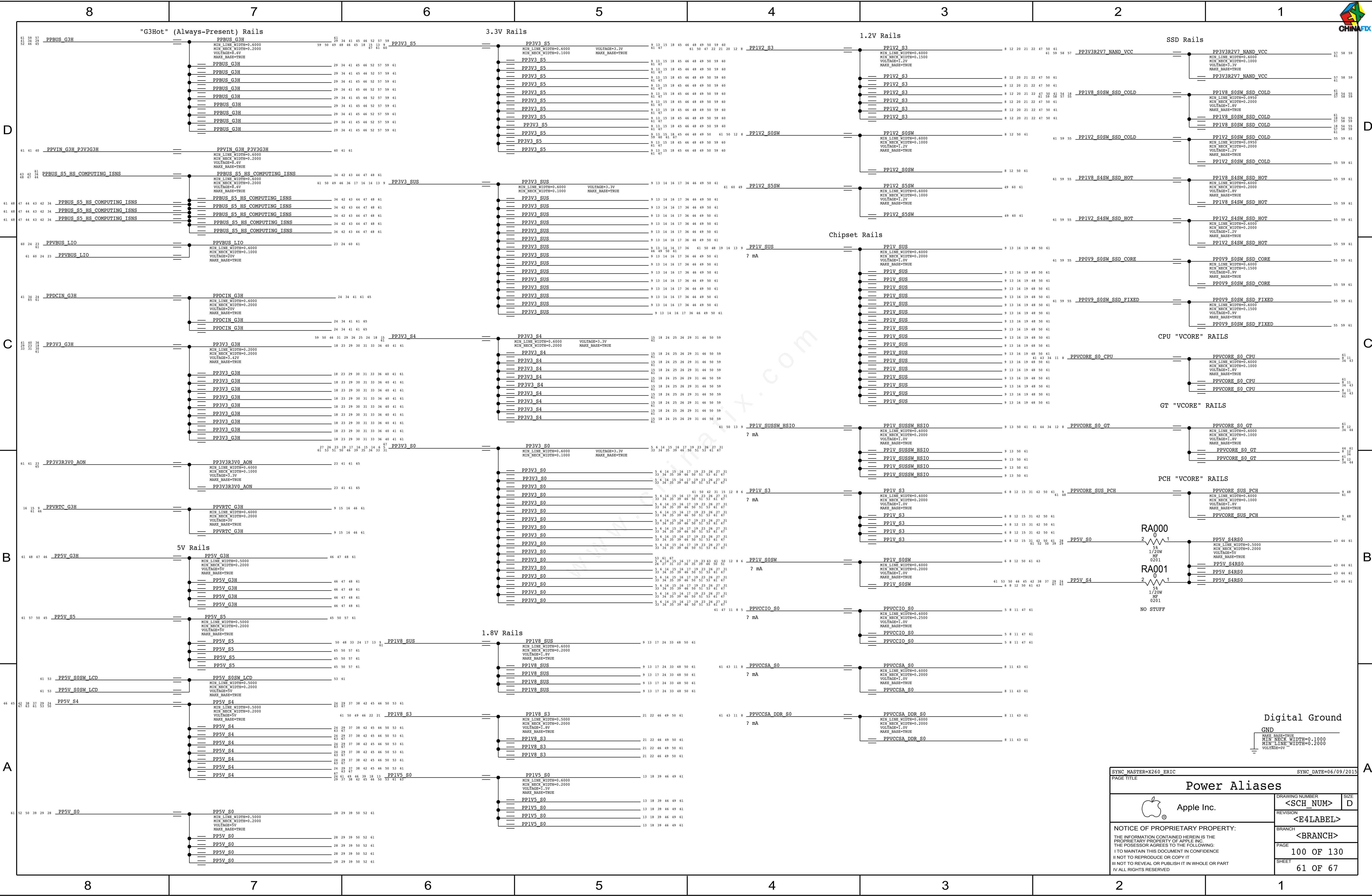
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DP SNK0 AC COUPLING



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		SHEET 60 OF 67	



SSD Rails

PP3V3R2V7 NAND_VCC

PP3V3R2V7 NAND_VCC

PP1V8_S0SW_SSD_COLD

PP1V8_S0SW_SSD_COLD

PP1V2_S0SW_SSD_COLD

PP1V2_S0SW_SSD_COLD

PP1V8_S4SW_SSD_HOT

PP1V8_S4SW_SSD_HOT

PP1V2_S4SW_SSD_HOT

PP1V2_S4SW_SSD_HOT

PP0V9_S0SW_SSD_CORE

PP0V9_S0SW_SSD_CORE

PP0V9_S0SW_SSD_FIXED

PP0V9_S0SW_SSD_FIXED

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CPU "VCORE" RAILS

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PCH "VCORE" RAILS

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PPVCORE_SUS_PCH

Digital Ground

GND

GND

GND

GND

Power Aliases

Apple Inc.

Apple Inc.

Apple Inc.

Apple Inc.

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Memory Bit/Byte Swizzle

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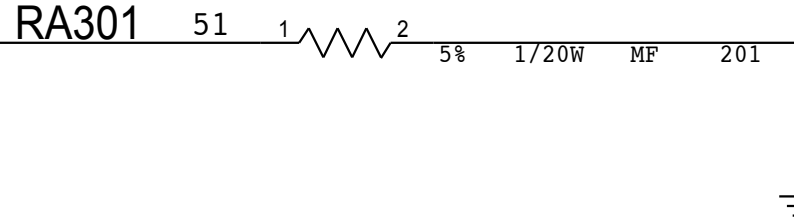
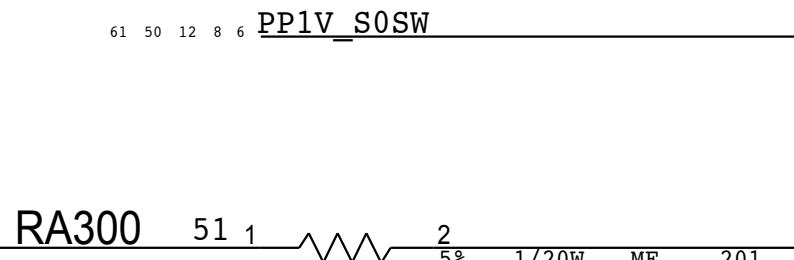
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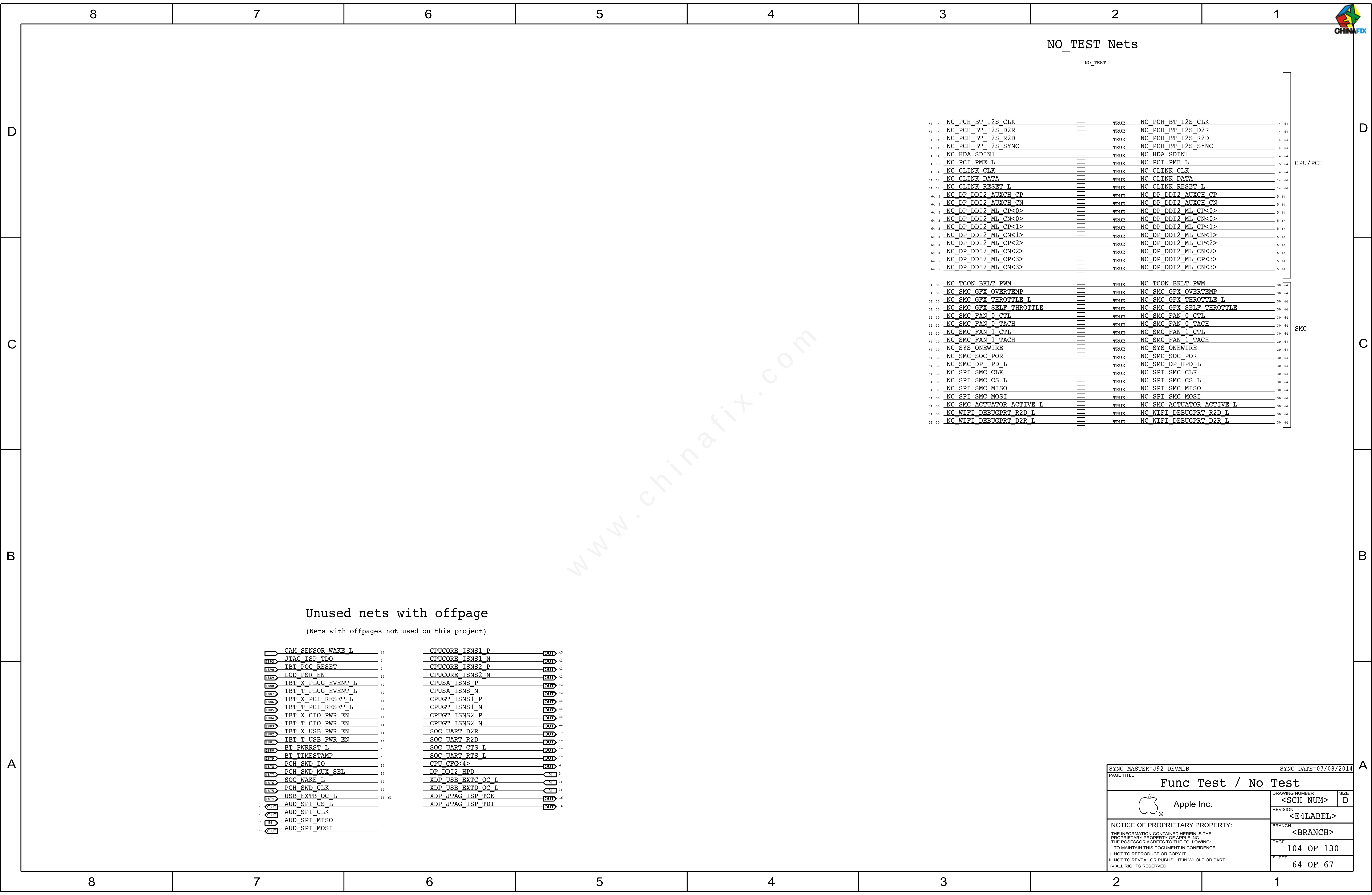
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SHEET		63 OF 67	

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=PCIE_SSD_R2D_P<3..2>				XDP_CPUPCH_TCK			
=PCIE_SSD_R2D_N<3..2>				XDP_CPUPCH_TDI			
PCIE_SSD_R2D_C_P<3..2>				XDP_CPUPCH_TDO			
PCIE_SSD_R2D_C_N<3..2>				XDP_CPUPCH_TMS			
=PCIE_SSD_D2R_P<1..0>				XDP_CPUPCH_TRST_L			
=PCIE_SSD_D2R_N<1..0>				TP_XDP_PCH_OBSDATA_A2			
PCIE_SSD_D2R_C_P<3..2>				TP_XDP_PCH_OBSDATA_A3			
PCIE_SSD_D2R_C_N<3..2>				TP_XDP_PCH_OBSDATA_B0			
PCIE_SSD_D2R_P<3..2>				TP_XDP_PCH_OBSDATA_C0			
PCIE_SSD_D2R_N<3..2>				TP_XDP_PCH_OBSDATA_C1			
PCIE_CLK100M_SSD_P				TP_XDP_PCH_OBSDATA_C2			
PCIE_CLK100M_SSD_N				TP_XDP_PCH_OBSDATA_C3			
S3X_DEBUG_UART_R2D				TP_XDP_PCH_OBSDATA_D0			
SSD_DEBUG_UART_D2R				TP_XDP_PCH_OBSDATA_D1			
SSD_DEBUG_UART_R2D				TP_XDP_PCH_OBSDATA_D2			
PM_SLP_S0S3_L				TP_XDP_PCH_OBSDATA_D3			
PM_SLP_S4_L				TP_XDP_PCH_OBSFN_C1			
SYSCLK_CLK24M_PCH				TP_XDP_PCH_TCK			
NC_PCH_CLK24M_XTALOUT				XDP_CPUPCH_TDI			
NC_PCH_DDPC_CTRLDATA				XDP_CPUPCH_TDO			
				XDP_CPUPCH_TMS			
				XDP_CPUPCH_TRST_L			
				XDP_CPUPCH_TCK			
				TP_ITP_PMODE			
				TP_CPU_MEMVTT_PWR_EN			
				TP_SMC_DEV_MLB			
				S3X_JTAG_TCK			
				S3X_JTAG_TMS			
				TP_SMC_DEBUGPRT_EN_L			
				TP_SPKR_ID0			
				TP_SPKR_ID1			





8		7		6		5		4		3		2		1	
EE Chaz Probe Points															
D	PPA504	P2MM1 PP SH	SMC_CBC_ON	30	41	PPA577	P2MM1 PP SH	USB3_EXT_A_D2R_N	16	60	TPA500	A SH	PPDCIN_G3H	24	34 41 61 65
	PPA505	P2MM1 PP SH	SMC_CHGR_INT_L	30	41	PPA578	P2MM1 PP SH	USB3_EXT_A_D2R_P	16	60	TPA501	A SH	PPDCIN_G3H	24	34 41 61 65
	PPA506	P2MM1 PP SH	MEM_A_CAA<0>	7	21	PPA579	P2MM1 PP SH	CPU_CATERR_L	6	30	TPA502	A SH	PPDCIN_G3H	24	34 41 61 65
	PPA507	P2MM1 PP SH	MEM_A_CLK_P<0>	7	21	PPA581	P2MM1 PP SH	MEM_B_CS_L<1>	7	22	TPA503	A SH	PPDCIN_G3H	24	34 41 61 65
	PPA508	P2MM1 PP SH	MEM_A_CLK_N<0>	7	21	PPA582	P2MM1 PP SH	MEM_B_CKE<1>	7	22	TPA504	A SH	GND		
	PPA509	P2MM1 PP SH	MEM_A_CS_L<0>	7	21	PPA583	P2MM1 PP SH	MEM_B_CAB<1>	7	22	TPA505	A SH	GND		
	PPA510	P2MM1 PP SH	MEM_B_DQ<12>	7	22 62	PPA584	P2MM1 PP SH	MEM_B_CAA<2>	7	22	TPA506	A SH	SMC_ONOFF_L	29	30 31 32
	PPA511	P2MM1 PP SH	MEM_A_DQS_P<1>	7	21 62	PPA587	P2MM1 PP SH	MEM_B_CAA<3>	7	22	TPA507	A SH	SSD_BOOT_L	17	54
	PPA512	P2MM1 PP SH	MEM_A_DQS_N<1>	7	21 62	PPA594	P2MM1 PP SH	XDP_USB_EXT_A_OC_L	16	23 24 63	TPA508	A SH	SMC_DEV_SUPPLY_L	30	
	PPA513	P2MM1 PP SH	AP_DEV_WAKE	14	25	PPA596	P2MM1 PP SH	USB_EXT_A_P	16	23	TPA509	A SH	GND		
C	PPA514	P2MM1 PP SH	AP_S0IX_WAKE_L	17	18	PPA597	P2MM1 PP SH	USB_EXT_A_N	16	23	TPA510	A SH	SMC_DEBUGPRT_TX_L	30	31 65
	PPA515	P2MM1 PP SH	AP_S0IX_WAKE_SEL	17	18	PPA598	P2MM1 PP SH	PM_DSW_PWRGD	15	30	TPA511	A SH	SMC_DEBUGPRT_RX_L	30	31 65
	PPA517	P2MM1 PP SH	MIPI_CLK_P	27	28	PPA500	P2MM1 PP SH	AP_CLKREQ_L	16	25	TPA512	A SH	SSD_UART_R2D	17	63
	PPA518	P2MM1 PP SH	MIPI_DATA_P	27	28	PPA501	P2MM1 PP SH	AP_RESET_L	14	25	TPA513	A SH	SSD_UART_D2R	17	63
	PPA519	P2MM1 PP SH	PCH_BT_UART_D2R	17	26	PPA502	P2MM1 PP SH	CAMERA_CLKREQ_L	16	27	TPA514	A SH	PCIE_TEST_D2R_P	16	65
	PPA520	P2MM1 PP SH	PCH_BT_UART_RTS_L	17	26	PPA503	P2MM1 PP SH	CAMERA_RESET_L	14	27	TPA515	A SH	PCIE_TEST_D2R_N	16	65
	PPA522	P2MM1 PP SH	PCIE_CAMERA_D2R_P	16	28	PPA521	P2MM1 PP SH	CHGR_BOOST_DH	41		TPA516	A SH	PCIE_TEST_R2D_P	16	65
	PPA523	P2MM1 PP SH	PCIE_CLK100M_SSD_P	16	54 63	PPA586	P2MM1 PP SH	CHGR_BOOST_DL	41		TPA517	A SH	PCIE_TEST_R2D_N	16	65
	PPA529	P2MM1 PP SH	AUD_PWR_EN	17	39	PPA524	P2MM1 PP SH	CHGR_BUCK_DH	41		TPA518	A SH	PCIE_CLK100M_TEST_P	16	65
	PPA530	P2MM1 PP SH	BT_UART_RTS_L	25	26	PPA525	P2MM1 PP SH	CHGR_BUCK_DL	41		TPA519	A SH	PCIE_CLK100M_TEST_N	16	65
B	PPA531	P2MM1 PP SH	CAMERA_PWR_EN	14	27	PPA526	P2MM1 PP SH	CPUCORE_FCCM	42	43	TPA520	A SH	TEST_CLKREQ_L	16	65
	PPA533	P2MM1 PP SH	EDP_BKLT_PWM	5	53	PPA527	P2MM1 PP SH	CPUGT_FCCM	42	44	TPA521	A SH	PLT_RST_L	5 14 15 19 26 30	
	PPA534	P2MM1 PP SH	HDA_SDIN0	14	39	PPA528	P2MM1 PP SH	CPUSA_FCCM	42	43	TPA522	A SH	USB_TEST_P	16	65
	PPA537	P2MM1 PP SH	LCD_IRQ_L	17	53	PPA532	P2MM1 PP SH	CPUCORE_PWM1	42	43	TPA523	A SH	USB_TEST_N	16	65
	PPA538	P2MM1 PP SH	MEM_CAM_A<8>	27	28	PPA535	P2MM1 PP SH	CPUCORE_PWM2	42	43	TPA524	A SH	GND		
	PPA539	P2MM1 PP SH	MEM_CAM_CS_L	27	28	PPA536	P2MM1 PP SH	CPUGT_PWM1	42	44	TPA525	A SH	GND		
	PPA543	P2MM1 PP SH	MIPI_CLK_N	27	28	PPA540	P2MM1 PP SH	CPUGT_PWM2	42	44	TPA526	A SH	POWER_GATE_EN	23 24 41	
	PPA544	P2MM1 PP SH	MIPI_DATA_N	27	28	PPA541	P2MM1 PP SH	CPUSA_PWM	42	43	TPA527	A SH	PP3V3R3V0_AON	23 41 61	
	PPA545	P2MM1 PP SH	PCH_BT_UART_CTS_L	17	26	PPA542	P2MM1 PP SH	P1V2_DRVH	47		TPA528	A SH	HIPWR_EN	23 41	
	PPA546	P2MM1 PP SH	PCH_BT_UART_R2D	17	26	PPA547	P2MM1 PP SH	P1V2_DRVL	47		TPA529	A SH	PM_SLP_S3_L	15 30 46 49 50 51	
A	PPA548	P2MM1 PP SH	PCIE_CAMERA_D2R_N	16	28	PPA589	P2MM1 PP SH	PVCCIO_DRVH	47		TPA529	A SH	PM_SLP_S3_L	15 30 46 49 50 51	
	PPA549	P2MM1 PP SH	PCIE_CLK100M_SSD_N	16	54 63	PPA588	P2MM1 PP SH	PVCCIO_DRVL	47		TPA529	A SH	PM_SLP_S3_L	15 30 46 49 50 51	
	PPA551	P2MM1 PP SH	PCIE_SSD_D2R_N<0>	16	63	PPA550	P2MM1 PP SH	P1VSUS_DRVH	48		TPA529	A SH	PM_SLP_S3_L	15 30 46 49 50 51	
	PPA552	P2MM1 PP SH	PCIE_SSD_D2R_P<0>	16	63	PPA590	P2MM1 PP SH	P1VSUS_DRVL	48		TPA529	A SH	PM_SLP_S3_L	15 30 46 49 50 51	
	PPA560	P2MM1 PP SH	SMC_BT_PWR_EN	25	30	PPA553	P2MM1 PP SH	LPC_SERIRQ	14	30	TPA529	A SH	PM_SLP_S3_L	15 30 46 49 50 51	
	PPA562	P2MM1 PP SH	SMC_OOB1_R2D_L	30	59	PPA554	P2MM1 PP SH	SMC_CLK32K	30	31	TPA529	A SH	PM_SLP_S3_L	15 30 46 49 50 51	
	PPA563	P2MM1 PP SH	SMC_PROCHOT	30	31	PPA555	P2MM1 PP SH	SYSCLK_CLK32K_PCH	16	18	TPA529	A SH	PM_SLP_S3_L	15 30 46 49 50 51	
	PPA564	P2MM1 PP SH	SMC_SENSOR_PWR_EN	30	32 34	PPA556	P2MM1 PP SH	SYSCLK_CLK32K_CAMERA_BT_AP	18 25		TPA529	A SH	PM_SLP_S3_L	15 30 46 49 50 51	
	PPA565	P2MM1 PP SH	SMC_WAKE_SCI_L	15	30	PPA557	P2MM1 PP SH	SYSCLK_CLK24M_PCH	16	18 63	TPA529	A SH	PM_SLP_S3_L	15 30 46 49 50 51	
	PPA566	P2MM1 PP SH	SMC_WIFI_PWR_EN	25	30	PPA558	P2MM1 PP SH	TPAD_SPI_IF_EN	17 29		TPA529	A SH	PM_SLP_S3_L	15 30 46 49 50 51	
A	PPA567	P2MM1 PP SH	AP_PCIE_WAKE_L	18	25	PPA559	P2MM1 PP SH	CPUCORE_GL1	43		TPA529	A SH	PM_SLP_S3_L	15 30 46 49 50 51	
	PPA569	P2MM1 PP SH	SSD_SR_EN_L	15	59	PPA561	P2MM1 PP SH	CPUCORE_GL2	43		TPA529	A SH	PM_SLP_S3_L	15 30 46 49 50 51	
	PPA570	P2MM1 PP SH	SYSCLK_CLK12M_SMC	18	30	PPA568	P2MM1 PP SH	CPUGT_GL1	44		TPA529	A SH	PM_SLP_S3_L	15 30 46 49 50 51	
	PPA571	P2MM1 PP SH	SYSCLK_CLK24M_SSD	18	54	PPA573	P2MM1 PP SH	CPUGT_GL2	44		TPA529	A SH	PM_SLP_S3_L	15 30 46 49 50 51	
	PPA572	P2MM1 PP SH	SYSCLK_CLK24M_CAMERA	18 27 65		PPA574	P2MM1 PP SH	SMC_BC_ACOK	30 31 34 41 63		TPA529	A SH	PM_SLP_S3_L	15 30 46 49 50 51	
	PPA575	P2MM1 PP SH	TPAD_SPI_CS_L	17 29		PPA580	P2MM1 PP SH	P2V7NAND_DRVL	57		TPA529	A SH	PM_SLP_S3_L	15 30 46 49 50 51	
	PPA576	P2MM1 PP SH	TPAD_SPI_INT_L	17 29		PPA585	P2MM1 PP SH	P1V8SUS_PGOOD	46 48		TPA529	A SH	PM_SLP_S3_L	15 30 46 49 50 51	
	PPA576	P2MM1 PP SH	TPAD_SPI_INT_L	17 29		PPA585	P2MM1 PP SH	P1V8SUS_PGOOD	46 48		TPA529	A SH	PM_SLP_S3_L	15 30 46 49 50 51	
	PPA576	P2MM1 PP SH	TPAD_SPI_INT_L	17 29		PPA585	P2MM1 PP SH	P1V8SUS_PGOOD	46 48		TPA529	A SH	PM_SLP_S3_L	15 30 46 49 50 51	
	PPA576	P2MM1 PP SH	TPAD_SPI_INT_L	17 29		PPA585	P2MM1 PP SH	P1V8SUS_PGOOD	46 48		TPA529	A SH	PM_SLP_S3_L	15 30 46 49 50 51	
SSD & WiFi Chamber Test Points															
TPA500 PPDCIN_G3H 24 34 41 61 65															
TPA501 PPDCIN_G3H 24 34 41 61 65															
TPA502 PPDCIN_G3H 24 34 41 61 65															
TPA503 PPDCIN_G3H 24 34 41 61 65															
TPA504 GND															
TPA505 GND															
TPA506 SMC_ONOFF_L 29 30 31 32															
TPA507 SSD_BOOT_L 17 54															
TPA508 SMC_DEV_SUPPLY_L 30															
TPA509 GND															
TPA510 SMC_DEBUGPRT_TX_L 30 31 65															
TPA511 SMC_DEBUGPRT_RX_L 30 31 65															
TPA512 SSD_UART_R2D 17 63															
TPA513 SSD_UART_D2R 17 63															
TPA514 PCIE_TEST_D2R_P 16 65															
TPA515 PCIE_TEST_D2R_N 16 65															
TPA516 PCIE_TEST_R2D_P 16 65															
TPA517 PCIE_TEST_R2D_N 16 65															
TPA518 PCIE_CLK100M_TEST_P 16 65															
TPA519 PCIE_CLK100M_TEST_N 16 65															
TPA520 TEST_CLKREQ_L 16 65															
TPA521 PLT_RST_L 5 14 15 19 26 30															
TPA522 USB_TEST_P 16 65															
TPA523 USB_TEST_N 16 65															
TPA524 GND															
TPA525 GND															
TPA526 POWER_GATE_EN 23 24 41															
TPA527 PP3V3R3V0_AON 23 41 61															
TPA528 HIPWR_EN 23 41															
TPA529 PM_SLP_S3_L 15 30 46 49 50 51</															



D

C

B

A

SSD & WiFi Chamber Test Points

65 27 18	SYSCLK_CLK24M_CAMERA	==	SYSCLK_CLK24M_CAMERA	18 27 65
			MAKE_BASE=TRUE	
24 23	SMC_UART_TX	==	TRUE	SMC_DEBUGPRT_TX_L 30 31 65
24 23	SMC_UART_RX	==	TRUE	SMC_DEBUGPRT_RX_L 30 31 65

NO_TEST Nets

		NO_TEST				MAKE_BASE					
65	16	NC_USB3_EXTB_D2RP	==	TRUE	NC_USB3_EXTB_D2RP	16	65	CPU/PCIE			
65	16	NC_USB3_EXTB_D2RN	==	TRUE	NC_USB3_EXTB_D2RN	16	65				
65	16	NC_USB3_EXTB_R2D_CP	==	TRUE	NC_USB3_EXTB_R2D_CP	16	65				
65	16	NC_USB3_EXTB_R2D_CN	==	TRUE	NC_USB3_EXTB_R2D_CN	16	65				
65	16	NC_USB_EXTBP	==	TRUE	NC_USB_EXTBP	16	65				
65	16	NC_USB_EXTBN	==	TRUE	NC_USB_EXTBN	16	65				
65	16	NC_USB3_EXTC_D2RP	==	TRUE	NC_USB3_EXTC_D2RP	16	65				
65	16	NC_USB3_EXTC_D2RN	==	TRUE	NC_USB3_EXTC_D2RN	16	65				
65	16	NC_USB3_EXTC_R2D_CP	==	TRUE	NC_USB3_EXTC_R2D_CP	16	65				
65	16	NC_USB3_EXTC_R2D_CN	==	TRUE	NC_USB3_EXTC_R2D_CN	16	65				
65	16	NC_USB_EXTCP	==	TRUE	NC_USB_EXTCP	16	65				
65	16	NC_USB_EXTCN	==	TRUE	NC_USB_EXTCN	16	65				
65	16	NC_USB3_EXTD_D2RP	==	TRUE	NC_USB3_EXTD_D2RP	16	65				
65	16	NC_USB3_EXTD_D2RN	==	TRUE	NC_USB3_EXTD_D2RN	16	65				
65	16	NC_USB3_EXTD_R2D_CP	==	TRUE	NC_USB3_EXTD_R2D_CP	16	65				
65	16	NC_USB3_EXTD_R2D_CN	==	TRUE	NC_USB3_EXTD_R2D_CN	16	65				
65	16	NC_USB_EXTDP	==	TRUE	NC_USB_EXTDP	16	65				
65	16	NC_USB_EXTDN	==	TRUE	NC_USB_EXTDN	16	65				



Reserved for Constraints!

Change List:

<radar://component/633819>	X260	HW	EE	Schematic	Proto 0
<radar://component/634034>	X260	HW	EE	Schematic	Proto 1
<radar://component/634038>	X260	HW	EE	Schematic	Proto 2
<radar://component/634042>	X260	HW	EE	Schematic	EVT
<radar://component/634046>	X260	HW	EE	Schematic	DVT

Kismet:

<afp://kismet.apple.com/Kismet-Projects/>

Useful Wiki Links:

Schematic Conventions - <<https://hmts.ecs.apple.com/wiki/index.php/User:Wferry/SchConventions>>
Schematic Design Wiki - <[https://hmts.ecs.apple.com/wiki/index.php/Schematic Design](https://hmts.ecs.apple.com/wiki/index.php/Schematic_Design)>

Other Info:

Page Allocations - <radar:19817053> 2015 Mobile Mac Schematic Page Allocations

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		SHEET		66 OF 67	

Power State Debug LEDs

